National Institute for Medical Research

WHO INFLUENZA CENTRE

LONDON

Report prepared for the WHO annual consultation on the composition of influenza vaccine for the Northern Hemisphere 2015/16

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WHO Collaborating Centre for
Reference and Research on Influenza
National Institute for Medical Research
The Ridgeway
Mill Hill
London, NW7 1AA

Dr. John McCauley (Director)
Dr. Rodney Daniels (Deputy Director)
Dr. Yi Pu Lin (Assistant Director)
Dr. Zheng Xiang
Ms Victoria Gregory
Ms Lynne Whittaker
Ms Chandrika Halai
Dr. Karen Cross
Ms Aine Rattigan
Ms Burcu Ermetal
Ms Mian Dai

Tel: 020 8816 2149; 020 8816 2152
Fax: 020 8906 4477
Email: jmccaul@nimr.mrc.ac.uk; whocc@nimr.mrc.ac.uk

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Dr. Caroline Brown
Dr. Dmitriy Pereyaslov
Dr. Pernille Jorgensen.

CONTENTS

Northern Hemisphere Vaccine Recommendation Meeting for 2015-16 influenza season.

WHO CC, London: Final Report for WHO meeting: 23-25 February 2015.

Influenza Activity in the WHO European Region, October 2014 – February 2015

Weekly reporting on influenza activity started in week 40/2014. The course of the 2014-2015 season has seen levels of transmission increasing slowly following week 50/2014 **(Figure 1A)** with a distinct West to East pattern in the spread across the Region **(Figure 2)**.

The proportion of respiratory specimens from patients presenting with ILI or ARI symptoms that have tested positive for influenza virus has likely reached its peak in some countries in the Northern part of the Region, while increasing trends in ILI and ARI with medium and high intensity of influenza activity have been reported for most countries in Western, Southern and Eastern parts of the Region **(Figure 2)**. Countries in the West and Central Asian parts of the Region have, so far, experienced low intensity of influenza activity.

For the whole WHO European Region, over 21,500 influenza detections have been reported to the European Surveillance System (TESSy) - with type A detections (81%) predominating over type B (19%) **(Table 1)**. Within type A, the A(H3N2) subtype (76%) has predominated over A(H1NI)pdm09 viruses (24%) **(Table 1)** and this pattern is evident for both sentinel and non-sentinel influenza detections **(Figure 1 B/C)**. Within type B, 99% of the viruses for which lineage-determination was performed were of the Yamagata lineage **(Table 1).** The majority of countries in the Region have reported influenza A virus detections to outnumber those of influenza B; however, the proportion of influenza B detections has been considerably greater than that observed over the same period for the 2013-2014 season. Influenza A(H3N2) detections have greatly outnumbered A(H1N1)pdm09 detections in most countries, with the exception of some in Southern Europe **(Figures 2 & 3)**.

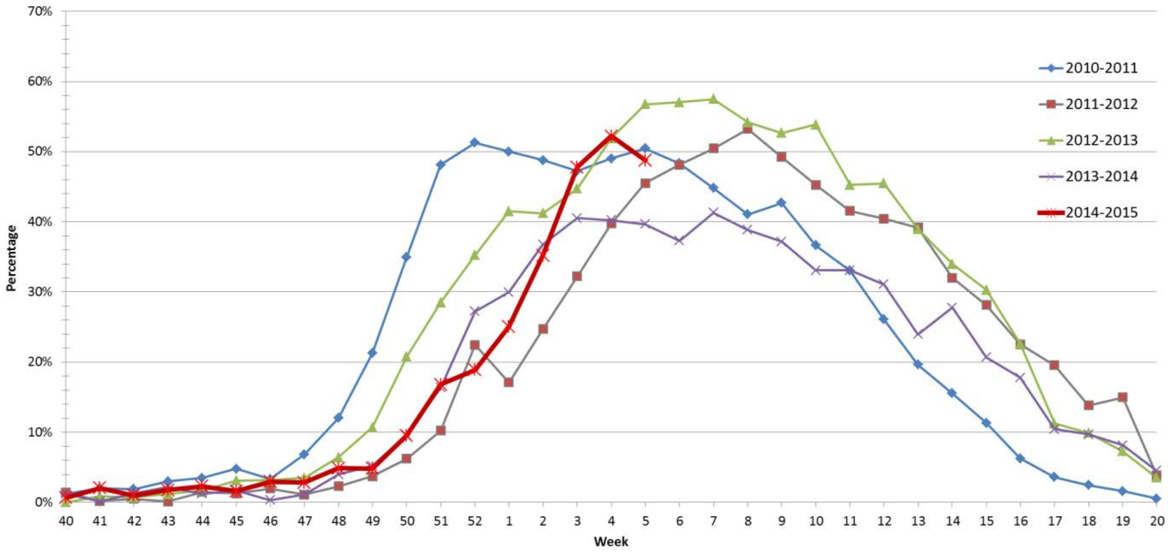
Since week 40/2014, eight countries (Finland, France, Ireland, Romania, Slovakia, Spain, Sweden and the UK) have reported a total of 1651 laboratory-confirmed hospitalized influenza cases **(Figure 4A)**, with type A detections (92%) predominating over type B (8%). Within type A, the A(H3N2) subtype (80%) has predominated over A(H1N1)pdm09 viruses (20%), in proportions similar to that observed in ILI and ARI patients **(Table 1 & Figure 1B/C)**. For hospitalized patients where age was reported, the group aged ≥65 years accounted for the greatest number of cases **(Figure 4B).**

Since week 40/2014, TESSy has received reports indicating that all influenza viruses examined for neuraminidase inhibitor susceptibility (two A (unsubtyped), 404 A(H3N2), 38 A(H1N1)pdm09 and 25 B viruses) showed no evidence of reduced susceptibility to the neuraminidase inhibitors oseltamivir and zanamivir. All viruses assessed for susceptibility to M2 ion channel blockers (adamantanes) were resistant due to the S31N amino acid substitution in the M2 protein.

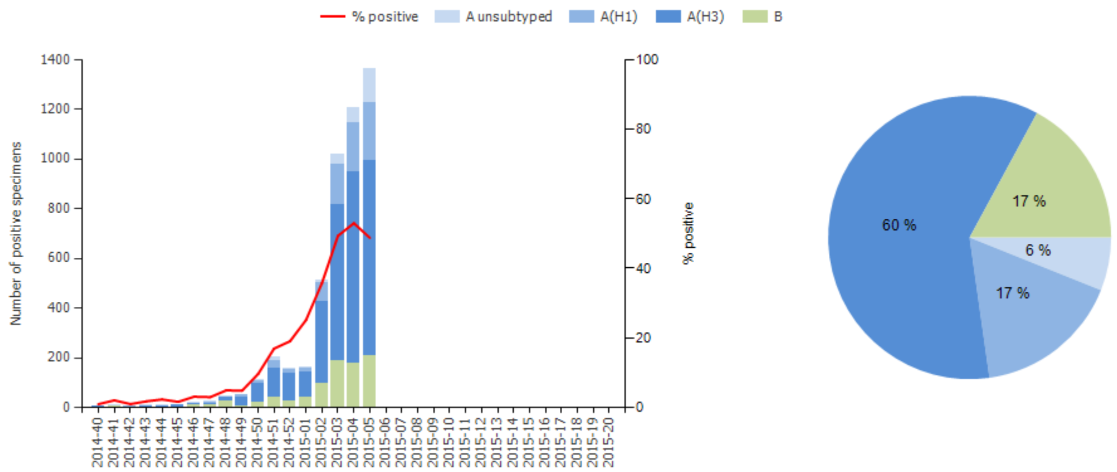
Excess all-cause mortality among the elderly (aged ≥65 years), concomitant with increased influenza activity and the predominance of A(H3N2) viruses, has been observed in recent weeks. Across all countries, a pooled analysis shows a higher level of mortality among elderly people than in the four previous influenza seasons (see: the European project for monitoring excess mortality for public health action (EuroMOMO) – http://www.euromomo.eu).

Figure 1. Influenza Activity in the WHO European Region 2014-2015 (weeks 40/2014-05/2015)

A. Percentage of sentinel ILI/ARI specimens testing positive for influenza by season



B. Sentinel Influenza Detections



C. Non-sentinel Influenza Detections

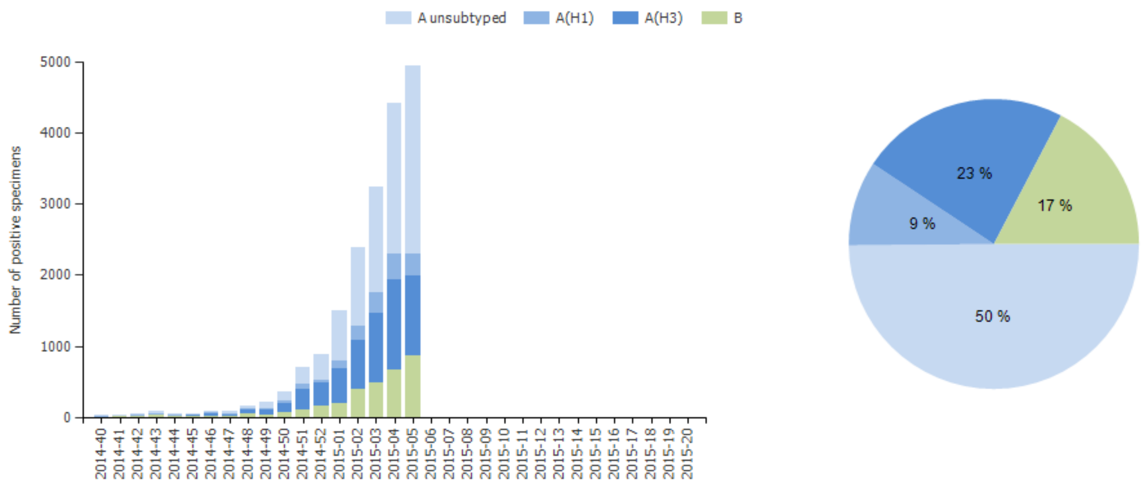


Figure 2. Development of the 2014-2015 Influenza Season in the WHO European Region

Intensity of Influenza Activity and Dominant Virus (sub)type

Year	2014													2015				
Country / Week	40	41	42	43	44	45	46	47	48	49	50	51	52	1	2	3	4	5
Iceland									-							AH3	AH3	AH3
Ireland															AH3	AH3	AH3	AH3
Portugal														B	B	B	A/B	B
Spain								AH3	AH3	A	AH3	AH3	A	AH3	AH3	AH3	AH3	AH3
United Kingdom (Northern Ireland)	AH3	B		A/B	B	B	B		AH3	B	AH3	-	A	AH3	AH3	AH3	AH3	AH3
United Kingdom (Wales)	-		AH3					-		AH3	AH3		-	AH3	AH3	AH3		AH3
United Kingdom (Scotland)		A				A	A	A	A/B	A	AH3	AH3	A	AH3	AH3	A	AH3	A
United Kingdom (England)	-	-						-										
France																		-
Belgium												AH3		AH3	AH3	AH3	AH3	AH3
Netherlands				A/B						AH3	AH3	AH3	AH3	AH3	AH3	AH3	A	A
Luxembourg												B/AH1				AH3	AH3	AH3
Switzerland	A/B			A		B	B/AH3	A	A/B	A/B	A/B	A	A	AB	AH3	AH3	AH3	AH3
Germany												AH3	AH3	AH3	AH3	AH3	AH3	AH3
Denmark														A	A	AH3	AH3	AH3
Norway						A	A	AB	B	B/AH3	B/AH3	B/AH3	B/AH3	B/AH3	B/AH3	B/AH3	AH3	AH3
Italy	-	-						A		A	A	A	A	A	AH1	AH1	AH1	A
Austria													-	-				
Malta	-														AH3	AH3	AH3	AH3
Slovenia												AH1	AH1	AH1	AH1	AH1	AH1	AH1
Sweden											AH3	AH3	AH3	AH3	AH3	AH3	AH3	AH3
Croatia	-	-												AH3	AH1	A	A	A
Czech Republic															B/AH3	AH3	AH3	AH3
Slovakia																		
Albania									-		AH3		-	AH3		AH3	AH3	
Hungary													-					
Poland																		
Serbia																		AH1
Greece																	-	
Lithuania																	AH3	AH3
Bulgaria																	AH3	AH3
Latvia									AH1	AH1		A	A	AH3	A	AH3	AH3	AH3
Romania													-		B	AH3	B	B
Estonia																AH3	AH3	AH3
Finland									AH3	AH3		AH3	AH3	AH3	AH3	AH3	AH3	AH3
Belarus																AH3	AH3	AH3
Republic of Moldova																		
Ukraine																		
Cyprus																		
Israel																AH3		AH3
Turkey		-	-									B		AH3		AH1/H3		
Georgia																		
Azerbaijan					B		B	B	B		AH3	AH3	AH3					
Uzbekistan										AH3			AH3					
Kazakhstan																		
Russian Federation									AH3	AH3	AH3	AH3	AH3	AH3		AH3	AH3	AH3

WEST

EAST

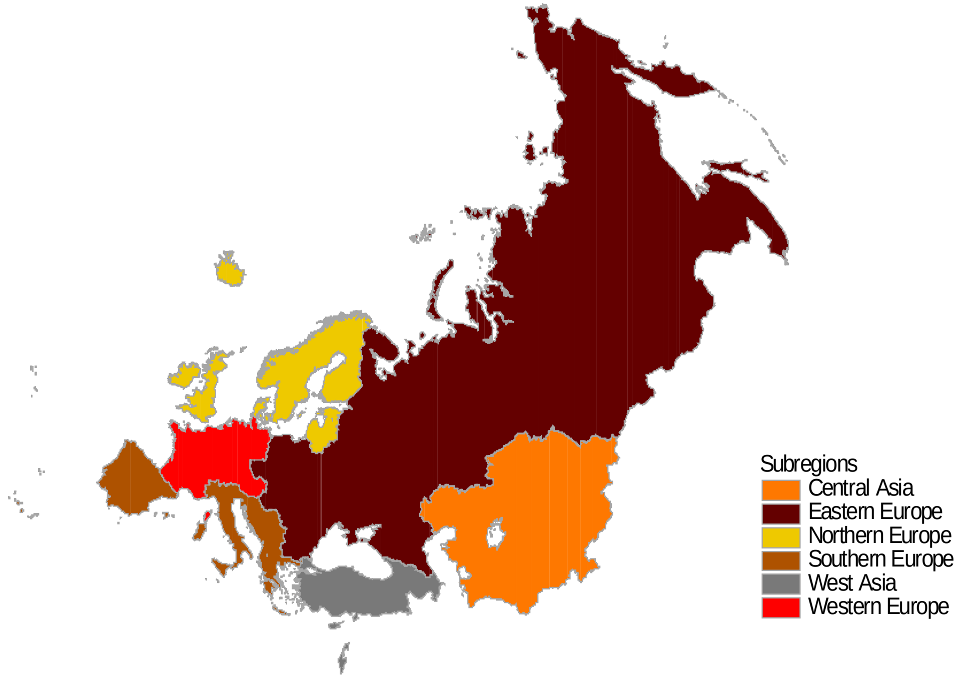
High = levels of activity higher than usual.

Medium = usual levels of activity.

Low = no activity or activity at baseline* level.

*** Baseline influenza activity is the level at which clinical influenza activity remains throughout the summer and most of the winter.**

Table 1. Distribution of Influenza Detections in the WHO European Region[1]: 2014-2015 Season (weeks 40/2014-05/2015)

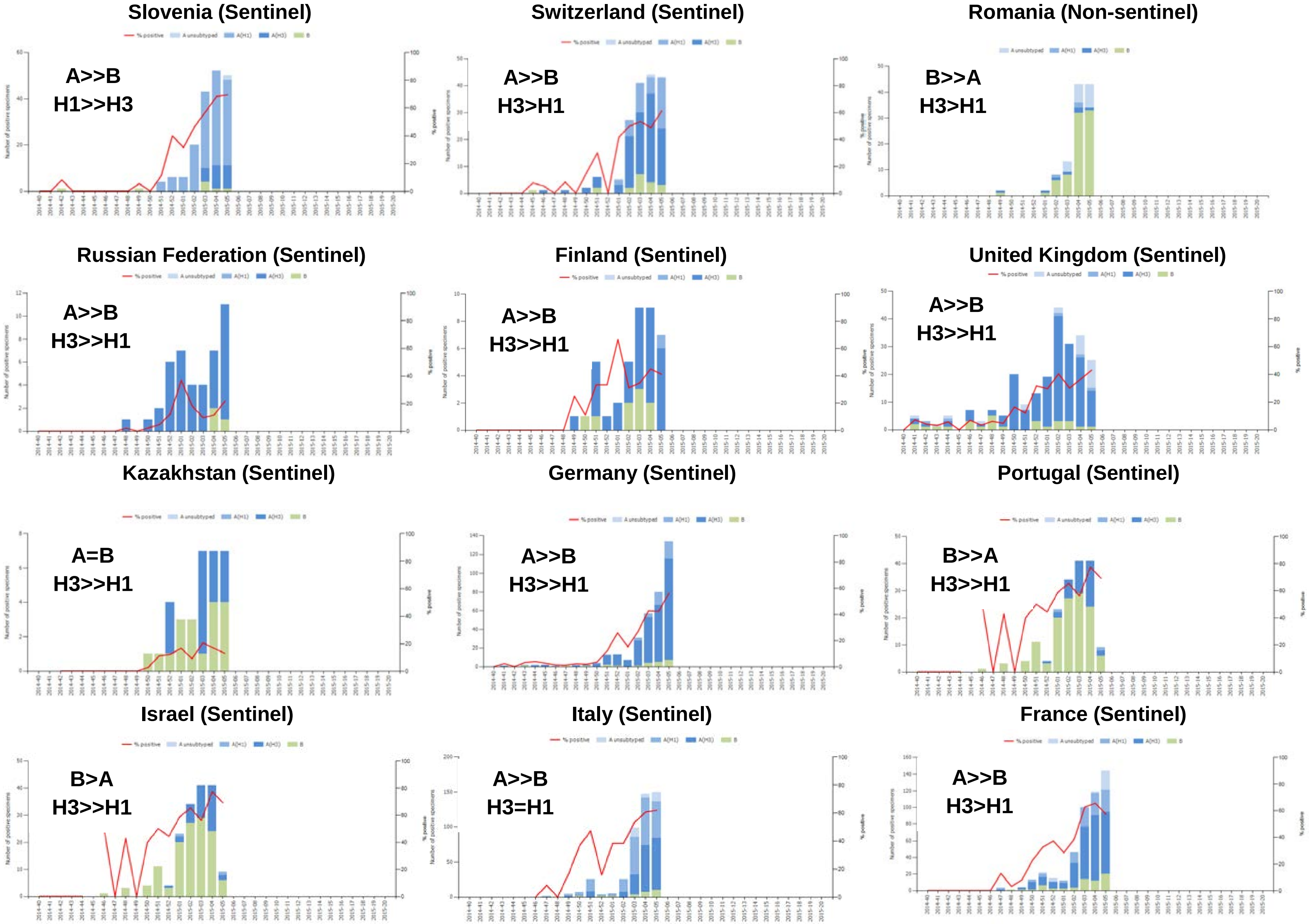


Influenza Detections*	Southern Europe		Western Europe		Eastern Europe		Northern Europe		West Asia		Central Asia		WHO European Region	
Influenza A	**4524**	**81%**	**6965**	**89%**	**1891**	**73%**	**3792**	**72%**	**106**	**57%**	**119**	**78%**	**17397**	**81%**
Influenza A subtyped	3933		3110		1882		2740		103		119		11887	
A (H1N1)pdm09	*1567*	*40%*	*836*	*27%*	*129*	*7%*	*244*	*9%*	*16*	*16%*	*8*	*7%*	*2800*	*24%*
A (H3N2)	*2366*	*60%*	*2274*	*73%*	*1753*	*93%*	*2496*	*91%*	*87*	*84%*	*111*	*93%*	*9087*	*76%*
Influenza B	**1059**	**19%**	**836**	**11%**	**687**	**27%**	**1453**	**28%**	**79**	**43%**	**33**	**22%**	**4147**	**19%**
B lineage determined	228		103		18		282		0		0		631	
B/Yamagata lineage	*228*	*100%*	*100*	*97%*	*18*	*100%*	*277*	*98%*	*0*		*0*		*623*	*99%*
B/Victoria lineage	*0*	*0%*	*3*	*3%*	*0*	*0%*	*5*	*2%*	*0*		*0*		*8*	*1%*
Total	***5583***		***7801***		***2578***		***5245***		***185***		***152***		***21544***	

*combined ILI, ARI sentinel and non-sentinel respiratory specimens positive for influenza

[1] WHO European Region Member States grouped into sub-regions according to United Nations geographic composition. Available from: http://unstats.un.org/UNSD/METHODS/M49/M49REGIN.HTM

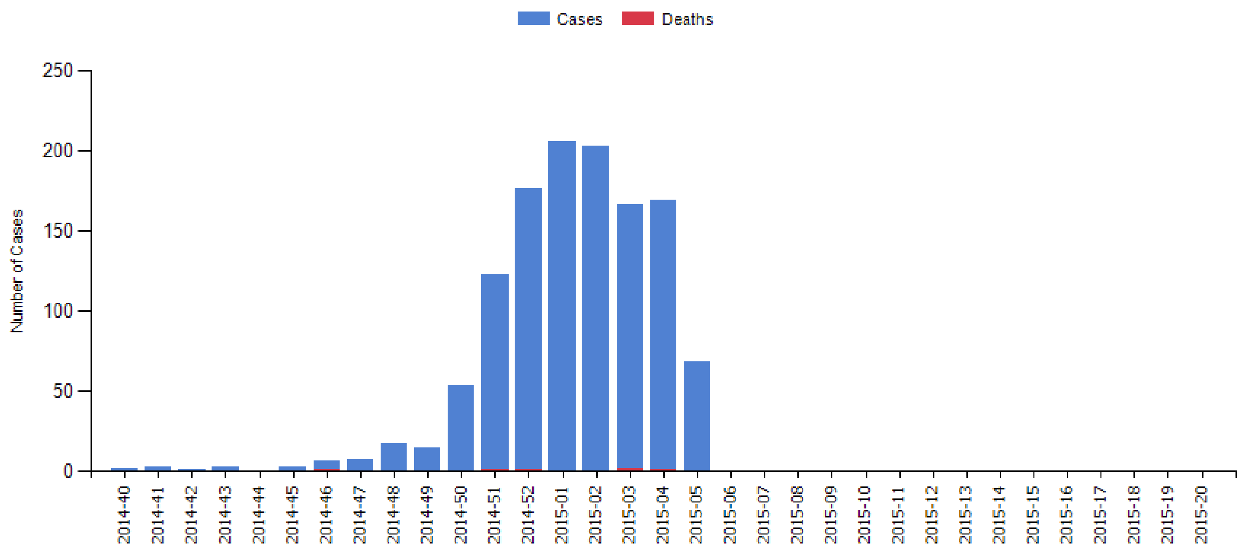
Figure 3. Patterns of Influenza Type and Subtype Circulation in the WHO European Region



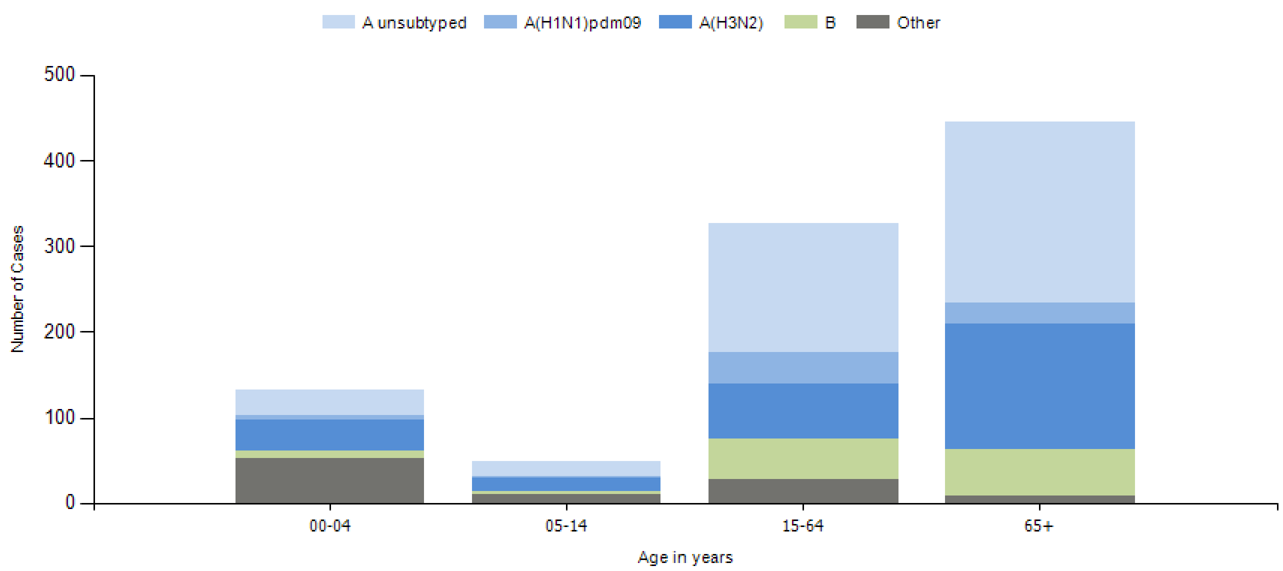
= represents similar ratio; > represents a ratio of at least 1.5:1; >> represents a ratio of ≥5

Figure 4. Laboratory-confirmed hospitalized influenza cases

A. Cases admitted by week and associated deaths



B. Distribution of virus (sub)type by age group in hospitalized cases



Influenza specimens received at NIMR

Samples, viruses or clinical specimens, with collection dates after 2014-08-31 have been received from 40 countries in Europe, Africa, the Middle East, Central Asia and the Far East. The large majority (~72%) were type A viruses, with A(H3N2) viruses predominating over A(H1N1)pdm09 viruses by a ratio of ~3:1 **(Table 2)**. Of the type B viruses (just over 27% of all specimens received) B/Yamagata lineage viruses predominated over those of the B/Victoria lineage by a ratio of ~7:1. These proportions of influenza types/subtypes/lineages are generally similar to those reported within the WHO European Region **(Figure 1 B/C** & **Table 1)**. **Table 3** shows the isolation rates from specimens received with collection dates after 2014-08-31. A(H1N1)pdm09 and influenza B viruses were recovered efficiently as assessed by being able to perform HI assays. However, for A(H3N2) while good recovery was achieved, as assessed by MUNANA-based neuraminidase activity, significant numbers could not be assessed by HI assay due to HA titres being too low (<4 in the presence of 20nM oseltamivir). **Table 4** summarises the number of viruses of the A(H1N1)pdm09 and A(H3N2) subtypes, and the two B/lineages, that showed ≤2-, 4- and ≥8-fold reductions in HI titres with post-infection ferret antisera compared with the titres against their respective homologous vaccine/reference viruses. For A(H3N2) viruses a summary of the results of plaque-reduction neutralisation assays is also shown.

Table 2. Summary of clinical samples and isolates received, with collection dates after 2014-08-31, by country

MONTH	TOTAL RECEIVED	A	H1N1pdm09		H3N2		B	B Victoria lineage		B Yamagata lineage	
Country			Number received	Number propagated[1]	Number received	Number propagated[2]		Number received	Number propagated[1]	Number received	Number propagated[1]
2014											
SEPTEMBER											
Belgium	1				1	1					
Cameroon	4						3	1	1		
Egypt	1		1	in process							
France	2				1	1				1	1
Ghana	4							4	in process		
Hong Kong	1		1	1							
Iran	1						1				
Madagascar	26				26	in process					
Oman	5		3	3						2	0
Senegal	3				2	in process				1	1
Spain	1				1	0 (1)					
Sweden	3				3	2 (1)					
Zambia	14	7			4	in process		1	in process	2	in process
OCTOBER											
Algeria	3				1	in process				2	in process
Belgium	5				5	1 (4)					
Cameroon	7						5	2	2		
Denmark	2				2	in process					
Egypt	6				5	0				1	1
Finland	1				1	in process					
France	6				5	0 (5)				1	1
Germany	5		3	3	1	1		1	1		
Ghana	1				1	in process					
Hong Kong	2		2	2							
Iran	2				1	0 (1)	1				
Israel	1						1				
Jordan	1						1				
Madagascar	3				1	1				2	2
Malta	4				4	in process					
Netherlands	5				4	2 (2)				1	1
Norway	8		5	3	3	1					
Oman	3				1	in process	2				
Slovenia	3				1	1	1			1	0
Senegal	4				4	4					
Spain	13				10	in process				3	3
Sweden	2				2	1 (1)					
Tunisia	1				1	in process					
Turkey	25	2	5	in process	6	in process	12				
Ukraine	1						1				
United Kingdom	2				1	1				1	1
Zambia	5	1								4	4
NOVEMBER											
Belgium	4		1	1	1	0 (1)				2	1
Cameroon	14						14				
Denmark	1				1	in process					
Egypt	12	2	7	1	2	0		1	0		
Finland	2				2	in process					
France	6		1	1	3	0 (3)				2	2
Germany	8		2	2	5	3 (2)				1	1
Ghana	8		3	in process	2	in process		3	in process		
Hong Kong	5				2	1 (1)				3	3
Iran	2		2	in process							
Israel	3				2	in process	1				
Jordan	2				1	in process	1				
Kazakhstan	4				4	in process					
Latvia	1		1	in process							
Luxembourg	1		1	in process							
Madagascar	1	1									
Morocco	3				1	in process				2	in process
Netherlands	3				3	0 (3)					
Norway	10				2	2				8	3
Oman	1				1	in process					
Portugal	2									2	2
Slovenia	1		1	in process							
Spain	15				13	in process	1			1	1
Sweden	3				3	3					
Switzerland	3				2	in process				1	0
Turkey	6	2	2	in process	2	in process					
United Kingdom	9				8	in process				1	1
Zambia	3				1	1	1			1	1

MONTH	TOTAL RECEIVED	A	H1N1pdm09		H3N2		B	B Victoria lineage		B Yamagata lineage	
Country			Number received	Number propagated[1]	Number received	Number propagated[2]		Number received	Number propagated[1]	Number received	Number propagated[1]
DECEMBER											
Albania	3		1	in process	2	in process					
Algeria	1				1	0 (1)					
Austria	8				7	1 (6)				1	1
Belgium	5		3	3	1	1				1	1
Cameroon	6						6				
Croatia	10		4	in process	2	in process	2			2	2
Denmark	5		2	2	3	in process					
Egypt	4	3	1	0							
Estonia	1				1	in process					
Finland	5		2	2	1	0 (1)				2	2
France	37		4	4	26	in process				7	7
Germany	29		4	4	23	11 (12)		1	1	1	1
Ghana	6		2	in process	1	in process		3	in process		
Greece	3				2	1 (1)				1	1
Hong Kong	2				2	1 (1)					
Iran	6		1	1	3	0 (3)				2	2
Israel	9		4	in process	5	3 (2)					
Italy	30		14	14	9	4 (5)				7	7
Jordan	9				7	in process				2	2
Kazakhstan	11		1	0	9	in process	1				
Latvia	8		1	1	5	3 (2)				2	in process
Luxembourg	11		6	3	3	1 (2)	2				
Malta	4				4	in process					
Morocco	10				2	in process	4			4	4
Netherlands	1		1	1							
Norway	25		4	in process	14	in process				7	in process
Oman	6				6	1 (5)					
Portugal	10				3	1 (2)				7	7
Russia	10				10	in process					
Slovenia	19		17	in process	1	in process	1				
Spain	39	6			26	in process				7	7
Switzerland	5		2	2	1	0 (1)				2	2
Tunisia	5						2			3	3
Turkey	6				2	in process	4				
Ukraine	11				1	0 (1)				10	10
United Kingdom	6		1	in process	4	in process				1	in process
Zambia	2	1								1	1
2015											
JANUARY											
Albania	12		2	in process	10	in process					
Algeria	3		2	1	1	0 (1)					
Croatia	1				1	in process					
Denmark	2				2	2					
Estonia	24	4	1	in process	18	in process				1	in process
Germany	11				11	7 (4)					
Ghana	3						1	1	in process	1	in process
Greece	58		11	in process	25	in process	22				
Italy	1		1	1							
Jordan	1		1	in process							
Kazakhstan	6				6	in process					
Latvia	2				2	0 (2)					
Luxembourg	1						1				
Malta	5				5	in process					
Morocco	2									2	2
Norway	4				4	in process					
Portugal	6		2	1	2	0 (2)				2	1
Slovenia	15	2	7	7	2	0 (2)	3			1	1
Spain	7	3			3	2 (1)				1	in process
Switzerland	2		1	1	1	0 (1)					
Turkey	15		5	in process	5	in process	5				
Ukraine	1									1	1
United Kingdom	11		2	in process	9	in process					
	881	34	151	65	453	66 (83)	100	18	5	125	95
			17.1%		51.4%			2.0%		14.2%	
40 Countries			72.4%				27.6%				

1. Propagated to sufficient titre to perform HI assay (the totalled number does not include any from batches that are in process)

2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir (the totalled number does not include any from batches that are in process); numbers in parentheses show the number of viruses either not recovered (no neuraminidase activity detected by MUNANA-based assay) or with HA titres too low to allow HI assay

Table 3. Isolation rates for specimens collected after 2014-08-31

	Clinical sample			Virus isolate		
	Number processed	Number propagated[1]	Percent recovered	Number processed	Number propagated[1]	Percent recovered
H1	25	17	68%	82	68	83%
H3	48	17 19[2]	75% (35%)[3]	236	118 100[2]	92% (50%)[3]
B Victoria	9	8	89%	7	6	86%
B Yamagata	34	20	59%	87	87	100%

1. Propagated to sufficient HA titre to perform HI assay
2. Viruses recovered as assessed by MUNANA (NA activity) assay, but HA titre too low to allow HI testing
3. The percentages for viruses that were assayed by HI are shown in parentheses

Table 4. Summary of isolates with collection dates after 2014-08-31 showing reduction in reactivity with antisera raised against reference viruses in HI & plaque-reduction neutralisation assays

	Number tested	≤2-fold[a]	4-fold[a]	≥8-fold[a]
H1[1]	85	85 (100%)		
H3[2a]	135	24 (17.8%)	60 (44.4%)	51 (37.8%)
H3[2b]	135	1 (0.7%)	7 (5.2%)	127 (94.1%)
H3[2c]	135	48 (35.6%)	55 (40.7%)	32 (23.7%)
H3[2d]	135			135 (100%)
H3[d1]	61	7 (11.5%)	30 (49.2%)	24 (39.3%)
H3[d2]	61	14 (23.0%)	28 (45.9%)	19 (31.1%)
H3[d3]	61	54 (88.5%)	7 (11.5%)	
H3[d4]	61	59 (96.7%)	2 (3.3%)	
H3[d5]	28	1 (3.6%)	1 (3.6%)	26 (92.8%)
H3[d6]	49	49 (100%)		
H3[d7]	33	24 (72.7%)	8 (24.2%)	1 (3.1%)
B/Victoria[3]	14	9 (64.3%)	1 (7.1%)	4 (28.6%)
B/Yamagata[4]	107	85 (79.4%)	22 (20.6%)	

a. Reduction in titre, compared to the homologous titre with post-infection ferret antisera indicated below:

1. Compared to homologous titres for ferret antisera raised against vaccine virus A/California/7/2009

2a. Compared to homologous titres for ferret antisera raised against cell-propagated A/Victoria/361/2011 (genetic subgroup 3C.1)

2b. Compared to homologous titres for ferret antisera raised against egg-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a)

2c. Compared to homologous titres for ferret antisera raised against cell-propagated A/Switzerland/9715293/2013 (genetic subgroup 3C.3a)

2d. Compared to homologous titres for ferret antisera raised against egg-propagated A/Texas/50/2012 (genetic subgroup 3C.1)

d1. Compared to homologous titres for ferret antisera raised against cell-propagated A/Victoria/361/2011 in plaque-reduction neutralisation assays

d2. Compared to homologous titres for ferret antisera raised against egg-propagated A/Switzerland/9715293/2013 in plaque-reduction neutralisation assays

d3. Compared to homologous titres for ferret antisera raised against cell-propagated A/Switzerland/9715293/2013 in plaque-reduction neutralisation assays

d4. Compared to homologous titres for ferret antisera raised against cell-propagated A/Hong Kong/5738/2014 (N/KYT) in plaque-reduction neutralisation assays

d5. Compared to homologous titres for ferret antisera raised against egg-propagated A/Hong Kong/5738/2014 (NYK) in plaque-reduction neutralisation assays

d6. Compared to homologous titres for ferret antisera raised against cell-propagated A/Hong Kong/7295/2014 (NYT) in plaque-reduction neutralisation assays

d7. Compared to homologous titres for ferret antisera raised against cell-propagated A/England/530/2014 (NYK) in plaque-reduction neutralisation assays

3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

4. Compared to homologous titres for ferret antisera raised against vaccine virus B/Phuket/3073/2013

Influenza A(H1N1) virus analysis.

H1N1pdm09 viruses were received from NICs in Europe, Africa (Algeria, Egypt, Ghana), the Middle East (Iran, Jordan, Oman), Central Asia (Kazakhstan) and Hong Kong SAR. **Table 5** summarises the antigenic (HI) results for viruses that were propagated successfully at NIMR.

All of the H1N1 viruses collected after 2014-08-31 were antigenically similar to the vaccine virus **(Table 5)**; **Tables 6-1** to **6-8** show the results of HI assays of H1N1 viruses carried out since the September 2014 report. Test viruses in each table are sorted by date of collection and viruses with collection dates after 2014-08-31 are shown below red lines in **Tables 6-2**, and **6-4** to **6-8**. All viruses were recognised by the panel of antisera at titres within 4-fold of the titres for the homologous viruses, including that raised against the A/California/7/2009 vaccine virus, with the exception of the antiserum raised against A/Christchurch/16/2010. This antiserum recognised just under half (45%) of the test viruses collected since 2014-08-31 at a titre within 4-fold of the titre for the homologous virus. Note also that none of the antisera raised against reference viruses collected since 2011 recognised the egg-propagated vaccine virus A/California/7/2009 at a titre within 4-fold of the titres of the antisera for their homologous viruses.

Figure 5 shows a phylogenetic tree for the HA genes of representative H1N1 viruses. Over the last five years the HA genes have evolved and eight genetic groups have been designated, with A/California/7/2009 representing group 1. Viruses collected after 2014-08-31 fell into genetic subgroup 6B **(Figure 5)**. Subgroup 6B is defined by the following amino acid substitutions in **HA1** and **HA2**:

- **D97N**, **S185T, S203T K163Q, A256T** and **K283E** in **HA1**, and **E47K, S124N** and **E172K** in **HA2** compared with A/California/7/2009 e.g. A/South Africa/3626/2013.

An amino acid sequence alignment of the HAs for a selected group of viruses is shown in **Figure 6**; the vaccine virus is shown in red, reference viruses are coloured blue, glycosylation motifs are highlighted and the genetic group is shown in green.

Figure 7 illustrates the location on an H1-HA structure of amino acid residues defining genetic subgroup 6B.

Figure 8 shows a phylogenetic tree for NA genes of representative H1N1 viruses and **Figure 9** shows an amino acid alignment for a selected group of these viruses. The HA and NA phylogenetic trees are generally congruent.

Antiviral resistance.

Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on close to 80 viruses with collection dates after 2014-08-31. All viruses showed normal inhibition (NI) to both neuraminidase inhibitors **(Table 15)**.

Conclusion.

The A(H1N1)pdm09 viruses received were antigenically similar to the vaccine virus, A/California/7/2009, and retain susceptibility to oseltamivir and zanamivir.

Table 5. Summary of influenza A(H1N1)pdm09 viruses analysed, collected after 2014-08-31

MONTH Country	H1N1pdm09				
	Number received[1]	Number propagated[2]	Fold reduction in HI titre		
			≤2 fold[3]	4 fold[3]	≥8 fold[3]
2014					
SEPTEMBER					
Hong Kong	1	1	1		
Oman	3	3	3		
OCTOBER					
Germany	3	3	3		
Hong Kong	2	2	2		
Norway	5	3	3		
NOVEMBER					
Belgium	1	1	1		
France	1	1	1		
Germany	2	2	2		
Ghana	3	0	0		
DECEMBER					
Albania	1	1	1		
Belgium	3	3	3		
Croatia	3	3	3		
Denmark	2	2	2		
England	1	1	1		
Finland	2	2	2		
France	4	4	4		
Germany	4	4	4		
Ghana	1	1	1		
Iran	1	1	1		
Israel	4	2	2		
Italy	14	14	14		
Latvia	1	1	1		
Luxembourg	6	3	3		
Norway	4	2	2		
Netherlands	1	1	1		
Slovenia	17	8	8		
Switzerland	2	2	2		
2015					
JANUARY					
Albania	2	0	0		
Algeria	2	1	1		
England	2	2	2		
Italy	1	1	1		
Jordan	1	1	1		
Portugal	2	1	1		
Slovenia	7	7	7		
Switzerland	1	1	1		
	110	85	85	0	0
23 Countries			100.0%	0.0%	0.0%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2. Propagated to sufficient titre to perform HI assays
3. Reduction in HI titre with ferret antiserum raised against A/California/7/2009 (egg grown vaccine virus)

Table 6-1 . Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-09-17)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre: Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	X-243 NIB F48/14	
	Genetic group						4	3	5	6	7	6A	6B	6B	
REFERENCE VIRUSES															
A/California/7/2009		2009-04-09	E1/E2	1280	1280	1280	320	320	320	640	1280	320	640	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	320	640	640	80	80	80	160	160	80	160	40	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	2560	160	80	160	320	160	320	160	80	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	2560	5120	2560	2560	2560	5120	2560	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	320	640	640	1280	1280	1280	2560	1280	1280	1280	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	2560	1280	1280	1280	2560	2560	2560	5120	5120	2560	2560	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	1280	1280	2560	2560	5120	2560	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	1280	1280	1280	2560	2560	5120	2560	1280	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	640	320	640	640	1280	2560	2560	5120	2560	1280	1280	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	1280	1280	1280	640	1280	1280	1280	5120	1280	2560	2560	
X-243 (A/South Africa/3626/2013)	6B		EX/E1	2560	1280	2560	2560	5120	5120	5120	5120	5120	2560	5120	
TEST VIRUSES															
A/Estonia/84426/2014	6B	2014-02-04	MDCK3/SIAT2	640	320	1280	2560	1280	2560	1280	5120	1280	640	5120	
A/Estonia/84490/2014		2014-02-06	MDCK2/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	5120	
A/Estonia/84627/2014		2014-02-10	MDCK2/MDCK1	2560	640	1280	2560	2560	2560	5120	5120	5120	5120	2560	
A/Estonia/84639/2014		2014-02-10	MDCK2/MDCK1	1280	320	640	640	1280	2560	2560	5120	2560	2560	2560	
A/Estonia/84981/2014		2014-02-20	MDCK2/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	2560	
A/Estonia/85066/2014		2014-02-21	MDCK2/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	5120	
A/Estonia/85108/2014		2014-02-25	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	2560	
A/Estonia/85212/2014		2014-02-27	MDCK2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	2560	
A/Estonia/85246/2014		2014-02-28	MDCK1/MDCK1	2560	640	1280	2560	2560	2560	2560	5120	5120	2560	2560	
A/Estonia/85353/2014		2014-03-04	MDCK1/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	5120	
A/Estonia/85408/2014		2014-03-05	MDCK2/MDCK1	2560	1280	2560	2560	2560	5120	5120	5120	5120	2560	5120	
A/Estonia/85422/2014		2014-03-06	MDCK2/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	5120	
A/Estonia/85519/2014	6B	2014-03-10	MDCK1/MDCK1	1280	640	1280	1280	1280	2560	2560	5120	2560	2560	1280	
A/Estonia/85629/2014		2014-03-13	MDCK2/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	2560	
A/Estonia/85660/2014		2014-03-13	MDCK2/MDCK1	1280	1280	1280	1280	2560	5120	5120	5120	5120	5120	2560	
A/Estonia/85729/2014		2014-03-17	MDCK1/MDCK1	2560	1280	2560	1280	2560	2560	5120	5120	5120	5120	2560	
A/Estonia/85739/2014	6B	2014-03-18	MDCK2/MDCK1	1280	640	640	640	1280	1280	2560	5120	2560	2560	1280	
A/Estonia/85759/2014		2014-03-18	MDCK2/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	2560	
A/Estonia/85792/2014		2014-03-19	MDCK1/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	2560	
A/Estonia/85847/2014		2014-03-20	MDCK2/MDCK1	1280	1280	1280	1280	2560	5120	2560	5120	5120	2560	2560	
A/Estonia/85829/2014	6B	2014-03-21	MDCK2/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	1280	1280	
A/Estonia/85899/2014		2014-03-25	MDCK2/MDCK1	2560	1280	2560	1280	2560	2560	2560	5120	5120	2560	2560	
A/Estonia/86382/2014	6B	2014-04-16	MDCK2/MDCK1	2560	1280	2560	2560	5120	5120	5120	5120	5120	5120	5120	

Vaccine

Table 6-2. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-10-07)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
	Genetic group						4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	1280	1280	1280	320	320	320	320	640	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	320	640	640	80	80	80	160	160	80	160	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	1280	160	80	160	160	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	2560	2560	2560	5120	2560	5120	2560	5120	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	640	640	640	1280	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	1280	640	640	640	1280	1280	1280	2560	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	1280	1280	1280	5120	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	1280	1280	2560	2560	2560	5120	5120	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	320	320	1280	1280	1280	2560	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	1280	1280	1280	1280	2560	2560	1280	5120	1280	2560	
TEST VIRUSES														
A/Oman/897/2014	6B	2014-03-11	MDCK1	1280	640	640	640	2560	1280	1280	5120	2560	1280	
A/Oman/903/2014	6B	2014-03-11	MDCK1	1280	640	640	640	1280	2560	1280	5120	2560	2560	
A/Oman/945/2014	6B	2014-03-12	MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Oman/1123/2014	6B	2014-03-18	MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	1280	
A/Oman/1930/2014	6B	2014-04-18	MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	1280	
A/Oman/2602/2014	6B	2014-05-09	MDCK2	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Oman/2785/2014	6B	2014-05-15	MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Oman/3992/2014	6B	2014-07-08	MDCK1	1280	640	640	640	2560	1280	1280	5120	2560	1280	
A/Oman/4003/2014	6B	2014-07-10	MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Oman/4005/2014	6B	2014-07-11	MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560	1280	
A/Oman/4034/2014	6B	2014-07-11	MDCK1	2560	1280	1280	1280	2560	2560	5120	5120	5120	2560	
A/Oman/4230/2014	6B	2014-07-25	MDCK1	1280	640	640	1280	2560	2560	2560	5120	5120	2560	
A/Oman/4902/2014	6B	2014-09-09	MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Oman/4941/2014	6B	2014-09-13	MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560	2560	
A/Oman/5074/2014	6B	2014-09-18	MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	

Vaccine

Table 6-3. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-10-14 + 2014-11-12)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre: Post infection ferret antisera										
				A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F26/14	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
	Genetic group						4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	640	640	1280	160	160	160	160	320	160	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	320	40	40	40	40	80	40	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	320	1280	1280	160	80	80	80	160	160	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	2560	2560	640	5120	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	160	160	640	640	320	1280	640	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	640	640	640	320	1280	1280	640	2560	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	1280	640	320	1280	1280	640	2560	1280	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	640	320	1280	1280	640	2560	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	160	640	640	320	1280	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	320	640	640	320	640	640	640	2560	640	1280	
TEST VIRUSES														
A/Egypt/ALX4789/2014	6B	unknown	MDCK1	320	640	640	640	1280	1280	1280	2560	2560	1280	
A/Egypt/ASW4779/2014	6B	unknown	MDCK1	640	640	640	1280	2560	1280	2560	2560	2560	1280	
A/Egypt/ALX4048/2014	6B	unknown	MDCK1	1280	640	640	640	1280	1280	1280	2560	2560	1280	
A/Egypt/ALX4164/2014	6B	unknown	MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	2560	2560	
A/Jordan/30219/2014	6B	2014-04-26	MDCK1/MDCK1	640	160	640	320	640	640	320	1280	640	640	
A/Jordan/20314/2014	6B	2014-05-06	MDCK1/MDCK1	640	320	640	320	1280	1280	640	1280	1280	1280	
A/Jordan/20363/2014	6B	2014-05-11	MDCK1/MDCK1	1280	1280	1280	1280	2560	2560	1280	5120	5120	2560	
A/Jordan/20360/2014	6B	2014-05-17	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	5120	2560	
A/Jordan/20349/2014	6B	2014-05-22	MDCK1/MDCK1	1280	640	1280	640	2560	2560	1280	5120	2560	2560	
A/Jordan/30260/2014	6B	2014-05-26	MDCK1/MDCK1	1280	640	1280	1280	2560	2560	1280	5120	5120	2560	
A/Jordan/20407/2014	6B	2014-07-08	MDCK1/MDCK1	1280	640	640	640	2560	2560	640	5120	2560	2560	

Vaccine

Table 6-4. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2014-12-02 + 2014-12-19 + 2015-01-14)

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post infection ferret antisera: A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F26/14	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
Genetic group							4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	640	640	1280	160	160	160	160	320	160	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	320	40	40	40	40	80	40	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	320	1280	1280	160	80	80	80	160	160	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	2560	2560	640	5120	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	160	160	640	640	320	1280	640	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK4/MDCK1	640	640	640	320	1280	1280	640	2560	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	1280	640	320	1280	1280	640	2560	1280	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	640	640	640	320	1280	1280	640	2560	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	160	640	640	320	1280	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	320	640	640	320	640	640	640	2560	640	1280	
TEST VIRUSES														
A/Italy/FVG-27/2014	6B	unknown	Cx/MDCK1	640	320	320	640	1280	1280	1280	2560	1280	1280	
A/Italy/FVG-28/2014	6B	unknown	Cx/MDCK1	640	640	640	640	1280	1280	1280	2560	1280	1280	
A/Parma/84/2014	6B	unknown	MDCK2/MDCK2	1280	640	1280	640	2560	2560	2560	5120	2560	2560	
A/Norway/2432/2014	6B	2014-10-02	MDCK2	1280	320	320	640	1280	1280	640	5120	2560	1280	
A/Rheinland-Pfalz/12/2014	6B	2014-10-24	C2/MDCK1	640	320	320	640	1280	1280	640	5120	2560	1280	
A/Hamburg/6/2014	6B	2014-10-27	C2/MDCK1	1280	640	640	1280	2560	2560	1280	5120	2560	2560	
A/Rheinland-Pfalz/14/2014	6B	2014-10-27	C3/MDCK1	320	320	320	320	640	640	640	1280	1280	640	
A/Norway/3023/2014	6B	2014-10-28	MDCK1	640	320	640	640	1280	1280	1280	5120	2560	1280	
A/Norway/3024/2014	6B	2014-10-31	MDCK1	640	640	640	640	1280	2560	1280	5120	2560	1280	
A/Mecklenburg-Vorpommern/5/2014	6B	2014-11-08	C2/MDCK1	640	320	640	640	640	1280	640	2560	2560	1280	
A/Belgium/G651/2014	6B	2014-11-24	MDCK1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Hamburg/7/2014	6B	2014-11-27	C3/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Belgium/G675/2014	6B	2014-12-03	MDCK1/MDCK1	320	160	160	160	1280	1280	1280	2560	1280	640	
A/Belgium/G683/2014	6B	2014-12-05	MDCK1/MDCK2	640	320	640	320	1280	1280	640	2560	1280	1280	
A/Rheinland-Pfalz/15/2014	6B	2014-12-15	C2/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	2560	2560	
A/Berlin/81/2014	6B	2014-12-26	C2/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	2560	2560	

Sequences in phylogenetic trees

Vaccine

Table 6-5 . Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-01-21)

				Haemagglutination inhibition titre										
				Post infection ferret antisera										
Viruses		Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
	Genetic group						4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E2	1280	640	1280	320	320	320	320	640	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	320	320	640	80	80	80	160	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	640	1280	160	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	1280	2560	5120	2560	2560	2560	5120	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	640	160	640	320	1280	1280	1280	2560	2560	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	320	640	640	1280	2560	2560	5120	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	640	1280	640	1280	1280	2560	5120	2560	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	320	1280	640	1280	2560	2560	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	320	1280	640	1280	2560	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	320	640	640	640	2560	1280	1280	
TEST VIRUSES														
A/Hong Kong/7531/2014	6B	2014-09-28	MDCK2/MDCK1	1280	640	640	640	2560	2560	2560	5120	2560	2560	
A/Hong Kong/7532/2014	6B	2014-10-01	MDCK2/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	2560	2560	
A/Hong Kong/7555/2014	6B	2014-10-11	MDCK1/MDCK1	2560	1280	2560	1280	2560	5120	5120	5120	5120	2560	
A/Nice/2419/2014	6B	2014-11-22	MDCK2/MDCK1	640	320	640	320	1280	1280	1280	2560	1280	1280	
A/Marseille/ 2408/2014	6B	2014-12-02	MDCK2/MDCK1	640	640	640	640	1280	1280	1280	5120	2560	1280	
A/Lyon/2425/2014		2014-12-11	MDCK2/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	2560	2560	
A/Netherlands/466/14	6B	2014-12-14	MDCK1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	2560	2560	
A/Poitiers/2483/2014	6B	2014-12-14	MDCKX/MDCK1/MDCK1	1280	640	1280	640	2560	2560	2560	5120	2560	1280	
A/Lyon/2436/2014	6B	2014-12-16	MDCK2/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	2560	2560	
A/Tehran/64649/2014	6B	2014-12-20	MDCK1/MDCK1	1280	640	640	320	2560	2560	1280	5120	2560	2560	

Sequences in phylogenetic trees

Vaccine

Table 6-6. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-01-28)

				Haemagglutination inhibition titre										
				Post infection ferret antisera										
Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F30/10	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
							4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E3	640	1280	1280	160	160	320	320	320	160	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	320	320	40	40	80	80	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	1280	1280	160	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	2560	5120	2560	2560	2560	5120	2560	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	640	1280	640	1280	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	1280	640	640	640	1280	2560	1280	2560	2560	1280	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	640	1280	2560	2560	5120	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	1280	1280	1280	2560	2560	5120	5120	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	320	160	160	320	1280	1280	640	2560	1280	1280	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	320	640	1280	640	2560	1280	1280	
TEST VIRUSES														
A/Hormozgan/82762/2014	6B	2014-03-05	E2/E1	320	320	320	320	640	640	640	1280	1280	640	
A/Belgium/G671/2014		2014-12-02	MDCK2	1280	320	320	640	1280	2560	1280	5120	2560	2560	
A/Luxembourg/14068163/2014	6B	2014-12-09	MDCK1/MDCK1	1280	1280	640	1280	2560	5120	2560	5120	5120	2560	
A/Latvia/12-020598/2014	6B	2014-12-09	C1/MDCK1	640	320	320	640	1280	1280	1280	2560	1280	1280	
A/Italy/FVG-30/2014	6B	2014-12-10	SIAT1/MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560	2560	
A/Italy/FVG-32/2014		2014-12-11	SIAT1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Schleswig-Holstein/9/2014	6B	2014-12-11	C3/MDCK1	1280	640	1280	1280	2560	2560	2560	5120	5120	2560	
A/Milano/166/2014		2014-12-15	SIAT1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Italy/FVG-33/2014	6B	2014-12-15	SIAT1/MDCK1	1280	1280	1280	1280	2560	5120	2560	5120	5120	2560	
A/Italy/FVG-34/2014		2014-12-15	SIAT1/MDCK1	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Milano/167/2014	6B	2014-12-16	SIAT1/MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560	2560	
A/Luxembourg/14069579/2014	6B	2014-12-17	MDCK1/MDCK1	1280	640	640	1280	2560	2560	2560	5120	5120	2560	
A/Italy/FVG-38/2014		2014-12-17	SIAT1/MDCK1	2560	1280	1280	2560	2560	5120	5120	5120	5120	2560	
A/Luxembourg/14069580/2014	6B	2014-12-18	MDCK1/MDCK1	1280	640	640	640	1280	2560	2560	5120	2560	2560	
A/Italy/FVG-35/2014		2014-12-18	SIAT1/MDCK1	2560	1280	1280	2560	5120	5120	5120	5120	5120	2560	
A/Italy/FVG-37/2014	6B	2014-12-19	SIAT1/MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560	1280	
A/Italy/FVG-41/2014	6B	2014-12-24	SIAT1/MDCK1	2560	2560	1280	2560	5120	5120	5120	5120	5120	5120	
A/Italy/FVG-39/2014		2014-12-29	SIAT1/MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560	2560	
A/Italy/FVG-43/2014		2014-12-29	SIAT1/MDCK1	1280	640	640	1280	2560	2560	2560	5120	5120	2560	
A/Switzerland/955/2014		2014-12-30	MDCK1	640	320	320	640	1280	1280	1280	5120	2560	1280	
A/Switzerland/988/2014	6B	2014-12-30	MDCK1	1280	640	640	1280	2560	2560	2560	5120	5120	2560	
A/Italy/FVG-42/2014	6B	2014-12-30	SIAT1/MDCK1	1280	640	640	640	1280	2560	2560	5120	5120	2560	
A/Italy/FVG-01/2015		2014-12-31	SIAT1/MDCK1	1280	1280	640	1280	2560	2560	2560	5120	2560	2560	
A/Berlin/87/2014	6B	2014-12-31	C2/MDCK1	1280	1280	1280	1280	2560	5120	2560	5120	5120	2560	
A/Lisboa/EVA60/2015	6B	2015-01-05	MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560	2560	
A/Italy/FVG-02/2015	6B	2015-01-05	SIAT1/MDCK1	2560	1280	1280	2560	2560	5120	5120	5120	5120	5120	
A/Switzerland/485/2015	6B	2015-01-07	MDCK1	2560	1280	640	1280	2560	5120	5120	5120	5120	5120	

Sequences in phylogenetic trees

Vaccine

Table 6-7. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-02-04)

Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F1/15	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
				Haemagglutination inhibition titre Post infection ferret antisera											
	Genetic group							4	3	5	6	7	6A	6B	
REFERENCE VIRUSES															
A/California/7/2009		2009-04-09	EP1/E3	1280	640	1280	1280	320	160	320	320	320	320	320	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	160	640	640	160	40	80	160	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/S1/MDCK3	640	320	1280	1280	320	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	2560	2560	2560	5120	5120	2560	2560	2560	5120	5120	2560	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	640	320	320	640	1280	1280	1280	2560	1280	640	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	320	640	320	320	640	640	1280	1280	2560	1280	640	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	1280	1280	1280	2560	1280	1280	2560	2560	5120	2560	2560	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	1280	1280	1280	1280	2560	2560	2560	5120	5120	2560	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	160	320	160	160	320	640	640	640	1280	1280	640	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	640	640	640	1280	640	1280	1280	1280	2560	1280	1280	
TEST VIRUSES															
A/Finland/472/2014	6B	2014-12-04	MDCK2/MDCK1		1280	640	1280	1280	1280	2560	2560	5120	2560	2560	
A/Finland/473/2014	6B	2014-12-04	MDCK2/MDCK1		640	160	320	320	640	1280	640	1280	1280	640	
A/Slovenia/2631/2014		2014-12-13	MDCK1/MDCK1		640	320	640	640	1280	1280	1280	5120	2560	1280	
A/Slovenia/2738/2014		2014-12-13	MDCK1/MDCK1		640	640	640	640	1280	1280	1280	5120	2560	1280	
A/Slovenia/2739/2014		2014-12-13	MDCK1/MDCK1		1280	640	640	640	1280	1280	1280	5120	2560	1280	
A/Slovenia/2740/2014	6B	2014-12-13	MDCKx/MDCK1		1280	640	640	640	1280	2560	1280	5120	2560	1280	
A/Israel/P-352/2014	6B	2014-12-17	MDCKx/MDCK1		640	320	320	640	1280	1280	1280	2560	1280	1280	
A/Norway/3237/2014	6B	2014-12-19	MDCK2		1280	640	640	1280	1280	1280	1280	5120	2560	1280	
A/Slovenia/2774/2014	6B	2014-12-19	MDCKx/MDCK1		1280	640	1280	1280	1280	2560	2560	5120	2560	2560	
A/Slovenia/2804/2014	6B	2014-12-19	MDCKx/MDCK1		320	320	320	320	640	1280	640	1280	1280	640	
A/Norway/3246/2014	6B	2014-12-20	MDCK1		1280	640	640	1280	1280	2560	2560	5120	2560	1280	
A/Slovenia/2805/2014		2014-12-22	MDCKx/MDCK1		1280	640	1280	1280	1280	2560	2560	5120	2560	1280	
A/Israel/P-412/2014	6B	2014-12-22	MDCKx/MDCK1		640	320	640	640	1280	1280	1280	2560	1280	1280	
A/Denmark/50/2014	6B	2014-12-25	MDCK3/MDCK1		640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Denmark/51/2014	6B	2014-12-25	MDCK3/MDCK1		640	320	320	640	640	1280	1280	2560	1280	1280	
A/Croatia/1982/2014	6B	2014-12-28	MDCKx/MDCK1		640	320	320	640	1280	1280	1280	2560	1280	1280	
A/Croatia/1994/2014	6B	2014-12-29	MDCKx/MDCK1		640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Slovenia/2908/2014	6B	2014-12-30	MDCKx/MDCK1		1280	640	640	1280	2560	2560	2560	5120	2560	2560	
A/Slovenia/9/2015		2015-01-02	MDCKx/MDCK1		640	640	640	640	1280	1280	1280	5120	2560	1280	
A/Slovenia/33/2015	6B	2015-01-02	MDCKx/MDCK1		640	640	320	640	1280	1280	1280	2560	1280	1280	
A/Setif/02/2015	6B	2015-01-04	MDCK2		1280	1280	640	1280	2560	2560	2560	5120	2560	2560	
A/Slovenia/46/2015		2015-01-06	MDCKx/MDCK1		640	320	320	640	1280	1280	1280	2560	1280	1280	
A/Slovenia/52/2015	6B	2015-01-06	MDCKx/MDCK1		1280	640	640	1280	2560	2560	2560	5120	2560	1280	
A/Slovenia/119/2015	6B	2015-01-09	MDCKx/MDCK1		1280	640	640	1280	2560	2560	2560	5120	2560	1280	
A/Slovenia/226/2015		2015-01-14	MDCKx/MDCK1		640	320	320	640	1280	1280	1280	2560	1280	640	
A/Slovenia/233/2015		2015-01-14	MDCKx/MDCK1		640	320	320	640	1280	1280	1280	2560	1280	640	

Sequences in phylogenetic trees

Vaccine

Table 6-8. Antigenic analyses of influenza A(H1N1)pdm09 viruses (2015-02-11)

Viruses	Genetic group	Collection date	Passage History	A/Cal 7/09 F30/11	A/Bayern 69/09 F11/11	A/Lviv N6/09 F14/13	A/Chch 16/10 F15/14	A/HK 3934/11 F21/11	A/Astrak 1/11 F22/13	A/St. P 27/11 F23/11	A/St. P 100/11 F24/11	A/HK 5659/12 F30/12	A/Sth Afr 3626/13 F3/14	
				Haemagglutination inhibition titre										
				Post infection ferret antisera										
Genetic group							4	3	5	6	7	6A	6B	
REFERENCE VIRUSES														
A/California/7/2009		2009-04-09	E1/E3	640	640	640	160	80	160	160	320	160	160	
A/Bayern/69/2009		2009-07-01	MDCK5/MDCK2	160	640	320	80	40	80	160	80	80	80	G155E
A/Lviv/N6/2009		2009-10-27	MDCK4/SIAT1/MDCK3	640	1280	1280	320	80	160	320	160	320	160	G155E>G, D222G
A/Christchurch/16/2010	4	2010-07-12	E1/E3	1280	2560	2560	5120	1280	2560	1280	5120	2560	1280	
A/Hong Kong/3934/2011	3	2011-03-29	MDCK2/MDCK3	320	160	320	320	640	640	640	1280	1280	320	
A/Astrakhan/1/2011	5	2011-02-28	MDCK1/MDCK5	640	320	320	320	1280	1280	640	2560	1280	640	
A/St. Petersburg/27/2011	6	2011-02-14	E1/E3	640	640	1280	640	640	1280	1280	2560	1280	1280	
A/St. Petersburg/100/2011	7	2011-03-14	E1/E3	1280	640	1280	640	1280	2560	1280	5120	2560	1280	
A/Hong Kong/5659/2012	6A	2012-05-21	MDCK4/MDCK2	160	80	160	160	320	640	320	1280	1280	320	
A/South Africa/3626/2013	6B	2013-06-06	E1/E2	320	640	640	320	640	640	1280	1280	1280	1280	
TEST VIRUSES														
A/Kanagawa/163/2014	6B	2014-09-16	MDCK1/MDCK1/MDCK1	640	640	640	640	1280	1280	1280	2560	2560	1280	
A/Ghana/FS-14-1069/2014	6C	2014-12-19	MDCK1	1280	640	640	1280	2560	2560	2560	5120	2560	1280	
A/Albania/5687/2014	6B	2014-12-22	MDCK1	640	320	640	640	1280	1280	1280	2560	1280	1280	
A/England/579/2014	6B	2014-12-22	MDCK1/MDCK1	640	320	640	640	1280	1280	1280	2560	2560	1280	
A/Croatia/1982/2014	6B	2014-12-28	E1/E1	640	640	640	640	640	1280	1280	2560	1280	1280	
A/England/2/2015	6B	2015-01-05	MDCK1/MDCK1	1280	640	640	1280	1280	2560	2560	5120	2560	1280	
A/England/6/2015	6B	2015-01-08	MDCK1/MDCK1	1280	640	640	1280	1280	1280	2560	5120	2560	1280	
A/Jordan/20077/2015	6B	2015-01-18	MDCK1	640	640	640	320	1280	1280	1280	2560	1280	1280	

Sequences in phylogenetic trees

Vaccine

Figure 5. Phylogenetic comparison of A(H1N1)pdm09 HA genes

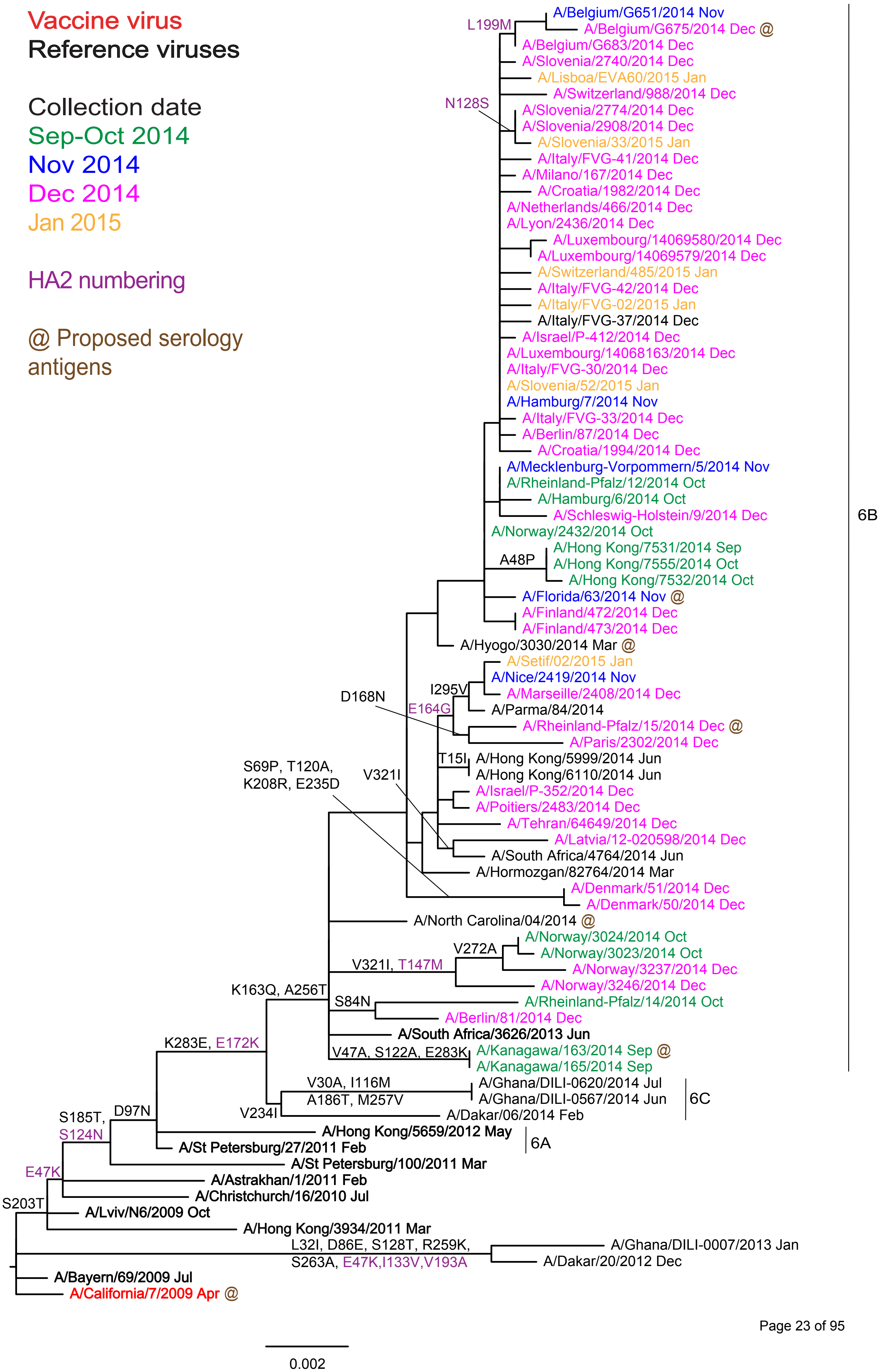


Figure 6. HA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

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                                   10        20        30        40        50        60        70        80        90       100
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009        DTLCIGYHANNSTDTVDTVLEKNVTVTHSVNLLEDKHNGKLCKLRGVAPLHLGKCNIAGWILGNPECESLSTASSWSYIVETPSSDNGTCYPGDFIDYEE
A/Bayern/69/2009           ..................................................................................S.................
A/Hong Kong/3934/2011      ..................................................................................S................. 3
A/Lviv/N6/2009             ..................................................................................S.................
A/Christchurch/16/2010     ..................................................................................S..........N...... 4
A/Astrakhan/1/2011         ..................................................................................S.............N... 5
A/St Petersburg/100/2011   ..................................................................................S................. 7
A/St Petersburg/27/2011    ..................................................................................S.............N... 6
A/Hong Kong/5659/2012      ..................................................................................S.............N... 6A
A/Ghana/DILI-0620/2014     .............................A....................................................S.............N... 6C
A/Kanagawa/165/2014        ..............................................A...................................S.............N... 6B
A/South Africa/3626/2013   ..................................................................................S.............N... 6B
A/Rheinland-Pfalz/14/2014  ..................................................................................SN............N... 6B
A/Norway/3246/2014         ..................................................................................S.............N... 6B
A/Norway/3237/2014         ..................................................................................S.............N... 6B
A/Denmark/51/2014          ....................................................................P.............S.............N... 6B
A/Latvia/12-020598/2014    ..................................................................................S.............N... 6B
A/Israel/P-352/2014        ..................................................................................S.............N... 6B
A/Paris/2302/2014          ..................................................................................S.............N... 6B
A/Marseille/2408/2014      ..................................................................................S.............N... 6B
A/Setif/02/2015            ..................................................................................S.............N... 6B
A/Finland/473/2014         ..................................................................................S.............N... 6B
A/Hong Kong/7555/2014      ...............................................P..................................S.............N... 6B
A/Schleswig-Holstein/9/2014 .................................................................................S.............N... 6B
A/Croatia/1994/2014        ..................................................................................S.............N... 6B
A/Switzerland/485/2015     ..................................................................................S.............N... 6B
A/Luxembourg/14069580/2014 ..................................................................................S.............N... 6B
A/Slovenia/33/2015         ..................................................................................S.............N... 6B
A/Lisboa/EVA60/2015        ..................................................................................S.............N... 6B
A/Belgium/G675/2014        ..................................................................................S.............N... 6B

                                  110       120       130       140       150       160       170       180       190       200
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009        LREQLSSVSSFERFEIFPKTSSWPNHDSNKGVTAACPHAGAKSFYKNLIWLVKKGNSYPKLSKSYINDKGKEVLVLWGIHHPSTSADQQSLYQNADAYVF
A/Bayern/69/2009           ......................................................E.............................................
A/Hong Kong/3934/2011      ..................X..............T......S.........................................P.................
A/Lviv/N6/2009             ....................................................................................................
A/Christchurch/16/2010     ........................D..................................T........................................
A/Astrakhan/1/2011         ....................................................................................................
A/St Petersburg/100/2011   ..........................................G.........................................T...........T...
A/St Petersburg/27/2011    ....................................................................................T...............
A/Hong Kong/5659/2012      ..................N..................R..............................................T...............
A/Ghana/DILI-0620/2014     ...............M....................................................................TT..............
A/Kanagawa/165/2014        .....................A........................................Q.....................T...............
A/South Africa/3626/2013   ..............................................................Q.....................T...............
A/Rheinland-Pfalz/14/2014  ...................A..........................................Q.....................T...............
A/Norway/3246/2014         ..............................................................Q.....................T...............
A/Norway/3237/2014         ...........K..................................................Q.....................T...............
A/Denmark/51/2014          .....X.............A..........................................Q.....................T...............
A/Latvia/12-020598/2014    .............................................R................Q.....................T...............
A/Israel/P-352/2014        ..............................................................Q.....................T...............
A/Paris/2302/2014          ..............................................................Q....N................T...............
A/Marseille/2408/2014      ..............................................................Q.....................T...............
A/Setif/02/2015            ..............................................................Q.....................T...............
A/Finland/473/2014         ..............................................................Q.....................T...............
A/Hong Kong/7555/2014      ..............................................................Q.....................T...............
A/Schleswig-Holstein/9/2014 .............................................................Q.....................T...............
A/Croatia/1994/2014        ..............................................................Q.....................T...............
A/Switzerland/485/2015     ..............................................................Q.....................T...............
A/Luxembourg/14069580/2014 ..............................................................Q.....................T...............
A/Slovenia/33/2015         ..............................................................Q.....................T...............
A/Lisboa/EVA60/2015        ..............................................................Q.....................T...............
A/Belgium/G675/2014        ..............................................................Q.....................T...............

                                  210       220       230       240       250       260       270       280       290       300
                           ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009        VGSSRYSKKFKPEIAIRPKVRDQEGRMNYYWTLVEPGDKITFEATGNLVVPRYAFAMERNAGSGIIISDTPVHDCNTTCQTPKGAINTSLPFQNIHPITI
A/Bayern/69/2009           ....................................................................................................
A/Hong Kong/3934/2011      ..T..................N..........................................................................V.....
A/Lviv/N6/2009             ..T..................G..............................................................................
A/Christchurch/16/2010     ..T..................N...........................A..................................................
A/Astrakhan/1/2011         ..T.K.........XV................................L...................................................
A/St Petersburg/100/2011   ..T...................R.............................................................................
A/St Petersburg/27/2011    ..T...................R.............................................................................
A/Hong Kong/5659/2012      ..T.............................................L...................................................
A/Ghana/DILI-0620/2014     ..T..............................I......................V.........................E.................
A/Kanagawa/165/2014        ..T....................................................T........................................X...
A/South Africa/3626/2013   ..T....................................................T..........................E.................
A/Rheinland-Pfalz/14/2014  ..T....................................................TI.........................E.................
A/Norway/3246/2014         ..T....................................................T..........................E.................
A/Norway/3237/2014         ..T....................................................T.....E.........A..........E.................
A/Denmark/51/2014          ..T....R..........................D....................T..........................E.................
A/Latvia/12-020598/2014    ..T....................................................T..........................E.................
A/Israel/P-352/2014        ..T....................................................T..........................E.................
A/Paris/2302/2014          ..T....................................................T..........................E.................
A/Marseille/2408/2014      ..T..........................H.........................T..........................E...........V.....
A/Setif/02/2015            ..T....................................................T..........................E...........V.....
A/Finland/473/2014         ..T....................................................T..........................E.................
A/Hong Kong/7555/2014      ..T....................................................T..........................E.................
A/Schleswig-Holstein/9/2014 .T....................................................T..........................E.................
A/Croatia/1994/2014        ..T....................................................T..............L...........E.................
A/Switzerland/485/2015     ..T....................................................T..........................E.................
A/Luxembourg/14069580/2014 ..T............M.......................................T..........................E.................
A/Slovenia/33/2015         ..T....................................................T..........................E.................
A/Lisboa/EVA60/2015        ..T....................................................T..........................E.................
A/Belgium/G675/2014        ..T....................................................T..........................E.................
```

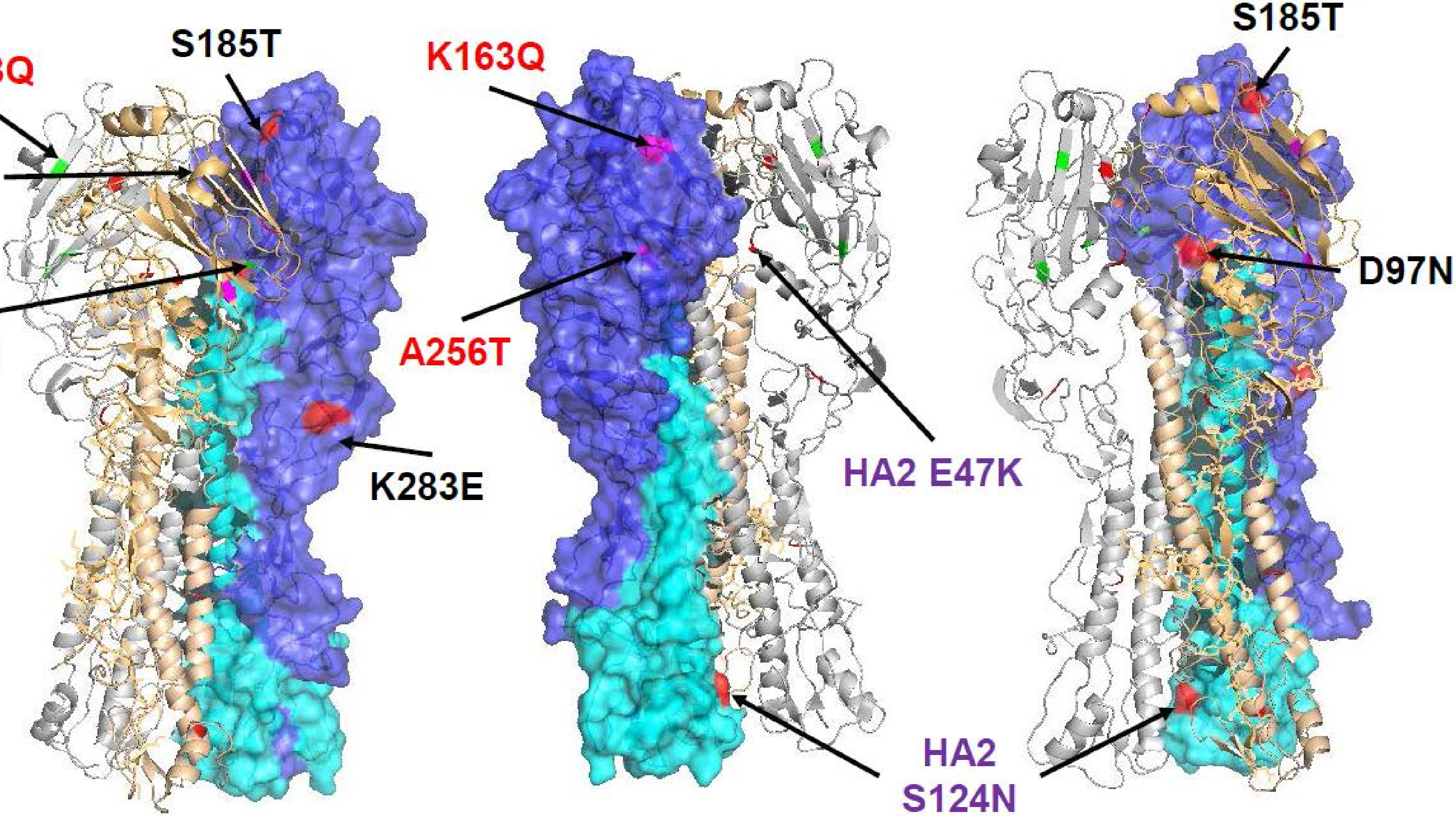
```
                                   310       320       330       340       350       360       370       380       390       400
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         GKCPKYVKSTKLRLATGLRNIPSIQSRGLFGAIAGFIEGGWTGMVDGWYGYHHQNEQGSGYAADLKSTQNAIDEITNKVNSVIEKMNTQFTAVGKEFNHL
A/Bayern/69/2009            ....................V...............................................................................
A/Hong Kong/3934/2011       ....................V...............................................................................
A/Lviv/N6/2009              ....................V...............................................................................
A/Christchurch/16/2010      ....................V....................................................K..........................
A/Astrakhan/1/2011          ....................V....................................................K..........................
A/St Petersburg/100/2011    ....................V....................................................K..........................
A/St Petersburg/27/2011     ....................V....................................................K..........................
A/Hong Kong/5659/2012       ....................V....................................................K..........................
A/Ghana/DILI-0620/2014      ....................V....................................................K..........................
A/Kanagawa/165/2014         ....................V....................................................K..........................
A/South Africa/3626/2013    ....................V....................................................K..........................
A/Rheinland-Pfalz/14/2014   ....................V....................................................K..........................
A/Norway/3246/2014          .........................................................................K..........................
A/Norway/3237/2014          .........................................................................K..........................
A/Denmark/51/2014           ....................V....................................................K..........................
A/Latvia/12-020598/2014     .........................................................................K..........................
A/Israel/P-352/2014         ....................V....................................................K..........................
A/Paris/2302/2014           ....................V....................................................K..........................
A/Marseille/2408/2014       ....................V....................................................K..........................
A/Setif/02/2015             ....................V....................................................K..........................
A/Finland/473/2014          ....................V....................................................K..........................
A/Hong Kong/7555/2014       ....................V....................................................K..........................
A/Schleswig-Holstein/9/2014 ....................V....................................................K..........................
A/Croatia/1994/2014         ....................V....................................................K..........................
A/Switzerland/485/2015      ....................V....................................................K..........................
A/Luxembourg/14069580/2014  ....................V....................................................K..........................
A/Slovenia/33/2015          ....................V....................................................K..........................
A/Lisboa/EVA60/2015         ....................V....................................................K..........................
A/Belgium/G675/2014         ....................V....................................................K..........................

                                   410       420       430       440       450       460       470       480       490       500
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         EKRIENLNKKVDDGFLDIWTYNAELLVLLENERTLDYHDSNVKNLYEKVRSQLKNNAKEIGNGCFEFYHKCDNTCMESVKNGTYDYPKYSEEAKLNREEI
A/Bayern/69/2009            ....................................................................................................
A/Hong Kong/3934/2011       ..............................................................................I.....................
A/Lviv/N6/2009              ....................................................................................................
A/Christchurch/16/2010      ....................................................................................................
A/Astrakhan/1/2011          ....................................................................................................
A/St Petersburg/100/2011    ..................................................N.................................................
A/St Petersburg/27/2011     ..................................................N.................................................
A/Hong Kong/5659/2012       ..................................................N.................................................
A/Ghana/DILI-0620/2014      ..........................................X.......N...............................................K.
A/Kanagawa/165/2014         ..................................................N...............................................K.
A/South Africa/3626/2013    ..................................................N...............................................K.
A/Rheinland-Pfalz/14/2014   ..................................................N...............................................K.
A/Norway/3246/2014          ..................................................N......................M........................K.
A/Norway/3237/2014          ..................................................N......................M........................K.
A/Denmark/51/2014           ..................................................N...............................................K.
A/Latvia/12-020598/2014     ..................................................N...............................................K.
A/Israel/P-352/2014         ..................................................N...............................................K.
A/Paris/2302/2014           ..................................................N.......................................G.......K.
A/Marseille/2408/2014       ..................................................N.......................................G.......K.
A/Setif/02/2015             ..................................................N.......................................G.......K.
A/Finland/473/2014          ..................................................N...............................................K.
A/Hong Kong/7555/2014       ..................................................N...............................................K.
A/Schleswig-Holstein/9/2014 ..................................................N...............................................K.
A/Croatia/1994/2014         ..................................................N...............................................K.
A/Switzerland/485/2015      ..................................................N...............................................K.
A/Luxembourg/14069580/2014  ..................................................N...............................................K.
A/Slovenia/33/2015          ..................................................N...S...........................................K.
A/Lisboa/EVA60/2015         ..................................................N...............................................K.
A/Belgium/G675/2014         ..................................................N...............................................K.

                                   510       520       530       540
                            ....|....|....|....|....|....|....|....|....|....
A/California/7/2009         DGVKLESTRIYQILAIYSTVASSLVLVVSLGAISFWMCSNGSLQCRICI
A/Bayern/69/2009            .................................................
A/Hong Kong/3934/2011       .................................................
A/Lviv/N6/2009              .................................................
A/Christchurch/16/2010      .................................................
A/Astrakhan/1/2011          .................................................
A/St Petersburg/100/2011    .................................................
A/St Petersburg/27/2011     .................................................
A/Hong Kong/5659/2012       .................................................
A/Ghana/DILI-0620/2014      .................................................
A/Kanagawa/165/2014         .................................................
A/South Africa/3626/2013    .......................V.........................
A/Rheinland-Pfalz/14/2014   ..I..............................................
A/Norway/3246/2014          .................................................
A/Norway/3237/2014          .................................................
A/Denmark/51/2014           .................................................
A/Latvia/12-020598/2014     .................................................
A/Israel/P-352/2014         .................................................
A/Paris/2302/2014           .................................................
A/Marseille/2408/2014       .................................................
A/Setif/02/2015             .................................................
A/Finland/473/2014          .................................................
A/Hong Kong/7555/2014       .................................................
A/Schleswig-Holstein/9/2014 .................................................
A/Croatia/1994/2014         .................................................
A/Switzerland/485/2015      .................................................
A/Luxembourg/14069580/2014  .................................................
A/Slovenia/33/2015          .................................................
A/Lisboa/EVA60/2015         .................................................
A/Belgium/G675/2014         ..I......................M.......................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 7. H1-HA locations of genetic group defining amino acid substitutions



HA1 substitutions D97N, S185T, S203T & K283E and HA2 substitutions E47K, S124N & E172K (not in structure) differentiate subgroup 6B and 6C viruses from A/California/7/2009. HA1 substitutions K163Q and A256T define subgroup 6B and V234I defines subgroup 6C.

Figure 8. Phylogenetic comparison of A(H1N1)pdm09 NA genes

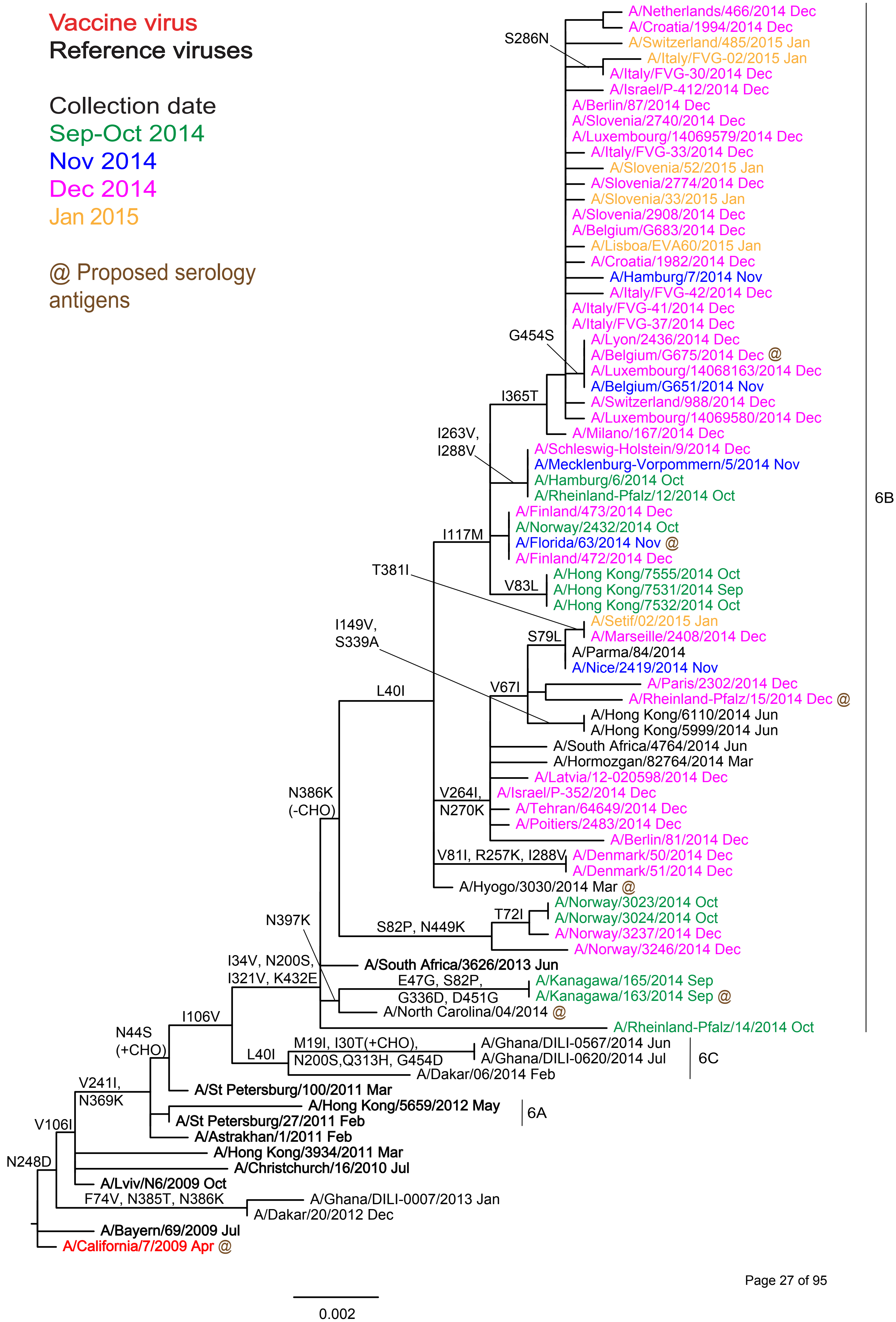


Figure 9. NA protein alignment for A(H1N1)pdm09 viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                    10        20        30        40        50        60        70        80        90       100
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         MNPNQKIITIGSVCMTIGMANLILQIGNIISIWISHSIQLGNQNQIETCNQSVITYENNTWVNQTYVNISNTNFAAGQSVVSVKLAGNSSLCPVSGWAIY
A/Bayern/69/2009            ....................................................................................................
A/Hong Kong/3934/2011       .............................................T...................................................... 3
A/Lviv/N6/2009              ....................................................................................................
A/Christchurch/16/2010      ...I..........I..................................................................................... 4
A/Astrakhan/1/2011          .................................................................................................... 5
A/St Petersburg/100/2011    ...........................................S........................................................ 7
A/St Petersburg/27/2011     .................................................................................................... 6
A/Hong Kong/5659/2012       ..................I................................................................................. 6A
A/Ghana/DILI-0620/2014      ..................I..........T.........I...S........................................................ 6C
A/Kanagawa/165/2014         .................................V.........S..G..................................P.................. 6B
A/South Africa/3626/2013    .................................V.........S........................................................ 6B
A/Rheinland-Pfalz/14/2014   ..................I..............V.........S......R.............................A................... 6B
A/Norway/3246/2014          .................................V.........S.....................................P.................. 6B
A/Norway/3237/2014          .................................V.........S...........................I.........P.................. 6B
A/Denmark/51/2014           .................................V.....I...S....................................I................... 6B
A/Latvia/12-020598/2014     .................................V.....I...S........................................................ 6B
A/Israel/P-352/2014         .................................V.....I...S........................................................ 6B
A/Paris/2302/2014           ..................T..............V.....I...S......................I................................. 6B
A/Marseille/2408/2014       .................................V.....I...S......................I...........L..................... 6B
A/Setif/02/2015             .................................V.....I...S......................I...........L..................... 6B
A/Finland/473/2014          .................................V.....I...S........................................................ 6B
A/Hong Kong/7555/2014       .................................V.....I...S......................................L................. 6B
A/Schleswig-Holstein/9/2014 .................................V.....I...S........................................................ 6B
A/Croatia/1994/2014         .................................V.....I...S........................................................ 6B
A/Switzerland/485/2015      .................................V.....I...S........................................................ 6B
A/Luxembourg/14069580/2014  .................................V.....I...S........................................................ 6B
A/Slovenia/33/2015          .................................V.....I...S........................................................ 6B
A/Lisboa/EVA60/2015         .................................V.....I...S........................................................ 6B
A/Belgium/G675/2014         .................................V.....I...S........................................................ 6B

                                   110       120       130       140       150       160       170       180       190       200
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         SKDNSVRIGSKGDVFVIREPFISCSPLECRTFFLTQGALLNDKHSNGTIKDRSPYRTLMSCPIGEVPSPYNSRFESVAWSASACHDGINWLTIGISGPDN
A/Bayern/69/2009            ....................................................................................................
A/Hong Kong/3934/2011       .....I..............................................................................................
A/Lviv/N6/2009              .....I..............................................................................................
A/Christchurch/16/2010      .....I..................................................................................S...........
A/Astrakhan/1/2011          .....I..............................................................................................
A/St Petersburg/100/2011    .....I..............................................................................................
A/St Petersburg/27/2011     .....I..............................................................................................
A/Hong Kong/5659/2012       .....I..............................................................................................
A/Ghana/DILI-0620/2014      ...................................................................................................S
A/Kanagawa/165/2014         ...................................................................................................S
A/South Africa/3626/2013    ...................................................................................................S
A/Rheinland-Pfalz/14/2014   .......................................................................................T...........S
A/Norway/3246/2014          ...................................................................................................S
A/Norway/3237/2014          .................................................................................................S.S
A/Denmark/51/2014           ...................................................................................................S
A/Latvia/12-020598/2014     ...................................................................................................S
A/Israel/P-352/2014         ...................................................................................................S
A/Paris/2302/2014           ...................................................................................................S
A/Marseille/2408/2014       ...................................................................................................S
A/Setif/02/2015             ...................................................................................................S
A/Finland/473/2014          ................M..................................................................................S
A/Hong Kong/7555/2014       ................M..................................................................................S
A/Schleswig-Holstein/9/2014 ................M..................................................................................S
A/Croatia/1994/2014         ................M..............................................................................V...S
A/Switzerland/485/2015      ................M..................................................................................S
A/Luxembourg/14069580/2014  ................M..................................................................................S
A/Slovenia/33/2015          ................M..................................................................................S
A/Lisboa/EVA60/2015         ................M..................................................................................S
A/Belgium/G675/2014         ................M..................................................................................S

                                   210       220       230       240       250       260       270       280       290       300
                            ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009         GAVAVLKYNGIITDTIKSWRNNILRTQESECACVNGSCFTVMTDGPSNGQASYKIFRIEKGKIVKSVEMNAPNYHYEECSCYPDSSEITCVCRDNWHGSN
A/Bayern/69/2009            .....................................................................................G..............
A/Hong Kong/3934/2011       ...............................................D....................................................
A/Lviv/N6/2009              ...............................................D....................................................
A/Christchurch/16/2010      ...............................................D....................................................
A/Astrakhan/1/2011          ........................................I......D....................................................
A/St Petersburg/100/2011    ........................................I......D....................................................
A/St Petersburg/27/2011     ........................................I......D....................................................
A/Hong Kong/5659/2012       ........................................I......D....................................................
A/Ghana/DILI-0620/2014      ........................................I......D....................................................
A/Kanagawa/165/2014         ........................................I......D....................................................
A/South Africa/3626/2013    ........................................I......D....................................................
A/Rheinland-Pfalz/14/2014   ........................................I......D.........................H..........................
A/Norway/3246/2014          ........................................I......D....................................................
A/Norway/3237/2014          ........................................I......D....................................................
A/Denmark/51/2014           ........................................I......D........K..............................V............
A/Latvia/12-020598/2014     ........................................I......D...............I.....K..............................
A/Israel/P-352/2014         ........................................I......D...............I.....K..............................
A/Paris/2302/2014           ........................................I......D...............I.....K..............................
A/Marseille/2408/2014       ........................................I......D...............I.....K..............................
A/Setif/02/2015             ........................................I......D...............I.....K..............................
A/Finland/473/2014          ........................................I......D....................................................
A/Hong Kong/7555/2014       ........................................I......D....................................................
A/Schleswig-Holstein/9/2014 ........................................I......D..............V........................V............
A/Croatia/1994/2014         ........................................I......D....................................................
A/Switzerland/485/2015      ........................................I......D....................................................
A/Luxembourg/14069580/2014  ........................................I......D....................................................
A/Slovenia/33/2015          ........................................I......D....................................................
A/Lisboa/EVA60/2015         ........................................I......D....................................................
A/Belgium/G675/2014         ........................................I......D....................................................
```

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                                    310       320       330       340       350       360       370       380       390       400
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/California/7/2009          RPWVSFNQNLEYQIGYICSGIFGDNPRPNDKTGSCGPVSSNGANGVKGFSFKYGNGVWIGRTKSISSRNGFEMIWDPNGWTGTDNNFSIKQDIVGINEWS
A/Bayern/69/2009             ....................................................................................................
A/Hong Kong/3934/2011        ............R................................................................................I......
A/Lviv/N6/2009               ....................................................................................................
A/Christchurch/16/2010       ........................................................................I...........................
A/Astrakhan/1/2011           ....................................................................K...............................
A/St Petersburg/100/2011     ........................................................I...........K...............................
A/St Petersburg/27/2011      ....................................................................K...............................
A/Hong Kong/5659/2012        ....................................................................K................S..............
A/Ghana/DILI-0620/2014       ............H.......................................................K...............................
A/Kanagawa/165/2014          ....................V..............D................................K...........................K...
A/South Africa/3626/2013     ....................V...............................................K...............................
A/Rheinland-Pfalz/14/2014    ....................V...............................................K...............................
A/Norway/3246/2014           ....................V...............................................K................K.........T....
A/Norway/3237/2014           ....................V...............................................K................K..............
A/Denmark/51/2014            ....................V...............................................K................K..............
A/Latvia/12-020598/2014      ....................V...............................................K................K..............
A/Israel/P-352/2014          ....................V...............................................K................K..............
A/Paris/2302/2014            ....................V...............................................K................K..M....I......
A/Marseille/2408/2014        ....................V...............................................K...........I....K..............
A/Setif/02/2015              ....................V...............................................K...........I....K..............
A/Finland/473/2014           ....................V...............................................K................K..............
A/Hong Kong/7555/2014        ....................V...............................................K................K..............
A/Schleswig-Holstein/9/2014  ....................V...............................................K................K..............
A/Croatia/1994/2014          ....................V...........................................T...K................K..............
A/Switzerland/485/2015       ....................V...........................................T...K................K.........M....
A/Luxembourg/14069580/2014   ....................V...........................................T...K................K..............
A/Slovenia/33/2015           ....................V...........................................T...K................K..............
A/Lisboa/EVA60/2015          ....................V...........................................T...K................K..............
A/Belgium/G675/2014          ....................V...........................................T...K................K..............

                                    410       420       430       440       450       460
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/California/7/2009          GYSGSFVQHPELTGLDCIRPCFWVELIRGRPKENTIWTSGSSISFCGVNSDTVGWSWPDGAELPFTIDK
A/Bayern/69/2009             .....................................................................
A/Hong Kong/3934/2011        ..................................................................V..
A/Lviv/N6/2009               .....................................................................
A/Christchurch/16/2010       .................................................................X---
A/Astrakhan/1/2011           .....................................................................
A/St Petersburg/100/2011     .....................................................................
A/St Petersburg/27/2011      .....................................................................
A/Hong Kong/5659/2012        ..................................................G..................
A/Ghana/DILI-0620/2014       .....................................................D...............
A/Kanagawa/165/2014          ...............................X..................G..................
A/South Africa/3626/2013     ...............................E.....................................
A/Rheinland-Pfalz/14/2014    ...............................E.....................................
A/Norway/3246/2014           ...............................E................K....................
A/Norway/3237/2014           ...............................E................K....................
A/Denmark/51/2014            ...............................E.....................................
A/Latvia/12-020598/2014      ...............................E.....................................
A/Israel/P-352/2014          ...............................E.....................................
A/Paris/2302/2014            ...............................E.....................................
A/Marseille/2408/2014        ...............................E.....................................
A/Setif/02/2015              ...............................E.....................................
A/Finland/473/2014           ...............................E.....................................
A/Hong Kong/7555/2014        ...............................E.....................................
A/Schleswig-Holstein/9/2014  ...............................E.....................................
A/Croatia/1994/2014          ...............................E.....................................
A/Switzerland/485/2015       ...............................E.S...................................
A/Luxembourg/14069580/2014   ...............................E.....................................
A/Slovenia/33/2015           ...............................E.....................................
A/Lisboa/EVA60/2015          ...............................E.....................................
A/Belgium/G675/2014          ...............................E.....................S...............
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Influenza A(H3N2) virus analysis.

Influenza A(H3N2) viruses collected since 2014-08-31 have been received from NICs in Europe, Africa/Indian Ocean (Ghana, Madagascar, Morocco, Senegal, Zambia), the Middle East (Israel, Jordan, Oman) and from Hong Kong SAR. These viruses have become increasingly difficult to characterise antigenically by HI assay due to either the variable agglutination of red blood cells from guinea pig, turkey and humans or the loss of the ability of viruses to agglutinate any of the red blood cells. Of the successfully propagated viruses, 52% had sufficient HA titre in assays conducted using guinea pig red blood cells in the presence of 20nM oseltamivir, added to circumvent any NA-mediated binding of H3N2 viruses to the red blood cells, to be analysed by HI assay **(Figure 10, Table 7)**. However the remainder were unable to agglutinate guinea pig red blood cells at all (23% of the total) or were unable to agglutinate red blood cells in the presence of 20nM oseltamivir (25% of the total). The vast majority (95%) of viruses in these last two cetegories belonged to genetic subgroup 3C.2a. Those viruses in genetic subgroup 3C.2a able to bind guinea pig red cells in the presence of oseltamivir showed either loss of or polymorphism at the 158/160 glycosylation motif acquired in the haemagglutinin of viruses in genetic subgroup 3C.2a **(Table 7, Figure 11)**.

Viruses were characterised using plaque reduction neutralisation assay (PRNA) to complement HI tests.

A total of 135 viruses could be analysed by the HI assay and summary of the HI results is shown in **Table 8** and the HI results are shown in **Tables 9-1** to **9-14**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in the tables are viruses with collection dates after 2014-08-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees presented here are highlighted.

All test viruses with collection dates after 2014-08-31 that were propagated in MDCK-SIAT1 cells reacted poorly in HI assays (≥8-fold decrease) with post-infection ferret antiserum raised against the egg-propagated vaccine virus, A/Texas/50/2012, compared with the titre of the antiserum with the homologous virus. Similarly, low levels of reactivity were seen with antisera raised against the egg-propagated reference virus A/Hong Kong/146/2013: only approximately 6% of test viruses reacted within 4-fold of the titre with the homologous egg-propagated virus. Better reactivity was seen with test viruses when analysed with an antiserum raised against the exclusively egg-propagated A/Stockholm/6/2014 isolate 2 **(Tables 9-1** to **9-13)**, a virus belonging to genetic subgroup 3C.3a. This antiserum showed a low titre for the homologous virus but recognised nearly 80% of test viruses at titres within 4-fold of the homologous titre. Antiserum raised against egg-propagated A/Switzerland/9715293/2013, the virus recommended for the Southern Hemisphere vaccine in 2015 and in genetic subgroup 3C.3a, had an homologous titre of 1280 in all but one HI assay and recognised only 13% of the test viruses at titres within 4-fold of the homologous titre. Antiserum raised against egg-propagated A/Hong Kong/5738/2014 clone 121, a virus in genetic subgroup 3C.2a, had an homologous titre of 640 or 1280 and failed to recognise any of the test viruses at titres within 4-fold of the homologous titre.

Ferret antisera raised against reference viruses propagated in tissue culture cells, A/Victoria/361/2011 and A/Samara/73/2013, recognised the test viruses somewhat more effectively. The antiserum raised against A/Victoria/361/2011 recognised ~60% of the test viruses at a titre within four-fold of the antiserum for the homologous virus **(Table 8)**, but the antiserum raised against A/Samara/73/2013 recognised only ~27% of test viruses at a titre within four-fold of the titre for the homologous virus. These reference viruses have HA genes from genetic groups 3C.1 and 3C.3 respectively. Antisera raised against reference viruses belonging to genetic subgroup 3C.3a that had been exclusively propagated in cell culture, A/Stockholm/6/2014 and A/Switzerland/9715293/2013, recognised 88% and 79%, respectively, of test viruses at titres within 4-fold of those with the corresponding homologous viruses. An antiserum raised against a reference virus belonging to genetic subgroup 3C.2a that had been exclusively propagated in cell culture, A/Hong Kong/5738/2014, recognised 74% of test viruses at titres within 4-fold of that for the homologous virus.

The results of PRNA performed with MDCK-SIAT1 cells are shown in **Tables 10-1** to **10-4**. A total of 61 test viruses with collection dates after 2014-08-31 were analysed with a panel of antisera raised against both cell culture-propagated and egg-propagated viruses from genetic subgroups 3C.1, 3C.2a and 3C.3a. The genetic groups to which the HA genes of the reference viruses and the test viruses belong are shown. The titres given are to the nearest 2-fold dilution factor with the numbers in parentheses being interpolated directly from the results using image-processing software. Compared to their cell-propagated cultivars, the egg-propagated cultivar of the reference virus A/Switzerland/9715923/2013 has the egg-adaptive changes I140R and G186V in HA1 and that of A/Hong Kong/5738/2014 has T160K (resulting in the loss of the potential glycosylation site motif at 158-160 in HA1), L194P and T203I. The cell culture-propagated cultivar of A/Hong Kong/5738/2014 is polymorphic at residue 160 leading to the partial loss of the potential glycosylation site at residues 158-160; the cell culture-propagated cultivar of A/England/530/2014 has lost this motif, but it is retained in the cell culture-propagated A/Hong Kong/7295/2014.

Both antisera raised against the egg-propagated reference viruses have higher homologous titres than those raised against the cell-propagated viruses. Two thirds of the viruses were recognised at titres within 4-fold of the titre for the homologous virus by the antiserum raised against the egg-propagated cultivar of A/Switzerland/9715293/2013 but the antiserum raised against the egg-propagated cultivar of A/Hong Kong/5738/2014 recognised only 2/28 (7%) of the test viruses at titres within 4-fold of the titre for the homologous virus. Nevertheless, there was little difference in the absolute titres for the test viruses seen between the two antisera used in the same tests **(Tables 10-1** and **10-2)**.

The antiserum raised against the cell culture-propagated A/Victoria/361/2011 recognised only 12% of the test viruses at a titre within 2-fold of the titre for the homologous virus, but each of the other antisera raised against cell culture-propagated reference viruses in genetic subgroups 3C.3a (A/Switzerland/9715293/2013) or 3C.2a (A/England/530/2014, A/Hong Kong/5738/2014, A/Hong Kong/7295/2014) recognised between 73% and 100% of the test viruses at titres within 2-fold of their titres for the homologous viruses: these percentages were for the antiserum raised against A/England/530/2014, 73%; A/Switzerland/9715293/2013, 89%; A/Hong Kong/5738/2014, 97%; A/Hong Kong/7295/2014, 100%.

Phylogenetic analysis of the HA genes of representative recently circulating H3N2 viruses is shown in **Figure 11**. The HA genes fell within genetic group 3C. This group has three subdivisions: 3C.1, 3C.2 and 3C.3 and three new genetic subgroups have emerged in 2014, one in subdivision 3C.2, 3C.2a, and two in 3C.3, 3C.3a and 3C.3b **(Figure 11)** (viruses in subgroup 3C.3b do not appear to be antigenic drift variants from viruses in 3C.3).

The previously used vaccine virus A/Texas/50/2012 belongs to genetic subdivision 3C.1. Most of the viruses received with collection dates since 2014-08-31 that have been sequenced fall into subgroups 3C.2a (60%), 3C.3b (27%) and 3C.3a (12.5%). Amino acid substitutions that define these subgroups compared with A/Texas/50/2012 are:

(**3C.2a**) **L3I**, **N144S** (resulting in the loss of a potential glycosylation site), **N145S, F159Y**, **K160T** (resulting in the gain of a potential glycosylation site), **V186G**[@], **N225D** and **Q311H** in **HA1**, e.g. A/Hong Kong/3579/2014.

(**3C.3a**) **T128A** (resulting in the loss of a potential glycosylation site), **A138S, R142G, N145S, F159S, V186G**[@] and **N225D** in **HA1**, e.g. A/Switzerland/9715293/2013.

(**3C.3b**) **E62K, K83R, N122D** (resulting in the loss of a potential glycosylation site), **T128A** (resulting in the loss of a potential glycosylation site), **R142G, N145S, L157S, V186G**[@] and **R261Q** in **HA1** with **M18K** in **HA2**.

[@]Note: the **G186V** substitution in HA1 occurred during adaptation of A/Texas/50/2012 to propagation in hens' eggs.

Figure 12 shows an HA amino acid sequence alignment for a representative selection of the viruses used to generate the phylogeny **(Figure 11)**. The vaccine virus A/Texas/50/2012 is shown in red, glycosylation motifs are highlighted and the reference viruses are coloured blue. The genetic group to which each of the sequences belongs is also shown.

Figure 13 shows the location on a 2005 H3 HA structure of amino acid substitutions that define genetic subgroups 3C.2a **(Figure 13A)**, 3C.3a **(Figure 13B)** and 3C.3b **(Figure 13C)** each compared with the vaccine virus A/Texas/50/2012.

Figure 14 shows a NA phylogenetic tree for representative viruses with the HA genetic groups indicated. **Figure 15** shows a NA amino acid sequence alignment for a representative selection of these viruses. As above, vaccine viruses are shown in red, glycosylation motifs are highlighted, the reference viruses are coloured blue and the genetic group to which the corresponding HA gene sequences belong is also shown.

The NA gene sequences of viruses with HA genes in genetic group 3C do not cluster in the same manner as the HA sequences. The NA substitutions **Y155F, D251V** and **S315G** define two major divisions among recently collected viruses. Recent viruses carrying these NA substitutions have HA genes in genetic subgroup 3C.3b. NA genes of viruses with HA genes in the 3C.2a subgroup fall into two distinct clusters within one of the major groups. Both clusters share the substitution **E221D**; one cluster is defined by the NA substitution **I392T**, the other cluster by the substitutions **T267K** and **I380V.** Viruses in genetic subgroup 3C.3a fall into three genetic clusters; one again is defined by the substitutions **E221D** and **I392T,** a second by **M51V, V143G, E221D, V263I** and **T434N**, and the third by **I77V, G93D, A110D, V313A** and **V421I**; viruses in this latter group are present in West Africa. **Figure 16** indicates the positions of the amino acid substitutions, on the structure of an N2 NA, that define viruses with HA genes falling into the 3C.2a, 3C.2b and 3C.3a genetic subgroups.

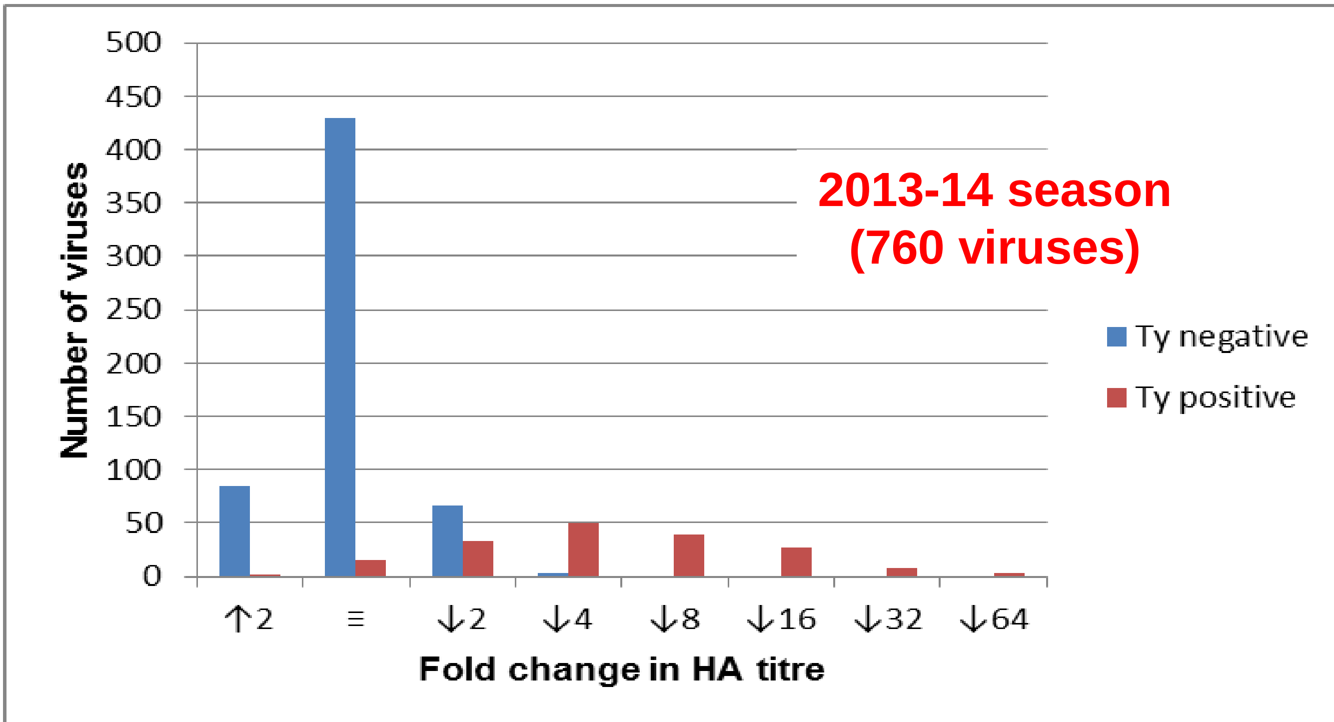
Antiviral resistance

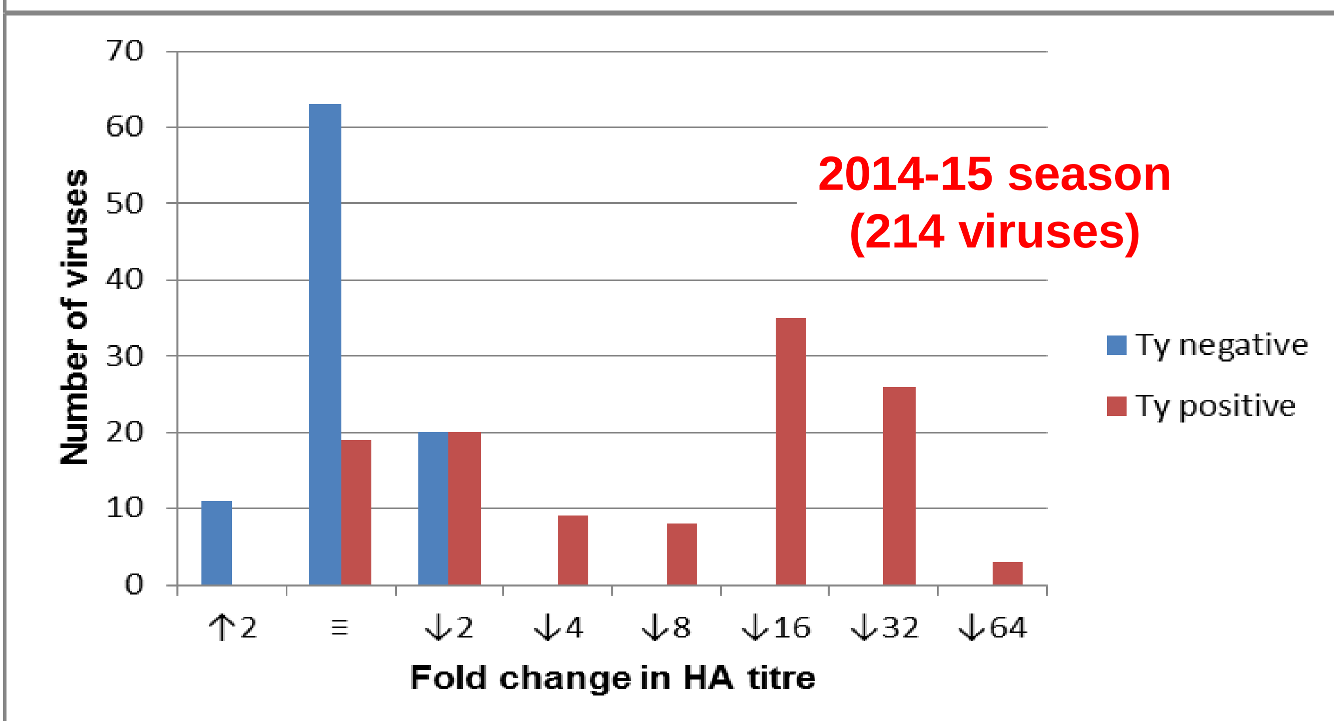
Phenotypic testing to assess susceptibility to oseltamivir and zanamivir was performed on 151 viruses recovered from samples with collection dates after 2014-08-31. All showed normal inhibition (NI) by both neuraminidase inhibitors (**Table 15**).

Conclusion.

All recent test viruses analysed showed poor reactivity with antisera raised against the egg-propagated vaccine virus A/Texas/50/2012 in HI assays. Viruses in genetic subgroup 3C.2a were the most common. Many viruses in subgroup 3C.2a did not agglutinate red blood cells and could not therefore be characterised by HI assay. The minority of viruses in genetic subgroup 3C.2a that were able to agglutinate red blood cells in the presence of oseltamivir had become a mixed population during culture resulting in the loss or partially loss of the glycosylation site motif at resides 158 to 160 in HA1. In the PRNA the subgroup 3C.2a viruses were not recognised well by an antiserum raised against cell culture-propagated A/Victoria/361/2011, but were recognised well by antisera raised against cell culture-propagated viruses in genetic subgroups 3C.2a and 3C.3a. However the egg-adaptive changes of viruses in 3C.2a and 3C.3a genetic subgroups resulted in the generation of antisera with poorer recognition of the cell-propagated test viruses compared to the homologous viruses. These egg-adapted viruses are antigenically distinct by HI assay. However, the corresponding cell culture-propagated cultivars are not antigenically distinct in the HI assay but are distinguishable in PRNA.

Figure 10. Analysis of H3N2 Influenza Virus Agglutination of Guinea Pig RBCs and the Effect of 20nM Oseltamivir





Fold shift in HA activity in the presence of 20nM oseltamivir:
[Enhancement (↑2) Equivalence (≡) Reduction (↓2-↓64)]

Table 7. A(H3N2) viruses recovered from specimens collected after 2014-08-31, as assessed by NA activity (MUNANA-based assay)

RBC binding phenotype[1]			HI assay possible	Number of isolates	Number sequenced[2]	Number in clade 3C.3(b)	Number in clade 3C.3a	Number in clade 3C.2a
Turkey	Guinea pig	Guinea pig + 20nM oseltamivir						
+	+	+	Yes	54	42 (78%)	8	19	15*
-	+	+	Yes	75	54 (72%)	38	6	10*
-	-	+	Yes	6	4 (67%)	4	0	0
+	-	+	Yes	0				
			Totals	135 (52%)	100 (74%)	50	25	25
+	+	-	No	60	50 (83%)	0	0	50
+	-	-	No	0				
-	+	-	No	7	5 (71%)	0	0	5
-	-	-	No	59	44 (75%)	4	0	40
			Totals	126 (48%)	99 (79%)	4	0	95
			Totals	261	199 (76%)	54	25	120

1. Negative (-) corresponds to HA titres of <4, while positive (+) corresponds to HA titres of ≥4
2. As of 2014-02-11
* These viruses showed either loss [NYK] or polymorphism [K/NYT or NYK/T] at the 158/160 glycosylation motif [NYT] in HA1

Table 8. Summary of influenza A(H3N2) viruses analysed, collected after 2014-08-31

MONTH	H3N2				
Country	Number received[1]	Number propagated[2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]
2014					
SEPTEMBER					
Belgium	1	1		1	
France	1	1			1
Madagascar	26	5		3	2
Sweden	3	2		1	1
OCTOBER					
Belgium	5	1		1	
Germany	1	1			1
Madagascar	1	1		1	
Netherlands	4	2	1	1	
Norway	3	1	1		
Senegal	4	3		1	2
Slovenia	2	2		1	1
Spain	10	4	2	2	
Sweden	2	1		1	
United Kingdom	1	1	1		
NOVEMBER					
Germany	5	3		2	1
Ghana	2	1			1
Hong Kong	2	1	1		
Norway	2	2		2	
Spain	13	6	2	4	
Sweden	3	3			3
United Kingdom	4	3		1	2
Zambia	1	1	1		
DECEMBER					
Austria	7	1	1		
Belgium	1	1	1		
Croatia	2	1	1		
France	26	19	5	6	8
Germany	23	11		7	4
Greece	2	1			1
Hong Kong	2	1		1	
Israel	5	3		2	1
Italy	9	4	2	2	
Jordon	7	1		1	
Latvia	6	3	1		2
Luxembourg	3	1			1
Malta	4	1		1	
Morocco	2	1			1
Norway	14	6		4	2
Oman	6	1			1
Portugal	3	1			1
Russia	10	8		2	6
Spain	26	5	3	2	
United Kingdom	8	2		1	1
2015					
JANUARY					
Denmark	2	2		1	1
Germany	11	7		1	6
Malta	5	1		1	
Norway	4	1		1	
Spain	3	2	1	1	
United Kingdom	9	4		4	
	296	135	24	60	51
28 Countries			17.8%	44.4%	37.8%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2. Propagated to sufficient titre to perform HI assay in presence of 20nM oseltamivir
3. Compared to homologous titres for ferret antisera raised against tissue culture propagated virus A/Victoria/361/2011

Table 9-1. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-11-05

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Nor 466/14 F13/14 3C.3a	A/Switz 9715293/13 NIB F13/14 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl123
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	40	80	80	80	40	80
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	160	640	320	80	320	160	160	80	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	1280	160	320	640	80	80	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	640	320	320	1280	640	160	640	320	320	160	640
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	640	640	640	1280	2560	160	320	640	160	80	1280
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	640	320	640	160	80	160	80	40
A/Stockholm/6/2014	3C.3	2014-02-06	SIAT1/SIAT4	<	80	80	320	160	80	640	320	320	320	160
A/Stockholm/6/2014	3C.3	2014-02-06	E4/E1 isolate 2	80	320	160	160	320	<	160	640	160	80	640
A/Norway/466/2014	3C.3a	2014-02-03	SIAT1/SIAT2	<	40	40	160	160	40	640	160	320	160	160
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	<	160	80	40	640	160	320	160	160
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	320	160	160	320	<	320	640	320	160	1280
TEST VIRUSES														
A/Egypt/GAB3932/2014	3C.3	unknown	SIAT1	80	320	320	640	320	160	640	160	320	160	320
A/Egypt/GAB3911/2014	3C.3	unknown	SIAT1	80	160	160	640	320	80	640	320	320	160	160
A/Egypt/GAB4019/2014		unknown	SIAT1	40	160	160	640	320	80	640	320	320	80	160
A/Egypt/IMB3452/2014	3C.3	unknown	SIAT1	320	640	640	1280	1280	320	1280	640	640	320	640
A/Egypt/Men4372/2014	3C.3	unknown	SIAT1	160	320	160	1280	640	160	640	320	640	160	320
A/Tasmania/11/2014	3C.3a	2014-03-16	E5/E1	<	640	40	80	80	<	160	160	80	80	160
A/Jiangxi-Donghu/11101/2014	3C.3a	2014-06-11	C3/SIAT1	<	<	40	80	40	<	160	80	80	<	40
A/Sichuan-Yanjiang/1239/2014	3C.3a	2014-06-14	C3/SIAT1	<	80	80	320	160	80	640	160	320	160	160
A/Oman/3676/2014	3C.3	2014-06-20	SIAT2	<	80	80	160	80	40	160	80	160	40	80
A/Guangdong-Haizhu/1510/2014	3C.3a	2014-06-25	C2/SIAT1	<	<	<	80	40	<	160	40	80	<	40
A/Sichuan-Jiangyang/1686/2014	3C.3a	2014-07-02	C2/SIAT1	<	40	40	160	80	40	320	160	160	80	160
A/Oman/4289/2014	3C.3a	2014-07-28	SIAT2	<	80	40	160	160	40	640	160	320	160	160
A/Slovenia/1944/2014	3C.3a	2014-10-09	SIAT1	<	40	40	160	80	40	640	160	320	160	160
						Vaccine SH2014 NH 2014/15								Vaccine SH2015

1. < = <40

Sequences in phylogenetic trees

Table 9-2. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-11-21

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Sth Afr 4655/13 F10/14 3C.3 cl 101-60	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Nor 466/14 F13/14 3C.3a	A/Switz 9715293/13 NIB F13/14 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl123
				Haemagglutination inhibition titre[1] – Post infection ferret antisera										
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	80	160	160	160	<	40	80	40	<	80
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	80	160	160	640	320	80	320	160	160	80	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	640	80	320	640	80	80	640
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	160	320	320	1280	640	160	640	320	320	160	320
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	640	640	640	1280	2560	160	160	640	80	80	640
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	40	40	80	320	320	320	80	40	80	40	<
A/Stockholm/6/2014	3C.3	2014-02-06	SIAT1/SIAT4	<	80	80	320	320	80	640	320	320	320	160
A/Stockholm/6/2014	3C.3	2014-02-06	E4/E1 isolate 2	80	160	160	160	320	<	160	320	160	80	640
A/Norway/466/2014	3C.3a	2014-02-03	SIAT1/SIAT2	<	40	40	160	160	40	640	160	160	160	160
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	40	160	160	40	320	160	160	160	160
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	160	160	640	<	320	320	160	160	1280
TEST VIRUSES														
A/Hong Kong/4244/2014	3C.2a	2014-02-16	MDCKx/SIAT1	40	80	80	320	160	80	320	80	160	80	160
A/Hong Kong/6013/2014	3C.2a	2014-06-05	MDCK2/SIAT1	<	<	40	160	80	<	160	80	80	40	80
A/Hong Kong/6033/2014	3C.3a	2014-06-05	MDCKx/SIAT1	<	40	80	320	160	80	640	160	320	160	160
A/Hong Kong/6086/2014	3C.3a	2014-06-08	MDCKx/SIAT1	<	40	40	160	160	40	640	160	320	160	160
A/Hong Kong/6315/2014	3C.3a	2014-06-14	MDCKx/SIAT1	<	40	40	160	80	40	320	160	320	160	160
A/Hong Kong/6421/2014	3C.3a	2014-06-19	MDCKx/SIAT1	<	40	40	160	80	40	320	160	160	160	160
A/Hong Kong/6872/2014	3C.3a	2014-07-14	MDCKx/SIAT1	<	80	80	160	160	80	640	160	320	160	160
A/Hong Kong/6873/2014	3C.3a	2014-07-15	MDCKx/SIAT1	<	80	80	320	160	80	640	160	320	160	160
A/Hong Kong/7229/2014	3C.3a	2014-07-31	MDCKx/SIAT1	<	40	40	160	160	40	320	160	320	160	160
A/Hong Kong/7243/2014	3C.3a	2014-08-02	MDCKx/SIAT1	<	40	40	160	160	40	640	160	160	160	160
A/Hong Kong/7276/2014	3C.3a	2014-08-02	MDCKx/SIAT1	<	40	40	160	160	40	640	160	320	160	160
A/Hong Kong/7272/2014	3C.3a	2014-08-04	MDCKx/SIAT1	<	40	40	160	80	40	320	80	160	160	160
A/Hong Kong/7313/2014	3C.3a	2014-08-07	MDCKx/SIAT1	<	<	40	80	80	40	320	80	160	80	80
A/Hong Kong/7329/2014	3C.3a	2014-08-11	MDCK2/SIAT1	<	40	40	160	160	40	640	160	320	160	160
A/Hong Kong/7347/2014	3C.2a	2014-08-14	MDCK2/SIAT1	40	80	80	320	160	40	320	80	160	40	80
A/Hong Kong/7364/2014	3C.2a	2014-08-18	MDCK2/SIAT3	<	<	<	80	40	<	160	40	80	40	80
A/Norway/3004/2014	3C.3b	2014-10-16	SIAT1	40	80	80	320	160	40	320	160	320	80	160

1. < = <40

Vaccine SH2014 NH 2014/15 (A/Texas 50/12 Egg F42/13)

Vaccine SH2015 (A/Switz 9715293/13 F25/14)

Table 9-3. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2014-11-27 + 2014-12-19

				Haemagglutination inhibition titre[1]										
				Post infection ferret antisera										
Viruses		Collection Date	Passage History	A/Perth 16/09	A/Vic 361/11	A/Texas 50/12	A/Samara 73/13	A/HK 146/13	A/Sth Afr 4655/13	A/Stock 6/14	A/Stock 6/14	A/Nor 466/14	A/Switz 9715293/13	A/Switz 9715293/13
				F18/11	T/C F09/12	Egg F42/13	F24/13	F40/13	F10/14	T/C F14/14	Egg F20/14	F13/14	NIB F13/14	F25/14
	Genetic group				3C.1	3C.1	3C.3	3C.2	3C.3 cl 101-60	3C.3a	3C.3a isolate 2	3C.3a	3C.3a	3C.3a cl123
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	40	80	80	80	40	80
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	160	640	320	80	320	160	160	80	160
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	1280	160	320	640	80	80	1280
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	640	320	320	1280	640	160	640	320	320	160	640
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	640	640	640	1280	2560	160	320	640	160	80	1280
A/South Africa/4655/2013	3C.3	2013-06-25	E7 clone 101-60	80	80	160	640	320	640	160	80	160	80	40
A/Stockholm/6/2014	3C.3	2014-02-06	SIAT1/SIAT4	<	80	80	320	160	80	640	320	320	320	160
A/Stockholm/6/2014	3C.3	2014-02-06	E4/E1 isolate 2	80	320	160	160	320	<	160	640	160	80	640
A/Norway/466/2014	3C.3a	2014-02-03	SIAT1/SIAT2	<	40	40	160	160	40	640	160	320	160	160
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	80	40	320	160	40	640	160	320	320	160
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	160	160	320	<	320	320	160	80	1280
TEST VIRUSES														
A/Hong Kong/7364/2014	3C.2a	2014-08-18	MDCK2/SIAT3	<	<	<	80	40	<	160	40	80	40	80
A/Niedersachsen/8/2014	3C.3a	2014-10-09	C2/SIAT1	<	40	40	160	80	<	320	80	160	80	80
A/England/516/2014	3C.3b	2014-10-24	SIAT1/SIAT1	160	160	160	320	160	40	320	80	160	80	160
A/Niedersachsen/9/2014	3C.3b	2014-11-03	C2/SIAT1	80	80	160	320	160	40	160	80	160	40	80
A/Rheinland-Pfalz/13/2014	3C.3b	2014-11-08	C2/SIAT1	80	80	80	320	160	<	160	80	160	40	80
A/England/527/2014	3C.3b	2014-11-18	SIAT1/SIAT1	80	80	80	320	160	40	160	160	160	40	80
A/England/528/2014	3C.2a	2014-11-20	SIAT2/SIAT1	<	<	<	80	80	<	40	<	<	<	<
A/England/530/2014	3C.2a	2014-11-26	MDCK1/SIAT1	<	<	40	160	80	<	80	80	80	40	80
						Vaccine SH2014 NH 2014/15								Vaccine SH2015

1. < = <40

Sequences in phylogenetic trees

Table 9-4. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-01-09

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 NIB F13/14 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121
				Haemagglutination inhibition titre[1] — Post infection ferret antisera										
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	80	160	80	160	40	80	<	80	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	320	640	320	320	160	80	160	160	40
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	640	640	160	640	80	640	160	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	640	640	320	1280	1280	320	640	160	640	320	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E6	320	320	320	640	1280	80	640	40	640	320	40
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	40	40	320	80	320	160	160	160	160	<
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	40	160	80	80	160	80	320	80	640	160	<
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	40	160	80	160	160	160	160	80	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	<	160	80	80	320	160	320	80	640	160	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK3	<	40	40	160	80	160	80	80	80	160	<
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5 clone121	<	<	40	80	80	160	80	40	<	320	640
TEST VIRUSES														
A/Stockholm/17/2014	3C.2a	2014-07-23	MDCK2/SIAT1	<	40	80	80	80	80	80	40	80	80	<
A/Stockholm/18/2014	3C.2a	2014-08-16	MDCK2/SIAT1	<	40	40	80	80	80	80	40	80	160	<
A/Stockholm/19/2014	3C.3b	2014-09-07	MDCK1/SIAT1	40	80	40	160	80	160	80	40	80	80	<
A/Stockholm/20/2014	3C.3a	2014-09-28	MDCK1/SIAT1	<	<	80	80	80	160	80	80	80	80	<
A/Belgium/H80/2014	3C.3b	2014-09-29	SIAT1/SIAT1	80	80	80	320	160	160	160	80	80	80	<
A/La Rioja/2053/2014	3C.3b	2014-10-08	SIAT1/SIAT1	80	160	80	320	160	160	160	40	160	160	<
A/Stockholm/22/2014	3C.2a	2014-10-18	MDCK2/SIAT1	<	80	<	160	80	160	80	40	80	160	<
A/Pais Vasco/1979/2014	3C.3	2014-10-22	SIAT1/SIAT1	40	160	40	320	160	160	160	80	160	160	40
A/Belgium/G589/2014	3C.3b	2014-10-22	SIAT1/SIAT1	40	80	80	160	80	160	160	40	80	80	<
A/La Rioja/2051/2014	3C.3b	2014-10-29	MDCKx/SIAT1	40	80	40	320	160	160	160	40	80	80	<
A/Sweden/76/2014	3C.3a	2014-11-02	MDCK1/SIAT1	<	40	40	160	80	320	160	80	160	80	<
A/Stockholm/26/2014	3C.3a	2014-11-03	MDCK1/SIAT1	<	<	<	80	40	80	40	<	80	<	<
A/Aragon/2031/2014	3C.3b	2014-11-07	SIAT1/SIAT1	40	80	40	160	160	160	160	40	80	80	<
A/La Rioja/2050/2014	3C.3b	2014-11-15	SIAT1/SIAT1	80	80	80	320	160	160	160	40	80	80	<
A/Stockholm/29/2014	3C.2a	2014-11-18	MDCK1/SIAT1	<	<	<	40	<	40	<	<	<	40	<
A/Galicia/2055/2014	3C.3	2014-11-20	SIAT2/SIAT1	80	160	80	320	160	160	160	40	160	160	<
A/Galicia/2054/2014	3C.3	2014-11-24	SIAT1/SIAT1	40	160	80	320	160	160	160	40	80	80	<
A/Extremadura/2061/2014	3C.3a	2014-11-25	SIAT1/SIAT1	<	80	80	160	160	320	160	160	160	160	40
A/Aragon/2062/2014	3C.3b	2014-11-27	SIAT1/SIAT1	80	80	40	320	160	160	160	40	80	80	<
						Vaccine SH2014 NH 2014/15						Vaccine SH2015		

1. < = <40

Sequences in phylogenetic trees

Table 9-5. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-01-13

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1]										
				Post infection ferret antisera										
				A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/HK 146/13 F40/13	A/Stock 6/14 T/C F14/14	A/Stock 6/14 Egg F20/14	A/Switz 9715293/13 NIB F13/14	A/Switz 9715293/13 F25/14	A/HK 5738/14 T/C F30/14	A/HK 5738/14 NIB F53/14
					3C.1	3C.1	3C.3	3C.2	3C.3a	3C.3a isolate 2	3C.3a	3C.3a cl123	3C.2a	3C.2a cl 121
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	160	160	320	40	160	40	160	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	160	640	320	320	160	160	160	160	40
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	640	160	640	80	640	160	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT3	320	640	320	1280	1280	320	640	160	640	640	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	80	640	320	320	640	160	320	40	320	320	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	80	320	160	320	160	160	160	160	40
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	320	160	160	320	160	320	80	640	320	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	40	160	160	320	160	160	160	80	<
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	320	160	160	640	320	320	160	1280	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	40	40	160	80	80	80	80	80	160	<
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	80	40	80	160	160	160	80	80	40	640	1280
TEST VIRUSES														
A/Italy/FVG-29/2014	3C.2a	unknown	Cx/SIAT2	<	40	40	80	40	80	40	40	40	80	<
A/La Rioja/2052/2014	3C.3b	2014-10-22	MDCKx/SIAT2	40	80	80	160	80	160	80	40	80	40	<
A/Baden-Wurttemberg/85/2014	3C.2a	2014-11-10	C2/SIAT1	<	40	40	160	80	80	80	40	80	80	<
A/Norway/3082/2014		2014-11-28	SIAT1	40	80	80	160	80	160	160	40	80	40	<
A/Berlin/80/2014	3C.3b	2014-12-17	C2/SIAT1	40	80	80	160	80	320	80	40	80	80	<
A/Berlin/82/2014		2014-12-18	C3/SIAT1	<	40	40	160	80	80	80	<	80	80	<
A/Rheinland-Pfalz/16/2014	3C.3b	2014-12-21	C2/SIAT1	40	80	80	160	80	160	80	40	80	80	<
A/Nordrhein-Westfalen/18/2014	3C.3b	2014-12-22	C2/SIAT1	80	80	80	320	160	160	160	40	80	80	<
A/Baden-Wurttemberg/87/2014	3C.3b	2014-12-22	C2/SIAT1	40	80	80	160	80	160	80	40	80	40	<
A/Bayern/45/2014	3C.3b	2014-12-23	C1/SIAT1	40	80	40	160	80	160	80	40	80	40	<
						Vaccine SH2014 NH 2014/15						Vaccine SH2015		

1. < = <40

Sequences in phylogenetic trees

Table 9-6. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-01-16

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] / Post infection ferret antisera: A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 NIB F13/14 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	1280	160	160	160	320	40	160	40	160	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	160	640	320	320	160	160	160	160	40
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	640	160	640	80	640	160	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT3	320	640	320	1280	1280	320	640	160	640	640	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	80	640	320	320	640	160	320	40	320	320	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	80	320	160	320	160	160	160	160	40
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	320	160	160	320	160	320	80	640	320	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	40	160	80	320	160	160	160	80	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	320	160	160	640	320	320	160	1280	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	80	80	320	160	160	160	160	160	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	80	40	80	160	160	160	80	80	40	640	1280
TEST VIRUSES														
A/Beijing-Xicheng/13100/2014		2014-08-04	MDCK2/SIAT1/SIAT1	<	80	80	320	160	320	160	80	80	80	<
A/Shanxi-Xinghualing/1663/2014		2014-08-12	MDCK3/SIAT1/SIAT1	<	80	80	320	160	320	160	320	160	160	40
A/Netherlands/395/2014	3C.3	2014-10-02	SIAT3/SIAT1	80	160	80	320	160	160	160	80	80	160	40
A/Guizhou-Xinyi/1799/2014		2014-10-07	MDCK3/SIAT1/SIAT1	<	80	80	320	160	320	320	320	160	160	40
A/Liaoning-Heping/1430/2014		2014-10-12	MDCK3/SIAT1/SIAT1	<	40	40	160	80	160	160	80	80	40	<
A/Shaanxi-Changan/1452/2014		2014-11-03	MDCK2/SIAT1/SIAT1	<	160	80	320	160	640	320	320	320	160	40
A/Xinjiang-Tianshan/11190/2014		2014-11-04	MDCK2/SIAT1/SIAT1	<	160	80	320	160	640	320	320	320	160	40
A/Henan-Xigong/11041/2014	3C.3a	2014-11-11	MDCK2/SIAT1/SIAT1	<	160	80	320	160	640	320	320	320	160	40
A/Norway/3079/2014	3C.3b	2014-11-30	SIAT2	40	80	80	160	80	160	80	40	40	40	<
A/Hong Kong/7622/2014	3C.3b	2014-11-30	MDCK1/SIAT1	80	160	80	320	80	160	80	40	40	160	<
A/Hong Kong/7624/2014	3C.3a	2014-12-01	MDCK1/SIAT1	<	80	40	160	80	320	160	80	80	80	<
A/Norway/3107/2014		2014-12-02	SIAT1	<	40	40	160	80	160	80	40	80	80	<
A/Belgium/G678/2014	3C.3b	2014-12-04	SIAT1/SIAT2	80	160	80	160	80	160	80	80	80	80	<
						Vaccine SH2014 NH 2014/15						Vaccine SH2015		

1. < = <40

Sequences in phylogenetic trees

Table 9-7. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-01-20

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 NIB F13/14 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121
				Haemagglutination inhibition titre[1]										
				Post infection ferret antisera										
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	40	160	<	80	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	80	160	160	320	160	320	160	80	160	160	40
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	640	640	160	640	80	640	160	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	160	320	320	640	640	320	320	160	320	320	40
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	640	640	640	1280	80	640	40	640	320	40
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	40	40	160	80	320	160	160	160	160	40
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	160	160	80	160	160	320	80	640	160	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	40	160	80	320	160	160	160	80	<
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	160	160	320	320	320	80	1280	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	40	40	160	160	160	80	80	80	160	<
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	40	40	40	160	80	160	80	40	40	320	640
TEST VIRUSES														
A/Marseille/2180/2014	3C.2a	2014-09-11	MDCK3/SIAT1	<	<	<	80	40	80	40	<	40	80	<
A/Netherlands/398/2014	3C.3a	2014-10-27	SIAT3/SIAT1	<	40	40	160	80	320	160	160	160	80	<
A/Lyon/2397/2014		2014-12-08	MDCK2/SIAT1	<	<	40	80	80	320	80	40	80	80	<
A/Lyon/2398/2014		2014-12-08	MDCK2/SIAT1	80	160	80	320	160	320	80	40	80	80	<
A/Lyon-CHU/21.99/2014	3C.3	2014-12-10	MDCK2/SIAT1	80	160	160	320	320	320	160	80	160	160	<
A/Lyon/2429/2014	3C.3	2014-12-15	MDCK2/SIAT1	80	160	160	320	320	320	160	80	160	160	<
A/Nimes/2454/2014	3C.3a	2014-12-18	MDCK2/SIAT1	<	<	40	160	80	320	80	80	80	80	<
						Vaccine SH2014 NH 2014/15						Vaccine SH2015		

1. < = <40

Sequences in phylogenetic trees

Table 9-8. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-01-23

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] Post infection ferret antisera A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 NIB F13/14 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E3	640	160	160	160	320	40	160	<	80	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	80	160	160	320	160	320	160	80	160	160	40
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	640	640	160	640	80	640	160	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	160	320	320	640	640	320	320	160	320	320	40
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	320	640	640	640	1280	80	640	40	640	320	40
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	40	40	160	80	320	160	160	160	160	40
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	160	160	80	160	160	320	80	640	160	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	80	80	320	160	640	160	160	160	160	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	160	160	320	320	320	80	1280	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	40	160	80	320	160	640	160	160	160	320	80
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	40	40	40	160	80	160	80	40	40	320	640
TEST VIRUSES														
A/Genova/1614/2014	3C.2a	2014-12-09	Cx/SIAT1	<	40	40	160	80	320	80	40	80	160	<
A/Milano/162/2014	3C.3	2014-12-10	SIAT1/SIAT1	40	80	80	160	80	160	80	40	80	80	<
A/Austria/830719/2014	3C.3b	2014-12-10	SIAT2/SIAT1	80	160	160	320	160	320	160	80	160	160	<
A/Italy/FVG-31/2014	3C.3a	2014-12-12	SIAT1/SIAT1	<	80	80	320	160	640	160	160	160	160	40
A/Madrid/15053/2014	3C.3	2014-12-22	SIAT2/SIAT1	80	80	80	160	160	160	80	40	80	80	<
A/Madrid/SO13031/2014		2014-12-22	SIAT2	80	80	80	160	160	320	80	40	80	80	<
A/Castilla La Mancha/7/2015	3C.3	2015-01-02	SIAT2/SIAT1	80	80	80	320	160	80	80	40	80	80	<
						Vaccine SH2014 NH 2014/15						Vaccine SH2015		

1. < = <40

Sequences in phylogenetic trees

Table 9-9. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-01-27

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 NIB F13/14 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121
				Haemagglutination inhibition titre[1]										
				Post infection ferret antisera										
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	160	160	160	320	80	160	40	80	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	320	640	320	640	320	160	320	160	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	640	320	640	80	640	160	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	320	1280	1280	640	640	160	640	640	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	640	640	640	1280	2560	160	640	80	1280	320	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	80	320	160	640	160	160	160	160	80
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	320	160	160	320	320	320	80	1280	160	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	80	40	320	160	640	160	160	160	160	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	320	320	320	640	320	640	160	1280	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	80	80	320	160	320	160	80	160	160	80
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	40	40	40	160	80	320	80	80	40	320	1280
TEST VIRUSES														
A/Lisboa/SU34/2014	3C.2a	2014-12-15	SIAT1/SIAT2	<	40	40	80	40	160	40	40	40	80	<
A/Milano/169/2014	3C.2a	2014-12-16	SIAT1/SIAT2	<	80	40	160	80	320	80	80	40	160	<
A/Oman /6132/2014	3C.2a	2014-12-20	SIAT1/SIAT1	<	<	<	80	40	160	40	<	40	40	<
A/Rheinland-Pfalz/17/2014	3C.3	2014-12-23	C2/SIAT1	40	40	40	160	80	80	40	<	40	<	<
A/Baden-Wurttemberg/89/2014	3C.2a	2014-12-29	C2/SIAT1	<	<	<	80	80	160	40	<	40	80	<
A/Luxembourg/14070685/2014	3C.2a	2014-12-29	MDCK1/SIAT2	<	<	<	40	40	160	<	<	40	40	<
A/Nordrhein-Westfalen/2/2015	3C.3b	2015-01-05	C2/SIAT1	<	40	40	80	40	80	40	<	40	<	<
A/Niedersachsen/1/2015		2015-01-05	C2/SIAT1	40	40	40	80	40	160	40	<	40	40	<
A/Saarland/1/2015	3C.3	2015-01-05	C2/SIAT1	<	40	40	160	80	80	40	<	40	<	<
A/Mecklenburg-Vorpommern/1/2015	3C.3b	2015-01-05	C3/SIAT1	40	40	40	160	80	160	40	40	40	40	<
A/Berlin/1/2015	3C.2a	2015-01-06	C2/SIAT1	<	<	<	80	40	80	40	<	40	40	<
A/Niedersachsen/4/2015	3C.3	2015-01-07	C2/SIAT1	<	80	40	160	80	160	80	40	40	40	<
A/Niedersachsen/5/2015	3C.3b	2015-01-08	C2/SIAT1	<	40	40	80	40	160	40	<	40	<	<
						Vaccine SH2014 NH 2014/15						Vaccine SH2015		

1. < = <40

Sequences in phylogenetic trees

Table 9-10. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-01-30

Viruses	Genetic group	Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 NIB F13/14 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121
				Haemagglutination inhibition titre[1] – Post infection ferret antisera										
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	160	160	160	320	80	160	40	80	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	320	640	320	640	320	160	320	160	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	640	320	640	80	640	160	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	320	1280	1280	640	640	160	640	640	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	640	640	640	1280	2560	160	640	80	1280	320	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	80	320	160	640	160	160	160	160	80
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	320	160	160	320	320	320	80	1280	160	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	80	80	320	160	640	160	160	160	160	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	320	320	320	640	320	640	160	1280	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	<	80	80	320	160	320	160	80	160	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	40	40	40	160	80	320	80	80	40	320	1280
TEST VIRUSES														
A/North Carolina/13/2014	3C.3a	2014-04-15	SIAT2/SIAT1	<	40	40	160	160	640	40	160	80	80	40
A/Madagascar/2676/2014	3C.3a	2014-09-23	MDCK2/SIAT1	<	80	40	320	160	640	160	160	160	160	<
A/Madagascar/2714/2014	3C.3a	2014-09-23	MDCK2/SIAT1	<	40	40	160	80	320	80	160	160	80	<
A/Michigan/15/2014	3C.2a	2014-09-24	MDCK1/SIAT1/SIAT1	<	<	40	160	80	320	40	40	40	80	<
A/Madagascar/2735/2014	2C.3a	2014-09-29	MDCK2/SIAT1	<	40	40	160	80	320	80	160	160	160	<
A/Madagascar/2739/2014	3C.3a	2014-09-30	MDCK1/SIAT1	<	80	40	160	160	640	160	160	160	160	40
A/Madagascar/2742/2014		2014-09-30	MDCK1/SIAT1	<	80	80	320	160	640	320	320	160	160	40
A/Dakar/29/2014	3C.3a	2014-10-02	C2/SIAT1	<	40	40	160	160	640	160	160	160	160	40
A/Dakar/30/2014	3C.3a	2014-10-07	C2/SIAT1	<	80	40	160	160	640	160	160	160	160	40
A/Dakar/31/2014	3C.3a	2014-10-16	C2/SIAT1	<	40	40	160	80	640	160	160	160	80	<
A/Madagascar/3069/2014	3C.3a	2014-10-22	MDCK2/SIAT1	<	80	40	160	80	640	160	160	160	160	40
A/Zambia/04-303/2014	3C.3	2014-11-14	MDCK2/SIAT1	80	160	160	320	160	640	160	160	160	160	40
A/Aragón/2102/2014	3C.3b	2014-12-05	SIAT2/SIAT1	<	80	40	80	80	160	80	40	40	40	<
A/Latvia/12-030846/2014	3C.2a	2014-12-10	C2/SIAT1	<	<	<	80	40	160	40	<	40	80	<
A/Baden-Wurttemberg/86/2014		2014-12-15	C2/SIAT1	80	80	80	320	160	320	160	40	80	80	<
A/Niedersachsen/12/2014		2014-12-18	C3/SIAT1	80	80	80	320	160	320	80	40	80	80	<
A/Latvia/12-051160/2014	3C.3b	2014-12-19	C2/SIAT1	80	160	160	320	160	320	160	80	160	160	<
A/Bayern/1/2015	3C.3	2014-12-20	C2/SIAT1	<	40	40	160	80	80	40	<	40	<	<
A/Latvia/12-066660/2014	3C.2a	2014-12-28	C2/SIAT1	<	40	40	160	80	320	80	40	80	160	<
						Vaccine SH2014 NH 2014/15						Vaccine SH2015		

1. < = <40

Sequences in phylogenetic trees

Table 9-11. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-02-03

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1] / Post infection ferret antisera: A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 NIB F13/14 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	80	160	160	160	40	80	<	80	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	320	640	640	640	320	160	320	160	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	1280	320	640	80	640	160	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	320	1280	1280	640	640	160	640	640	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	640	640	640	1280	2560	160	640	80	1280	640	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	80	320	160	640	160	320	160	160	80
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	320	160	160	320	320	320	160	640	160	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	40	320	160	640	160	160	160	160	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	320	160	320	640	320	640	160	1280	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	40	80	80	320	160	320	160	160	160	320	80
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	40	40	80	320	160	320	160	80	40	640	1280
TEST VIRUSES														
A/North Carolina/13/2014	3C.3a	2014-04-15	E5/E1	40	320	320	320	320	320	640	160	1280	320	40
A/Canberra/82/2014	3C.2a	2014-07-31	E5/E3	40	40	40	160	80	80	80	40	<	160	640
A/Dakar/03/2014	3C.3a	2014-10-29	C2/MDCK1	<	80	80	160	160	640	320	160	320	80	<
A/Malta/MX001759/2014	3C.3a	2014-12-09	SIAT1	40	80	40	160	80	160	80	40	40	40	<
A/Israel/O-7051/2014		2014-12-15	SIATx/SIAT1	<	80	80	320	160	640	160	160	160	160	40
A/Bourgogne/2295/2014	3C.3b	2014-12-15	MDCK2/SIAT1	40	80	80	160	80	160	160	40	80	40	<
A/Croatia/1923/2014	3C.2a	2014-12-16	MDCKx/SIAT1	40	160	80	320	160	320	160	40	80	80	<
A/Marrakech/79/2015	3C.3a	2014-12-16	MDCK2/SIAT1	<	<	40	160	80	160	80	40	80	40	<
A/Nord Pas de Calais/2301/2014	3C.3	2014-12-16	MDCK1/SIAT1	160	160	160	640	160	320	160	80	160	80	<
A/Caen/2343/2014	3C.2a	2014-12-16	MDCK2/SIAT1	<	80	40	160	80	320	160	80	80	80	40
A/Nord Pas de Calais/2318/2014	3C.2a	2014-12-17	MDCK1/SIAT1	40	80	80	160	80	320	80	40	80	40	<
A/Paris/2332/2014	3C.3a	2014-12-22	MDCK1/SIAT1	<	80	40	160	80	640	160	160	160	80	40
A/Lorraine/2355/2014	3C.2a	2014-12-22	MDCK2/SIAT1	<	<	<	80	40	160	40	<	40	40	<
A/Jordan/30483/2014	3C.3a	2014-12-23	MDCK1/SIAT1	<	80	80	160	160	640	320	160	160	160	40
A/Greece/1/2014	3C.2a	2014-12-28	SIAT1	<	<	<	40	40	160	<	<	<	40	<
A/Norway/3224/2014	3C.3a	2014-12-28	SIAT1	40	80	80	160	80	160	80	40	80	40	<
A/Norway/3226/2014		2014-12-28	SIAT1	40	80	80	160	80	160	80	40	80	40	<
A/Norway/3229/2014	3C.3b	2014-12-28	SIAT1	<	40	<	80	<	80	<	<	<	<	<
A/Israel/O-7414/2014	3C.3b	2014-12-28	SIATx/SIAT1	<	80	80	160	160	320	80	40	160	160	40
A/Norway/3293/2014	3C.3b	2014-12-30	SIAT1	40	80	80	160	80	160	80	40	80	40	<
A/Norway/3298/2014		2014-12-30	SIAT1	40	80	80	160	80	160	80	40	40	40	<
A/Denmark/02/2015	3C.2a	2015-01-02	SIAT3/SIAT1	<	80	40	80	40	320	80	40	40	80	<
A/Malta/MX502071/2015	3C.3b	2015-01-05	SIAT1	<	80	40	160	80	160	80	<	40	40	<
A/Denmark/01/2015	3C.2a	2015-01-05	SIAT3/SIAT1	<	40	<	80	40	320	80	40	40	80	<
A/Castilla La Mancha/76/2015	3C.3a	2015-01-08	SIAT1	<	80	40	160	80	640	160	160	160	160	<
						Vaccine SH2014 NH 2014/15						Vaccine SH2015		

1. < = <40

Sequences in phylogenetic trees

Table 9-12. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-02-06

Viruses	Genetic group	Collection Date	Passage History	Haemagglutination inhibition titre[1]										
				Post infection ferret antisera										
				A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12 3C.1	A/Texas 50/12 Egg F42/13 3C.1	A/Samara 73/13 F24/13 3C.3	A/HK 146/13 F40/13 3C.2	A/Stock 6/14 T/C F14/14 3C.3a	A/Stock 6/14 Egg F20/14 3C.3a isolate 2	A/Switz 9715293/13 NIB F13/14 3C.3a	A/Switz 9715293/13 F25/14 3C.3a cl123	A/HK 5738/14 T/C F30/14 3C.2a	A/HK 5738/14 NIB F53/14 3C.2a cl 121
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	640	80	160	160	160	40	80	<	80	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	320	640	640	640	320	160	320	160	80
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	1280	320	640	80	640	160	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	320	1280	1280	640	640	160	640	640	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	640	640	640	1280	2560	160	640	80	1280	640	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	80	320	160	640	160	320	160	160	80
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	320	160	160	320	320	320	160	640	160	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	40	320	160	640	160	160	160	160	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	160	320	640	320	640	160	1280	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	40	80	80	320	160	320	160	160	160	320	80
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	40	40	80	320	160	320	160	80	40	640	1280
TEST VIRUSES														
A/Dakar/20/2014	3C.3a	2014-08-21	C3/MDCK1	<	40	40	320	160	320	160	160	160	80	<
A/Dakar/23/2014	3C.3a	2014-08-21	C3/MDCK1	<	40	40	160	80	320	160	80	160	80	<
A/Ghana/DILI-14-1014/2014		2014-11-04	Cx/SIAT1	<	<	<	80	40	320	80	80	80	80	<
A/Paris/2337/2014		2014-12-04	MDCK2/SIAT1	<	160	80	320	320	640	640	640	640	320	80
A/Rouen/2273/2014	3C.3a	2014-12-05	MDCK1/SIAT1	<	40	40	160	80	320	80	80	80	80	<
A/Paris/2304/2014		2014-12-06	MDCK2/SIAT1	40	40	40	160	80	320	160	160	80	80	40
A/Rouen/2291/2014	3C.2a	2014-12-09	MDCK1/SIAT1	<	80	80	160	80	160	80	40	80	80	<
A/Israel/O-6994/2014	3C.2a	2014-12-11	SIATx/SIAT2	40	40	<	80	40	160	40	<	40	80	<
A/Picardie/2297/2014	3C.3a	2014-12-13	MDCK2/SIAT1	40	<	<	80	40	80	<	<	40	80	<
A/Paris/2306/2014	3C.3a	2014-12-13	MDCK2/SIAT1	<	40	40	80	80	160	80	40	80	80	<
A/Caen/2354/2014	3C.3a	2014-12-14	MDCK2/SIAT1	40	80	80	320	80	160	80	40	80	80	<
A/Asturias/15059/2014		2014-12-15	SIAT2/SIAT3	40	80	40	320	80	160	80	40	80	80	<
A/Nord Pas de Calais/2317/2014		2014-12-15	MDCK2/SIAT1	<	40	40	80	40	160	40	<	40	<	<
A/Norway/0144/2015	3C.3b	2015-01-03	SIAT1	<	80	40	160	80	160	80	<	80	40	<
						Vaccine SH2014 NH 2014/15						Vaccine SH2015		

1. < = <40

Sequences in phylogenetic trees

Table 9-13. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-02-10

				Haemagglutination inhibition titre[1]										
				Post infection ferret antisera										
Viruses		Collection Date	Passage History	A/Perth 16/09 F18/11	A/Vic 361/11 T/C F09/12	A/Texas 50/12 Egg F42/13	A/Samara 73/13 F24/13	A/HK 146/13 F40/13	A/Stock 6/14 T/C F14/14	A/Stock 6/14 Egg F20/14	A/Switz 9715293/13 NIB F13/14	A/Switz 9715293/13 F25/14	A/HK 5738/14 T/C F30/14	A/HK 5738/14 NIB F53/14
	Genetic group				3C.1	3C.1	3C.3	3C.2	3C.3a	3C.3a isolate 2	3C.3a	3C.3a cl123	3C.2a	3C.2a cl 121
REFERENCE VIRUSES														
A/Perth/16/2009		2009-07-04	E3/E2	320	80	80	80	160	40	80	<	80	<	<
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	160	320	320	640	320	640	320	160	320	160	40
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	640	1280	1280	1280	640	320	640	80	640	160	80
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	320	320	320	1280	1280	640	640	160	640	320	80
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	640	640	640	1280	2560	160	640	80	640	320	80
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	<	80	40	320	160	640	160	160	160	160	40
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	80	320	160	160	320	320	320	80	640	160	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	<	40	80	320	160	640	160	160	160	160	40
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	40	160	160	320	320	320	640	160	1280	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	40	80	80	320	160	320	160	80	160	320	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	80	40	80	320	160	320	160	80	40	320	1280
TEST VIRUSES														
A/Chita/267/2014	3C.3a	2014-12-03	C1/SIAT1	<	80	40	160	80	640	160	160	160	160	40
A/Chita/265/2014		2014-12-05	C1/SIAT1	<	40	40	160	80	320	160	160	160	80	<
A/Chita/261/2014	3C.3a	2014-12-12	C1/SIAT1	<	40	40	160	80	320	160	160	160	160	<
A/Astrakhan/25/2014	3C.2a	2014-12-16	C2/SIAT1	<	<	<	80	40	160	40	<	40	40	<
A/Chita/255/2014		2014-12-18	C1/SIAT1	<	40	40	160	80	640	160	160	160	160	<
A/Chita/257/2014	3C.3a	2014-12-18	C1/SIAT1	<	40	40	160	80	640	160	80	160	80	<
A/Burgos/262/2014	3C.3b	2014-12-19	SIAT1	160	320	160	320	160	640	320	80	320	160	<
A/Chita/259/2014	3C.3a	2014-12-19	C1/SIAT1	<	80	80	160	160	640	160	160	160	160	40
A/Chita/263/2014		2014-12-19	C1/SIAT1	<	40	40	320	160	640	160	160	160	160	40
A/England/561/2014	3C.3	2014-12-21	MDCK1/SIAT1	40	80	40	160	80	160	80	40	80	40	<
A/England/607/2014	3C.2a	2014-12-27	MDCK2/SIAT1	<	40	<	80	80	160	40	<	40	40	<
A/England/15/2015	3C.3	2015-01-02	SIAT1/SIAT1	40	80	40	160	40	80	40	<	40	<	<
A/England/13/2015	3C.3	2015-01-05	MDCK1/SIAT1	<	80	80	320	160	640	320	160	160	160	40
A/England/18/2015		2015-01-07	SIAT1/SIAT1	40	80	40	160	80	160	40	<	40	40	<
A/England/32/2015	3C.3b	2015-01-09	SIAT2/SIAT1	40	80	80	160	80	160	80	40	80	40	<
						Vaccine SH2014 NH 2014/15						Vaccine SH2015		

1. < = <40

Table 9-14. Antigenic analyses of influenza A(H3N2) viruses (Guinea Pig RBC with 20nM Oseltamivir) 2015-02-03

Viruses		Collection Date	Passage History	Haemagglutination inhibition titre[1]						
				Post infection ferret antisera						
				[2]A/HK 7295/2014 F02/15	[2]A/HK 5576/2014 F03/15	[2]A/Eng 530/2014 F04/15	A/HK 5738/14 T/C F30/14	A/HK 5738/14 NIB F53/14	A/Nth Carol 13/14 CDC F24/15 SIAT	A/Nth Carol 13/14 CDC F11/15 Egg
Amino acid sequence at the HA1 158-160 glycosylation site				NYT	NYT	NYK	N/KYT	NYK	NSK	NSK
Genetic group				3C.2a	3C.2a	3C.2a	3C.2a	3C.2a cl121	3C.3a	3C.3a
REFERENCE VIRUSES										
A/Perth/16/2009		2009-07-04	E3/E2	40	40	40	<	40	<	40
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT5	320	640	320	160	80	160	640
A/Texas/50/2012	3C.1	2012-04-15	E5/E2	160	160	160	160	80	80	640
A/Samara/73/2013	3C.3	2013-03-12	C1/SIAT4	640	320	320	640	80	160	640
A/Hong Kong/146/2013	3C.2	2013-01-11	E3/E3	640	160	160	640	80	160	640
A/Stockholm/6/2014	3C.3a	2014-02-06	SIAT1/SIAT3	160	320	160	160	80	160	320
A/Stockholm/6/2014	3C.3a	2014-02-06	E4/E1 isolate 2	160	160	160	160	80	160	1280
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT3	320	320	160	160	40	40	320
A/Switzerland/9715293/2013	3C.3a	2013-12-06	E4/E1 clone 123	320	160	160	320	40	160	1280
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2/SIAT1	320	320	160	320	80	40	320
A/Hong Kong/5738/2014	3C.2a	2014-04-30	E5/E1 clone121	320	640	640	640	1280	40	80
TEST VIRUSES										
A/North Carolina/13/2014	3C.3a	2014-04-15	E5/E1	ND	ND	ND	320	40	160	1280
A/North Carolina/13/2014	3C.3a	2014-04-15	SIAT2/SIAT1	40	40	<	ND	ND	80	160
A/Michigan/15/2014	3C.2a	2014-09-24	MDCK1/SIAT1/SIAT1	80	80	40	ND	ND	<	80
A/England/530/2014	3C.2a	2014-11-26	MDCKx/SIAT2	ND	ND	160	ND	ND	ND	ND

1. < = <40; 2. Antisera raised against SIAT cell isolated viruses (homologous viruses not available at time of test); ND = Not Done

Table 10-1. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2015-01-14

Viruses		Collection Date	Passage History	Neutralisation titre[1]				
				Post infection ferret sera				
				A/Vic/ 361/11 T/C F09/12	A/Switz/ 9715923/13 cl123 Egg F25/14	A/Switz/ 9715923/13 T/C NIB F13/14	A/HK/ 5738/14 cl121 Egg NIB F53/14	A/HK/ 5738/14 T/C F30/14
Amino acid sequence at the HA1 158-160 glycosylation site				NFK	NSK	NSK	NYK	N/KYT
Genetic group				3C.1	3C.3a	3C.3a	3C.2a	3C.2a
REFERENCE VIRUSES								
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	320 (255)	320 (200)	80 (100)	ND	40 (26)
A/Switzerland/9715293/2013 cl123	3C.3a	2013-12-06	E5	80 (62)	640 (534)	80 (65)	ND	80 (90)
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	40 (45)	80 (79)	160 (138)	ND	40 (33)
A/Hong Kong/5738/2014 cl121	3C.2a	2014-04-30	E6	40 (48)	40 (36)	80 (62)	2560 (2237)	320 (422)
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2	40 (45)	80 (75)	80 (60)	80 (59)	320 (304)
TEST VIRUSES								
A/Buenos Aires/6358/2014	3C.2a	Unknown	SIAT1	40 (58)	80 (79)	160 (174)	320 (263)	640 (840)
A/Buenos Aires/10959404/2014	3C.2a	Unknown	SIAT1	80 (67)	160 (148)	320 (390)	160 (129)	1280 (1352)
A/Italy/FVG-29/2014	3C.2a	Unknown	Cx/SIAT2	80 (84)	80 (102)	40 (56)	80 (106)	320 (259)
A/Paris/1971/2014	3C.2a	2014-10-02	MDCK2/SIAT1	160 (145)	160 (224)	320 (349)	320 (302)	320 (315)
A/Stockholm/25/2014	3C.2a	2014-10-21	MDCK3/SIAT1	80 (79)	80 (75)	80 (69)	80 (78)	160 (217)
A/Belgium/H82/2014	3C.2a	2014-10-23	SIAT1/SIAT1	80 (73)	320 (300)	320 (346)	160 (189)	1280 (1579)
A/Belgium/G604/2014	3C.2a	2014-10-29	SIAT1/SIAT1	160 (155)	320 (450)	160 (137)	320 (272)	1280 (1034)
A/England/527/2014	3C.3	2014-11-18	SIAT1/SIAT1	320 (364)	320 (338)	320 (381)	40 (37)	80 (70)
A/Galicia/2056/2014	3C.2a	2014-11-19	MDCKx/SIAT1	80 (74)	160 (190)	80 (118)	320 (288)	1280 (1046)
A/England/528/2014	3C.2a	2014-11-20	SIAT2/SIAT1	40 (28)	160 (132)	80 (60)	160 (179)	1280 (1227)
A/England/530/2014	3C.2a	2014-11-26	MDCK1/SIAT1	40 (56)	160 (192)	80 (91)	80 (109)	160 (194)
A/Galicia/2084/2014	3C.3	2014-12-02	MDCKx/SIAT1	320 (325)	320 (274)	160 (190)	160 (159)	320 (427)
					Vaccine			

1. The titres associated with a 50% reduction in plaque formation are shown based on doubling dilution of antisera closest to the automated reading value (shown in parentheses)

Table 10-2. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 2015-02-05

Viruses		Collection Date	Passage History	Neutralisation titre[1] Post infection ferret sera A/Vic/ 361/11 T/C F09/12	A/Switz/ 9715923/13 cl123 Egg F25/14	A/Switz/ 9715923/13 T/C NIB F13/14	A/HK/ 5738/14 cl121 Egg NIB F53/14	A/HK/ 5738/14 T/C F30/14	A/HK 7295/14 T/C F02/15
Amino acid sequence at the HA1 158-160 glycosylation site				NFK	NSK	NSK	NYK	N/KYT	NYT
Genetic group				3C.1	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a
REFERENCE VIRUSES									
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	320 (255)	320 (200)	80 (100)	ND	40 (26)	40(47)
A/Switzerland/9715293/2013 cl123	3C.3a	2013-12-06	E5	80 (62)	640 (534)	80 (65)	ND	80 (90)	80(95)
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	40 (45)	80 (79)	160 (138)	ND	40 (33)	40 (48)
A/Hong Kong/5738/2014 cl121	3C.2a	2014-04-30	E6	40 (48)	40 (36)	80 (62)	2560 (2237)	320 (422)	160-320
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2	40 (45)	80 (75)	80 (60)	80 (59)	320 (304)	320
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80 (71)	160 (130)	80 (80)	80 (87)	320 (247)	320
TEST VIRUSES									
A/Finland/485/2014	3C.2a	2014-12-01	SIAT1/SIAT1	80 (69)	160 (133)	160 (149)	160 (143)	640 (624)	320-640
A/Andalucia/2098/2014	3C.2a	2014-12-04	SIAT2/SIAT1	160 (123)	160 (158)	160 (187)	160 (145)	640 (502)	160-320
A/Andalucia/2095/2014	3C.2a	2014-12-08	SIAT2/SIAT1	160 (179)	320 (256)	160 (225)	320 (314)	320 (426)	320
A/Andalucia/2096/2014	3C.2a	2014-12-10	SIAT2/SIAT1	80 (106)	320 (425)	1280 (1002)	640 (553)	1280 (1638)	320-640
A/Oman/6060/2014	3C.2a	2014-12-11	SIAT1/SIAT1	80 (74)	160 (216)	320 (246)	160 (199)	1280 (1158)	320-640
A/Oman/6078/2014	3C.2a	2014-12-12	SIAT3/SIAT1	40 (29)	320 (403)	80 (115)	80 (112)	1280 (1188)	320-640
A/Castilla La Mancha/2121/2014	3C.2a	2014-12-16	SIAT2/SIAT1	80 (77)	320 (440)	320 (283)	320 (359)	1280 (1388)	320-640
A/Oman/6150/2014		2014-12-21	SIAT1/SIAT1	40 (30)	160 (198)	160 (158)	80 (118)	1280 (967)	160
A/Latvia/12-060164/2014	3C.2a	2014-12-22	C2/SIAT1	40 (43)	40 (59)	40 (38)	40 (47)	80 (86)	160-320
A/Sachsen-Anhalt/25/2014	3C.2a	2014-12-22	C2/SIAT1	80 (70)	160 (150)	160 (153)	160 (116)	320 (279)	320
A/Bayern/47/2015		2014-12-22	C3/SIAT1	40 (41)	80 (87)	80 (84)	80 (90)	320 (286)	320
A/Roma/14/2014		2014-12-27	SIAT1/SIAT1	80 (64)	160 (136)	80 (107)	160 (123)	640 (598)	320
A/Slovenia/44/2015	3C.2a	2015-01-06	MDCKx/SIAT1	80 (83)	320 (307)	640 (787)	1280 (1071)	1280 (1388)	320
A/Nordrhein-Westfalen/1/2015	3C.2a	2015-01-06	C2/SIAT1	40 (55)	80 (92)	80 (97)	80 (78)	160 (215)	160
A/Bayern/2/2015	3C.2a	2015-01-07	C2/SIAT1	80 (69)	160 (148)	160 (135)	160 (120)	640 (533)	320
A/Slovenia/141/2015	3C.2a	2015-01-12	MDCKx/SIAT1	80 (67)	320 (310)	640 (942)	160 (164)	1280 (1297)	320-640

Vaccine

1. The titres associated with a 50% reduction in plaque formation are shown based on doubling dilution of antisera closest to the automated reading value (shown in parentheses)

Table 10-3. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 09-02-2015

Viruses		Collection Date	Passage History	Neutralisation titre[1] Post infection ferret sera A/Vic/ 361/11 T/C F09/12	A/Switz/ 9715923/13 cl123 Egg F25/14	A/Switz/ 9715923/13 T/C NIB F13/14	A/HK/ 5738/14 T/C F30/14	A/HK/ 7295/14 T/C F02/15	A/Eng/ 530/2014 T/C F04/15
Amino acid sequence at the HA1 158-160 glycosylation site				NFK	NSK	NSK	N/KYT	NYT	NYK
Genetic group				3C.1	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a
REFERENCE VIRUSES									
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	320 (255)	320 (200)	80 (100)	40 (26)	40 (47)	<
A/Switzerland/9715293/2013 cl123	3C.3a	2013-12-06	E5	80 (62)	640 (534)	80 (65)	80 (90)	80 (95)	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	40 (45)	80 (79)	160 (138)	40 (33)	40 (48)	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2	40 (45)	80 (75)	80 (60)	320 (304)	320 (308)	80
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80 (71)	160 (130)	80 (80)	320 (247)	320 (315)	80-160
A/England/530/2014	3C.2a	2014-11-26	MDCKx/SIAT2	40 (30)	80 (79)	40 (38)	80 (78)	80 (117)	160
TEST VIRUSES									
A/Austria/830096/2014	3C.2a	2014-12-04	SIAT4/SIAT1	40 (26)	40 (35)	40 (33)	160 (118)	160 (123)	<
A/Austria/830857/2014	3C.2a	2014-12-08	SIAT1/SIAT1	40 (40)	160 (140)	80 (90)	320 (252)	320 (269)	40
A/Austria/830724/2014	3C.2a	2014-12-09	SIAT2/SIAT1	80 (61)	160 (126)	160 (188)	320 (475)	640 (512)	40
A/Austria/830722/2014	3C.2a	2014-12-10	SIAT1/SIAT1	40 (44)	80 (83)	80 (73)	320 (266)	320 (317)	40
A/Austria/831131/2014	3C.2a	2014-12-12	SIAT2/SIAT1	80 (60)	160 (122)	80 (102)	320 (306)	320 (310)	80
A/Roma/13/2014	3C.2a	2014-12-16	SIAT1/SIAT1	40 (37)	160 (197)	160 (200)	1280 (1081)	640 (764)	40-80
A/Milano/169/2014	3C.2a	2014-12-16	SIAT1/SIAT1	40 (59)	80 (118)	80 (68)	640 (499)	640 (516)	40
A/Milano/168/2014	3C.2a	2014-12-17	SIAT1/SIAT1	40 (49)	160 (222)	320 (353)	1280 (1103)	640 (827)	80-160
A/Italy/FVG-36/2014		2014-12-19	SIAT1/SIAT1	40 (30)	80 (66)	40 (53)	320 (386)	320 (451)	80
A/Bremen/7/2014	3C.2a	2014-12-22	C3/SIAT1	40 (26)	40 (49)	40 (46)	160 (157)	160 (175)	40
A/Oman/6163/2014	3C.2a	2014-12-22	SIAT1/SIAT1	40 (40)	160 (191)	320 (287)	640 (837)	640 (590)	80
A/Berlin/90/2014	3C.2a	2014-12-26	C3/SIAT1	40 (31)	40 (55)	40 (40)	160 (146)	160 (178)	80
A/Latvia/12-066494/2014	3C.2a	2014-12-28	C2/SIAT1	40 (46)	80 (90)	80 (94)	320 (216)	320 (243)	40-80
A/Schleswig-Holstein/10/2014	3C.2a	2014-12-29	C3/SIAT1	40 (47)	80 (77)	80 (78)	320 (334)	320 (216)	160
A/Latvia/01-000487/2015	3C.2a	2015-01-01	C2/SIAT1	40 (38)	80 (65)	40 (40)	160 (150)	160 (209)	40-80
A/Latvia/01-000586/2015	3C.2a	2015-01-01	C1/SIAT1	40 (36)	80 (70)	80 (57)	160 (142)	160 (138)	80
A/Brandenburg/1/2015	3C.2a	2015-01-05	C2/SIAT1	80 (59)	80 (87)	80 (81)	320 (300)	160 (152)	80-160
					Vaccine				

1. The titres associated with a 50% reduction in plaque formation are shown based on doubling dilution of antisera closest to the automated reading value (shown in parentheses); < = <40

Table 10-4. Antigenic analysis of influenza A(H3N2) viruses - Plaque Reduction Neutralisation (MDCK-SIAT) 12-02-2015

Viruses		Collection Date	Passage History	Neutralisation titre[1] Post infection ferret sera A/Vic/ 361/11 T/C F09/12	A/Switz/ 9715923/13 cl123 Egg F25/14	A/Switz/ 9715923/13 T/C NIB F13/14	A/HK/ 5738/14 T/C F30/14	A/HK/ 7295/14 T/C F02/15	A/Eng/ 530/2014 T/C F04/15
Amino acid sequence at the HA1 158-160 glycosylation site				NFK	NSK	NSK	N/KYT	NYT	NYK
Genetic group				3C.1	3C.3a	3C.3a	3C.2a	3C.2a	3C.2a
REFERENCE VIRUSES									
A/Victoria/361/2011	3C.1	2011-10-24	MDCK2/SIAT4	320 (255)	320 (200)	80 (100)	40 (26)	40 (47)	<
A/Switzerland/9715293/2013 cl123	3C.3a	2013-12-06	E5	80 (62)	640 (534)	80 (65)	80 (90)	80 (95)	80
A/Switzerland/9715293/2013	3C.3a	2013-12-06	SIAT1/SIAT2	40 (45)	80 (79)	160 (138)	40 (33)	40 (48)	40
A/Hong Kong/5738/2014	3C.2a	2014-04-30	MDCK1/MDCK2	40 (45)	80 (75)	80 (60)	320 (304)	320 (308)	80
A/Hong Kong/7295/2014	3C.2a	2014-08-07	MDCK3	80 (71)	160 (130)	80 (80)	320 (247)	320 (315)	80-160
A/England/530/2014	3C.2a	2014-11-26	MDCKx/SIAT2	40 (30)	80 (79)	40 (38)	80 (78)	80 (117)	160
TEST VIRUSES									
A/Buenos Aires/6367/2014	3C.2a	Unknown	SIAT1	40 (58)	160 (152)	80 (97)	640 (586)	640 (603)	160
A/Buenos Aires/2274881/2014	3C.2a	Unknown	SIAT1	80 (110)	320 (362)	160 (187)	640 (587)	640 (703)	80-160
A/Buenos Aires/2274920/2014	3C.2a	Unknown	SIAT1	80 (116)	160 (159)	160 (150)	1280 (1012)	640 (628)	80
A/Buenos Aires/2275130/2014	3C.2a	Unknown	SIAT1	160 (145)	320 (288)	160 (163)	1280 (1165)	640 (974)	160
A/Santa Fe/85912/2014	3C.2a	Unknown	SIAT1	80 (117)	320 (330)	160 (193)	640 (644)	320 (450)	80
A/Stockholm/24/2014	3C.2a	2014-09-26	MDCK2/SIAT2	80 (89)	160 (129)	80 (90)	320 (343)	320 (377)	80-160
A/Norway/3086/2014	3C.2a	2014-12-02	SIAT1	80 (65)	160 (180)	80 (109)	640 (860)	640 (730)	160
A/Lisboa/SU34/2014	3C.2a	2014-12-15	SIAT1/SIAT1	40 (48)	80 (95)	80 (67)	640 (507)	640 (480)	80-160
A/Ukraine/501/2014	3C.2a	2014-12-18	SIAT1/SIAT1	80 (71)	80 (115)	80 (75)	640 (672)	1280 (1033)	80
A/Castilla La Mancha/2131/2014	3C.2a	2014-12-18	SIAT2/SIAT1	80 (83)	160 (154)	160 (148)	1280 (1077)	640 (760)	80-160
A/Italy/FVG-40/2014	3C.2a	2014-12-23	SIAT1/SIAT1	80 (81)	160 (137)	160 (120)	640 (629)	640 (643)	80
A/Madrid/15051/2014	3C.2a	2014-12-26	SIAT2/SIAT1	80 (94)	160 (135)	80 (116)	640 (709)	640 (592)	80
A/Madrid/15050/2014	3C.2a	2014-12-26	SIAT2/SIAT1	80 (94)	160 (224)	160 (135)	1280 (1018)	640 (630)	80-160
A/Luxembourg/14070685/2014	3C.2a	2014-12-29	MDCK1/SIAT1	80 (82)	320 (336)	160 (152)	640 (705)	640 (583)	160
A/Lisboa/SU63/2014	3C.2a	2014-12-29	SIAT1/SIAT1	80 (70)	160 (141)	160 (173)	640 (515)	640 (532)	80-160
A/Lisboa/SU61/2015	3C.2a	2015-01-04	SIAT1/SIAT1	80 (74)	160 (144)	160 (148)	640 (579)	640 (556)	160
					Vaccine				

1. The titres associated with a 50% reduction in plaque formation are shown based on doubling dilution of antisera closest to the automated reading value (shown in parentheses); < = <40

Phylogenetic comparison of influenza A(H3N2) HA genes

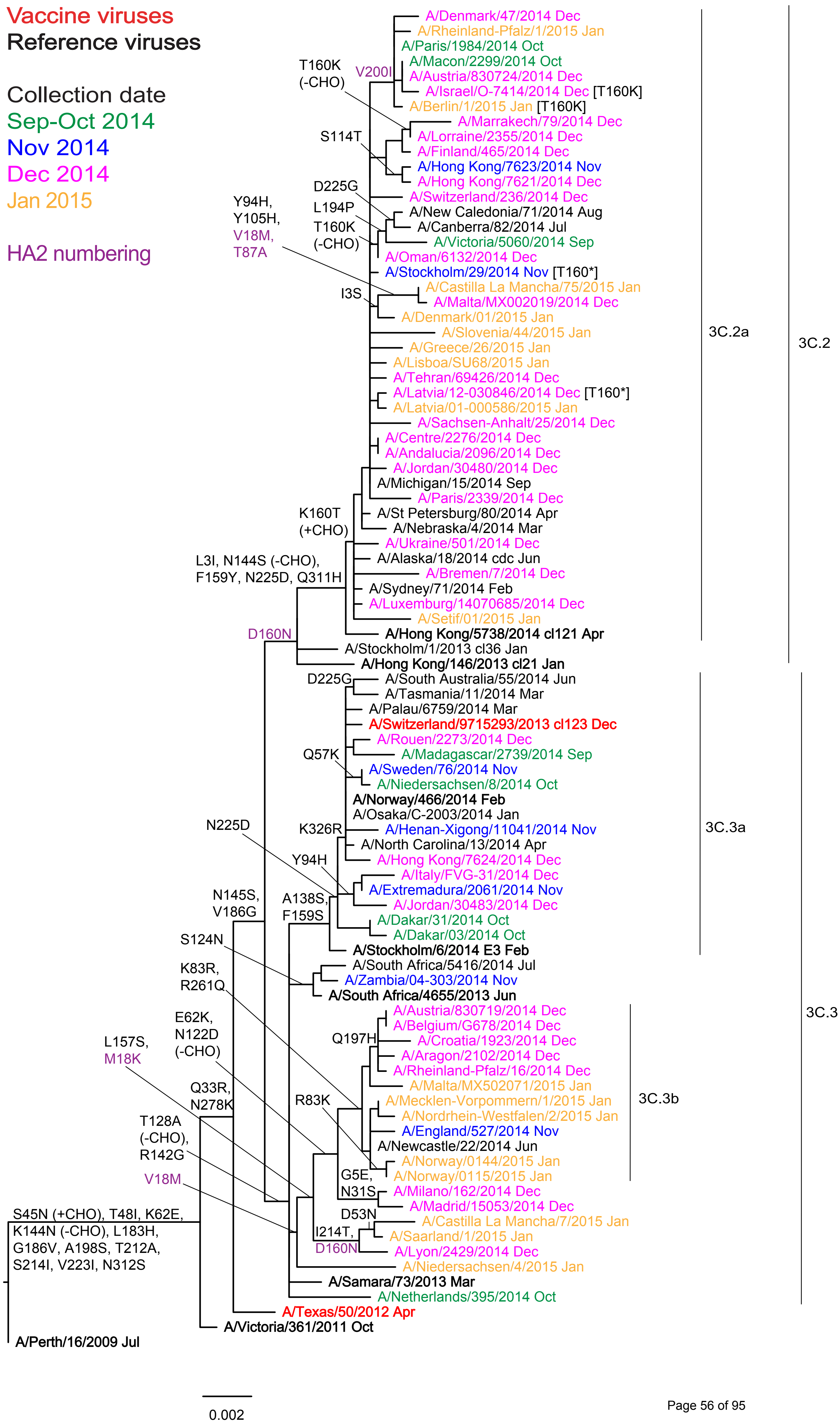


Figure 12. HA protein alignment for A(H3N2) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

```
                                       10        20        30        40        50        60        70        80        90       100
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009               QKLPGNDNSTATLCLGHHAVPNGTIVKTITNDQIEVTNATELVQSSSTGEICDSPHQILDGKNCTLIDALLGDPQCDGFQNKKWDLFVERSKAYSNCYPY
A/Victoria/361/2011           ............................................N..I.............E...................................... 3C.1
A/Texas/50/2012               ................................R...........N..I.............E...................................... 3C.1
A/Samara/73/2013              ................................R...........N..I.............E...................................... 3C.3
A/Niedersachsen/4/2015        ........................V.......R...........N..I.............E...........H.......................... 3C.3
A/Castilla La Mancha/7/2015   ................................R...........N..I....N........E...................................... 3C.3
A/Milano/162/2014             ....E.........................S.R...........N..I.................................................... 3C.3
A/Norway/0144/2015            ................................R...........N..I.................................................... 3C.3b
A/Malta/MX502071/2015         ............................................N..I..................................R................. 3C.3b
A/Croatia/1923/2014           ................................R...........N..I..................................R....I............ 3C.3b
A/South Africa/4655/2013      ................................R...........N..I.............E...................................... 3C.3
A/Stockholm/6/2014            ................................R...........N..I.............E...................................... 3C.3a
A/Jordan/30483/2014           ................................R...........N..I.............E................................H..... 3C.3a
A/Hong Kong/7624/2014         ................................R...........N..I.............E...................................... 3C.3a
A/Norway/466/2014             ................................R...........N..I.............E...................................... 3C.3a
A/Sweden/76/2014              ................................R...........N..I........K....E...................................... 3C.3a
A/Rouen/2273/2014             .E..............................R...........N..I.............E...................................... 3C.3a
A/Switzerland/9715293/2013    ................................R...........N..I.............E...................................... 3C.3a
A/Hong Kong/146/2013          ................................R...........N..I.............E...................................... 3C.2
A/Hong Kong/5738/2014         ..I.............................R...........N..I.............E...................................... 3C.2a
A/Setif/01/2015               ..I.............................R...........N..I.............E................................H..... 3C.2a
A/Ukraine/501/2014            ..I.............................R...........N..I.............E...........H.......................... 3C.2a
A/Paris/2339/2014             ..I.............................R...........N..I.............E...................................... 3C.2a
A/Latvia/01-000586/2015       ..I.E...........................R...........N..I.............E...................................... 3C.2a
A/Denmark/01/2015             ..S.............................R...........N..I.............E...................................... 3C.2a
A/Victoria/5060/2014          ..I.............................R...........N..I.............E...................................... 3C.2a
A/Hong Kong/7621/2014         ..I.............................R...........N..I....N........E...................................... 3C.2a
A/Marrakech/79/2014           ..I.............................R...........N..I.............E...................................... 3C.2a
A/Israel/O-7414/2014          .NI.............................R...........N..I.............E...................................... 3C.2a
A/Rheinland-Pfalz/1/2015      ..I.............................R...........N..I.............E...................................... 3C.2a

                                      110       120       130       140       150       160       170       180       190       200
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009               DVPDYASLRSLVASSGTLEFNNESFNWTGVTQNGTSSACIRRSKNSFFSRLNWLTHLNFKYPALNVTMPNNEQFDKLYIWGVLHPGTDKDQIFLYAQASG
A/Victoria/361/2011           ...........................................N..........R..........................H..V..........S..
A/Texas/50/2012               ...........................N...............N......................................H..V..........P..
A/Samara/73/2013              ...........................A...........K.G.NS.....................................H.............S..
A/Niedersachsen/4/2015        ...........................A.............G.NS.....................................H........S....S..
A/Castilla La Mancha/7/2015   ...........................A.............G.NS..........S..........................H.............S..
A/Milano/162/2014             .....................D.....A.............G.NS..........S..........................H.............S..
A/Norway/0144/2015            .....................D.....A.............G.NS..........S..........................H.............S..
A/Malta/MX502071/2015         .....................D.....A.............G.NS..........S..........................H......GP......S..
A/Croatia/1923/2014           .....................D.....A.............G.NS..........S..........................H............HS..
A/South Africa/4655/2013      .......................N...A.............G.NS.....................................H.............S..
A/Stockholm/6/2014            ...........................A.........S...G.NS............S........................H..V..........S..
A/Jordan/30483/2014           ...........................A.........S...G.NS............S........................H.............S..
A/Hong Kong/7624/2014         ...........................A.........S...GTNS............S........................H.............S..
A/Norway/466/2014             ...........................A.........S...G.NS............S........................H.............S..
A/Sweden/76/2014              ...........................A.........S...G.NS............S........................H.............S..
A/Rouen/2273/2014             ...........................A.........S...G.NS............S........................H.............S..
A/Switzerland/9715293/2013    ...........................A.........S.R.G.NS............S........................H..V..........S..
A/Hong Kong/146/2013          .......................................K...NS.....................................H..W..........S..
A/Hong Kong/5738/2014         ...........................................SS............Y........................H.........P...S..
A/Setif/01/2015               ..................................A........SS............YT.......................H.............S..
A/Ukraine/501/2014            ...........................................SS............YT..........K............H.............S..
A/Paris/2339/2014             ...........................................SS............YT.......................H.............S..
A/Latvia/01-000586/2015       ...........................................SS............YT.......................H.............S..
A/Denmark/01/2015             ...........................................SS............YT..........K............H.............S..
A/Victoria/5060/2014          .....................................S.....SS............Y........................H.........P...S..
A/Hong Kong/7621/2014         .............T.............................SS............YT.......................H.............S..
A/Marrakech/79/2014           ..................................K........SS............Y........................H.............S..
A/Israel/O-7414/2014          ...........................................SS............Y........................H.............S..
A/Rheinland-Pfalz/1/2015      ...........................................SS............YT.......................H.............S..

                                      210       220       230       240       250       260       270       280       290       300
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009               RITVSTKRSQQTVSPNIGSRPRVRNIPSRISIYWTIVKPGDILLINSTGNLIAPRGYFKIRSGKSSIMRSDAPIGKCNSECITPNGSIPNDKPFQNVNRI
A/Victoria/361/2011           ...........A.I........I.............................................................................
A/Texas/50/2012               ...........A.I....F...I..............................................................K..............
A/Samara/73/2013              ...........A.I........I..............................................................K..............
A/Niedersachsen/4/2015        ...........A.I........I..............................................................K..............
A/Castilla La Mancha/7/2015   ...........A.T........I..............................................................K..............
A/Milano/162/2014             ...........A.I........I..............................................................K..............
A/Norway/0144/2015            ...........A.I........I.....................................Q........................K..............
A/Malta/MX502071/2015         ...........A.I........I.....................................Q........................K..............
A/Croatia/1923/2014           ......Q....A.I........I.....................................Q........................K..............
A/South Africa/4655/2013      ...........A.I........I..............................................................K..............
A/Stockholm/6/2014            ...........A.I........I..............................................................K..............
A/Jordan/30483/2014           ...........A.I........I.D............................................................K..............
A/Hong Kong/7624/2014         ...........A.I........I.D............................................................K..............
A/Norway/466/2014             ...........A.I........I.D............................................................K..............
A/Sweden/76/2014              ...........A.I........I.D............................................................K..............
A/Rouen/2273/2014             ...........A.I........I.D............................................................K..............
A/Switzerland/9715293/2013    ...........A.I........I.D............................................................K..............
A/Hong Kong/146/2013          ...........A.I........I..............................................................K..............
A/Hong Kong/5738/2014         ..I........A.I........I.D............................................................K..............
A/Setif/01/2015               ...........A.I........I.D............................................................K..............
A/Ukraine/501/2014            ...........A.I........I.D............................................................K..............
A/Paris/2339/2014             ...........A.I.......KI.D............................................................K..............
A/Latvia/01-000586/2015       ...........A.I........I.D............................................................K..............
A/Denmark/01/2015             ...........A.I........I.D............................................................K..............
A/Victoria/5060/2014          ...........A.I........I.D............................................................K..............
A/Hong Kong/7621/2014         ...........A.I........I.D............................................................K..............
A/Marrakech/79/2014           ...........A.I........I.D............................................................K..............
A/Israel/O-7414/2014          ...........A.I........I.D............................................................K..............
A/Rheinland-Pfalz/1/2015      ...........A.I........I.D............................................................K..............
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                                    310       320       330       340       350       360       370       380       390       400
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009              TYGACPRYVKQNTLKLATGMRNVPEKQTRGIFGAIAGFIENGWEGMVDGWYGFRHQNSEGRGQAADLKSTQAAIDQINGKLNRLIGKTNEKFHQIEKEFS
A/Victoria/361/2011          ...........S........................................................................................
A/Texas/50/2012              ...........S........................................................................................
A/Samara/73/2013             ...........S........................................................................................
A/Niedersachsen/4/2015       ...........S..................................M.....................................................
A/Castilla La Mancha/7/2015  ...........S..................................K.....................................................
A/Milano/162/2014            ...........S..................................K.....................................................
A/Norway/0144/2015           ...........S..................................K.....................................................
A/Malta/MX502071/2015        ...........S..................................K.....................................................
A/Croatia/1923/2014          ...........S..................................K.....................................................
A/South Africa/4655/2013     ...........S........................................................................................
A/Stockholm/6/2014           ...........S........................................................................................
A/Jordan/30483/2014          ...........S........................................................................................
A/Hong Kong/7624/2014        ...........S.............R..........................................................................
A/Norway/466/2014            ...........S.............R..........................................................................
A/Sweden/76/2014             ...........S.............R..........................................................................
A/Rouen/2273/2014            ...........S.............R..........................................................................
A/Switzerland/9715293/2013   ...........S.............R..........................................................................
A/Hong Kong/146/2013         ....................................................................................................
A/Hong Kong/5738/2014        ..........H.........................................................................................
A/Setif/01/2015              ..........HS........................................................................................
A/Ukraine/501/2014           ..........HS........................................................................................
A/Paris/2339/2014            ..........HS........................................................................................
A/Latvia/01-000586/2015      ..........HS........................................................................................
A/Denmark/01/2015            ..........HS........................................................................................
A/Victoria/5060/2014         ..........HS........................................................................................
A/Hong Kong/7621/2014        ..........HS........................................................................................
A/Marrakech/79/2014          ..........HS........................................................................................
A/Israel/O-7414/2014         ..........HS........................................................................................
A/Rheinland-Pfalz/1/2015     ..........HS........................................................................................

                                    410       420       430       440       450       460       470       480       490       500
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009              EVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQHTIDLTDSEMNKLFEKTKKQLRENAEDMGNGCFKIYHKCDNACIGSIRNGTYDHDVYRDEALNNRF
A/Victoria/361/2011          ....................................................................................................
A/Texas/50/2012              ....................................................................................................
A/Samara/73/2013             ....................................................................................................
A/Niedersachsen/4/2015       ........................................................................................N...........
A/Castilla La Mancha/7/2015  ............................................S...............................................N.......
A/Milano/162/2014            ....................................................................................................
A/Norway/0144/2015           ....................................................................................................
A/Malta/MX502071/2015        ....................................................................................................
A/Croatia/1923/2014          ....................................................................................................
A/South Africa/4655/2013     .....V..............................................................................................
A/Stockholm/6/2014           ....................................................................................................
A/Jordan/30483/2014          ....................................................................................................
A/Hong Kong/7624/2014        ....................................................................................................
A/Norway/466/2014            ....................................................................................................
A/Sweden/76/2014             ....................................................................................................
A/Rouen/2273/2014            ....................................................................................................
A/Switzerland/9715293/2013   ....................................................................................................
A/Hong Kong/146/2013         ........................................................................................N...........
A/Hong Kong/5738/2014        ........................................................................................N...........
A/Setif/01/2015              ........................................................................................N...........
A/Ukraine/501/2014           ........................................................................................N...........
A/Paris/2339/2014            ........................................................................................N...........
A/Latvia/01-000586/2015      ........................................................................................N...........
A/Denmark/01/2015            ........................................................................................N...........
A/Victoria/5060/2014         ........................................................................................N...........
A/Hong Kong/7621/2014        ........................................................................................N...........
A/Marrakech/79/2014          ........................................................................................N...........
A/Israel/O-7414/2014         ........................................................................................N...........
A/Rheinland-Pfalz/1/2015     ........................................................................................N...........

                                    510       520       530       540       550
                             ....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009              QIKGVELKSGYKDWILWISFAISCFLLCVALLGFIMWACQKGNIRCNICI
A/Victoria/361/2011          ..................................................
A/Texas/50/2012              ..................................................
A/Samara/73/2013             ..................................................
A/Niedersachsen/4/2015       ..R...............................................
A/Castilla La Mancha/7/2015  ..................................................
A/Milano/162/2014            ..................................................
A/Norway/0144/2015           ..................................................
A/Malta/MX502071/2015        ..................................................
A/Croatia/1923/2014          ..................................................
A/South Africa/4655/2013     ..................................................
A/Stockholm/6/2014           ..................................................
A/Jordan/30483/2014          ..................................................
A/Hong Kong/7624/2014        ..................................................
A/Norway/466/2014            ..................................................
A/Sweden/76/2014             ..................................................
A/Rouen/2273/2014            ..................................................
A/Switzerland/9715293/2013   ..................................................
A/Hong Kong/146/2013         ..................................................
A/Hong Kong/5738/2014        ..................................................
A/Setif/01/2015              ..................................................
A/Ukraine/501/2014           ..................................................
A/Paris/2339/2014            ..................................................
A/Latvia/01-000586/2015      ..................................................
A/Denmark/01/2015            ..................................................
A/Victoria/5060/2014         ..................................................
A/Hong Kong/7621/2014        ..................................................
A/Marrakech/79/2014          ..................................................
A/Israel/O-7414/2014         ............................I.....................
A/Rheinland-Pfalz/1/2015     ............................I.....................
```
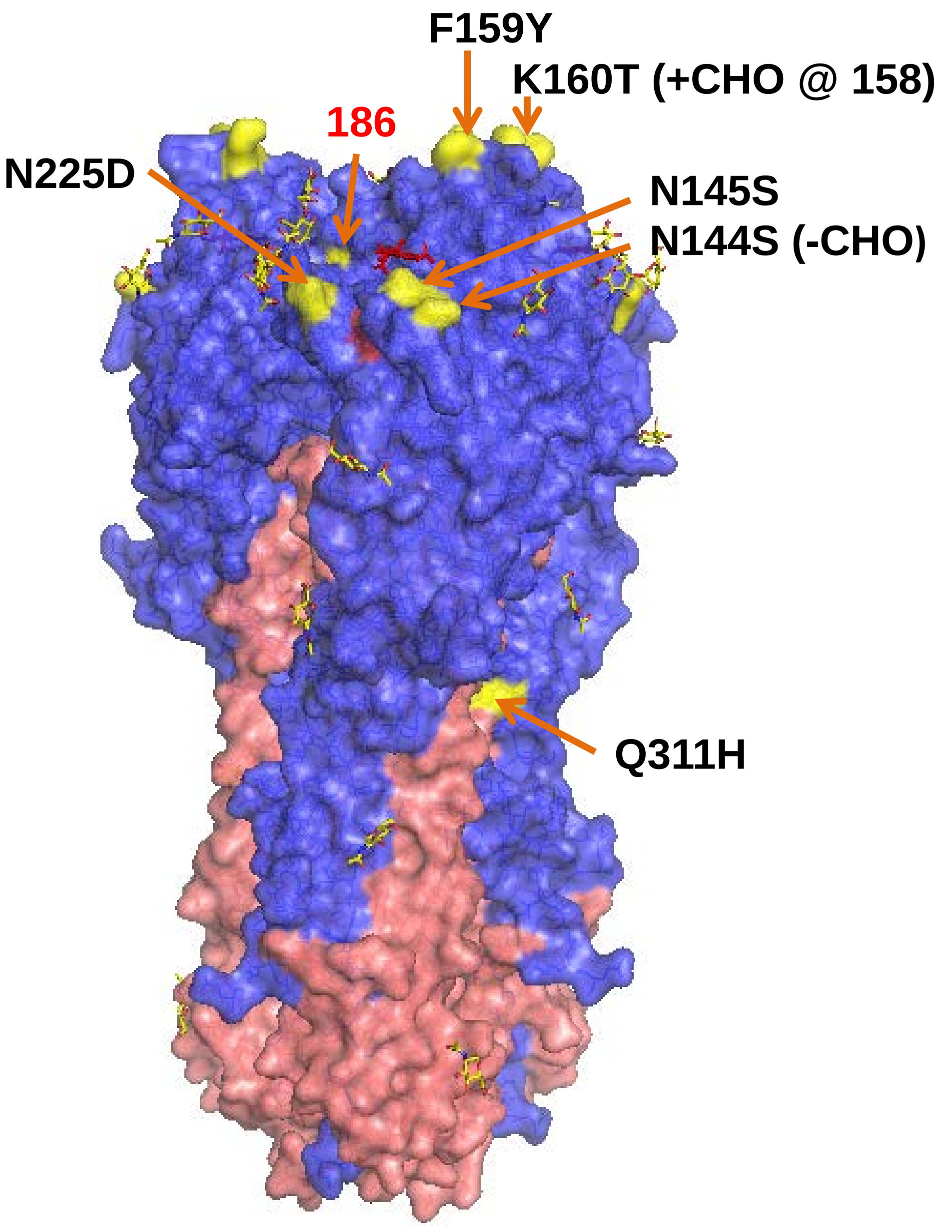
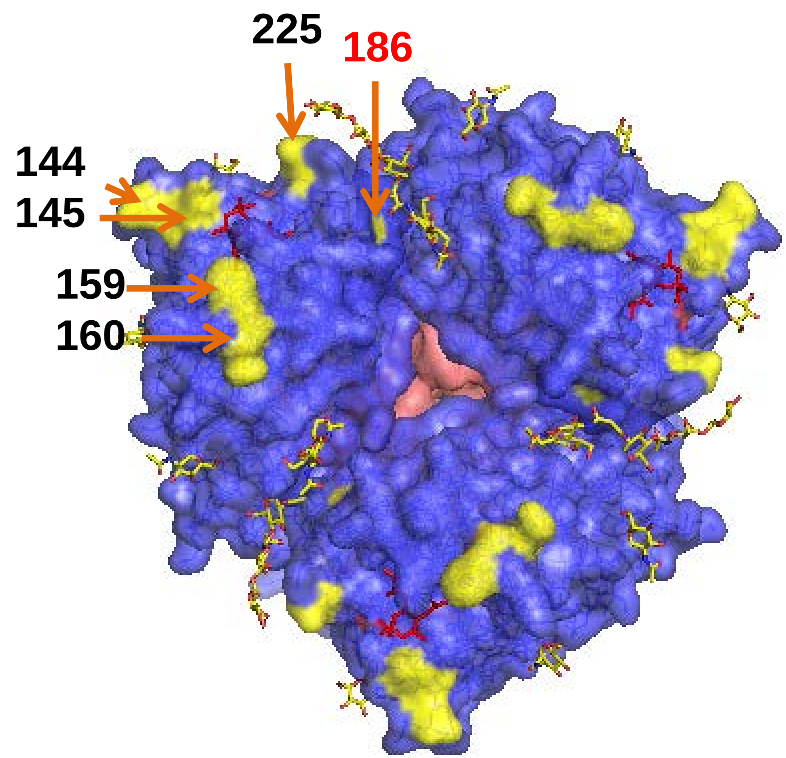

Vaccine virus: **Reference virus**: **Genetic group**: **HA1/HA2 boundary**: **Potential N-linked glycosylation site**

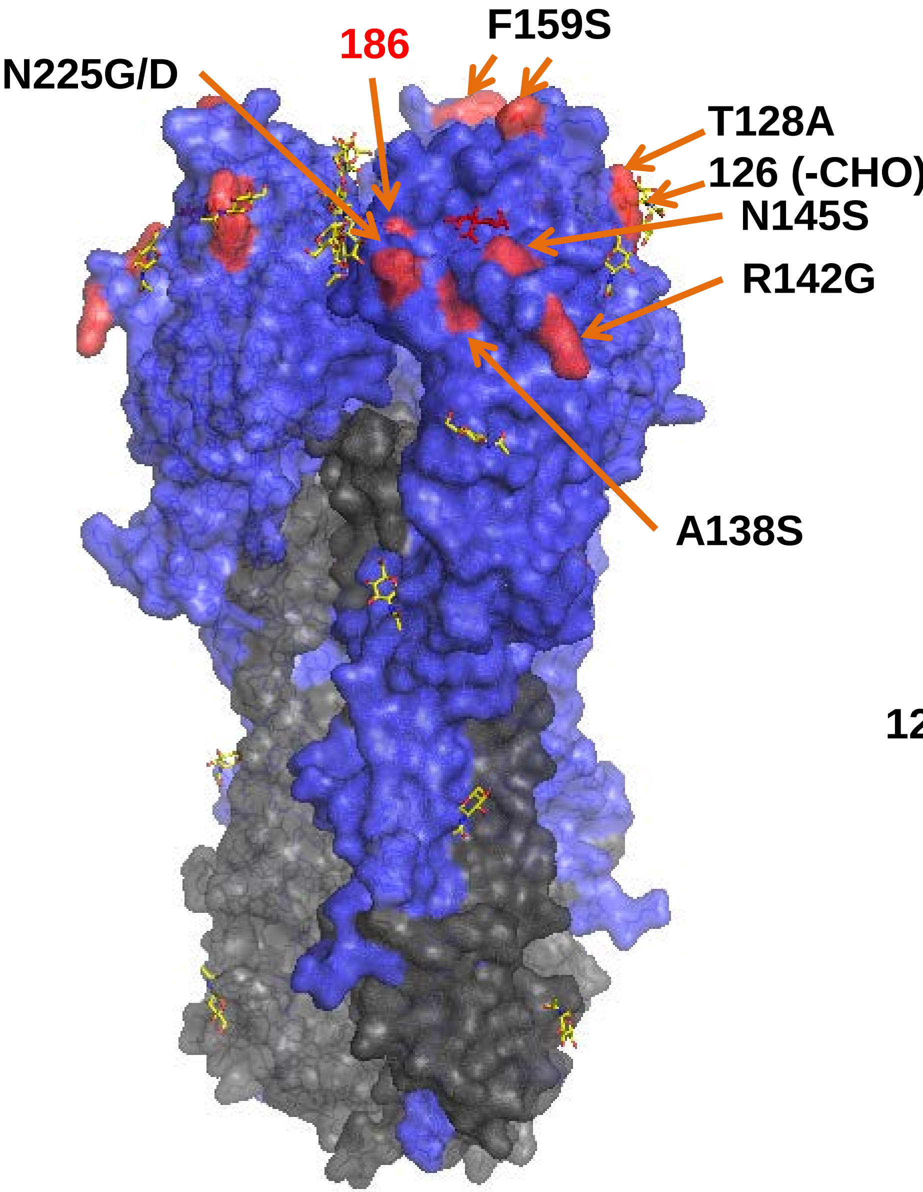
Figure 13A. HA amino acid substitutions, relative to A/Texas/50/2012, defining genetic subgroup 3C.2a

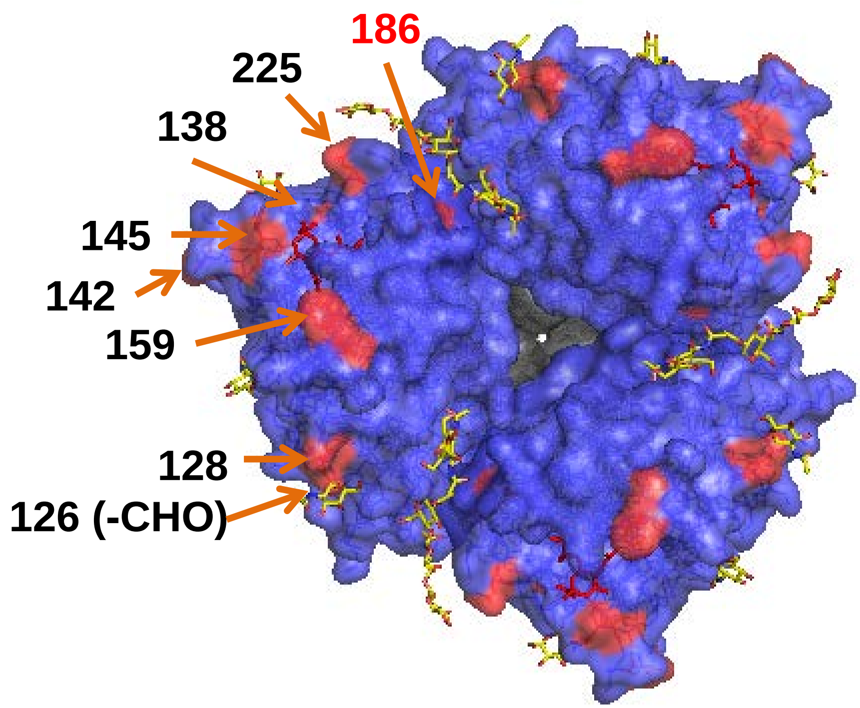




View from top, looking down the 3-fold axis

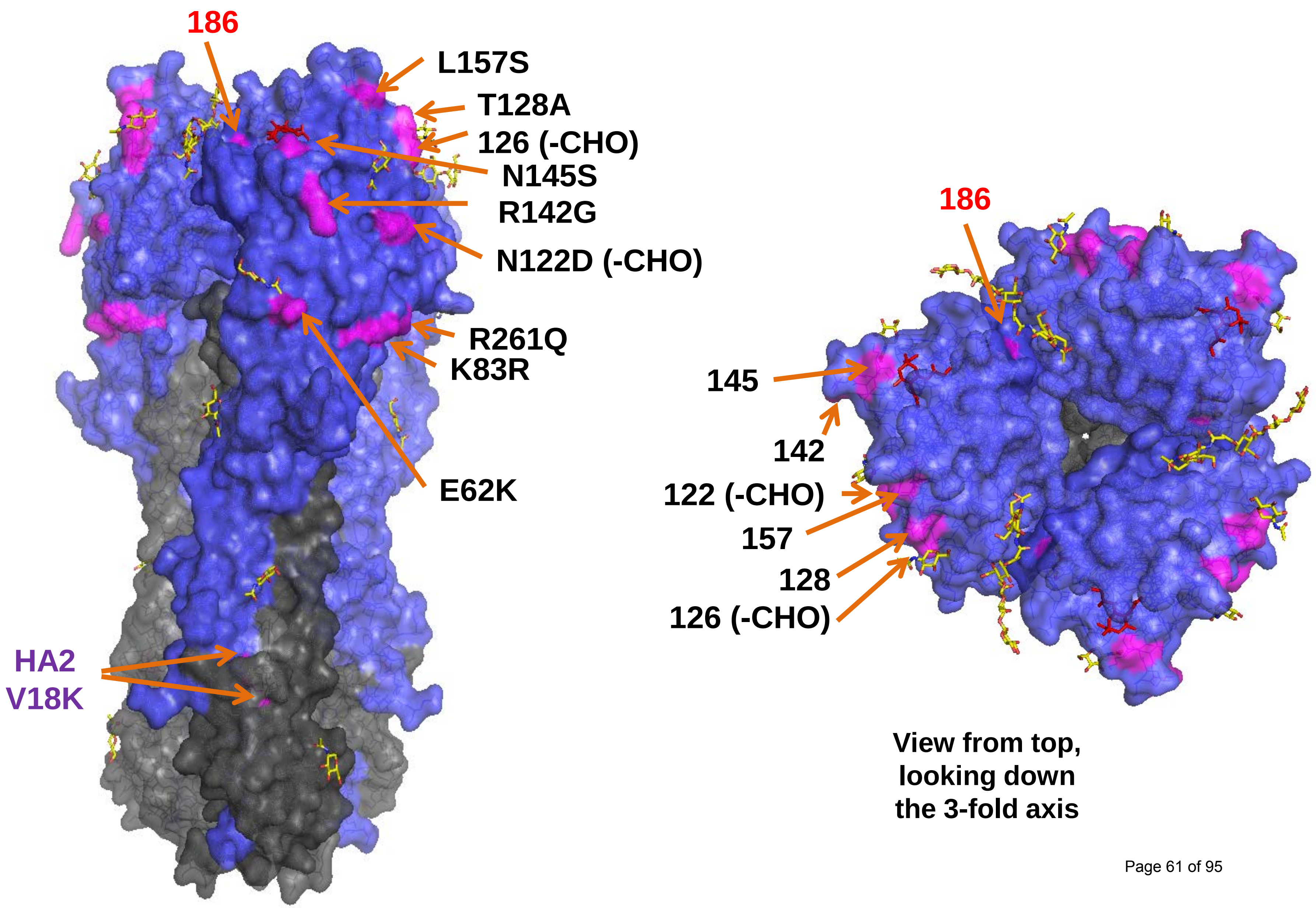
Figure 13B. HA amino acid substitutions, relative to A/Texas/50/2012, defining genetic subgroup 3C.3a





View from top, looking down the 3-fold axis

Figure 13C. HA amino acid substitutions, relative to A/Texas/50/2012, defining genetic subgroup 3C.3b



View from top, looking down the 3-fold axis

Figure 14. Phylogenetic comparison of influenza A(H3N2) NA genes

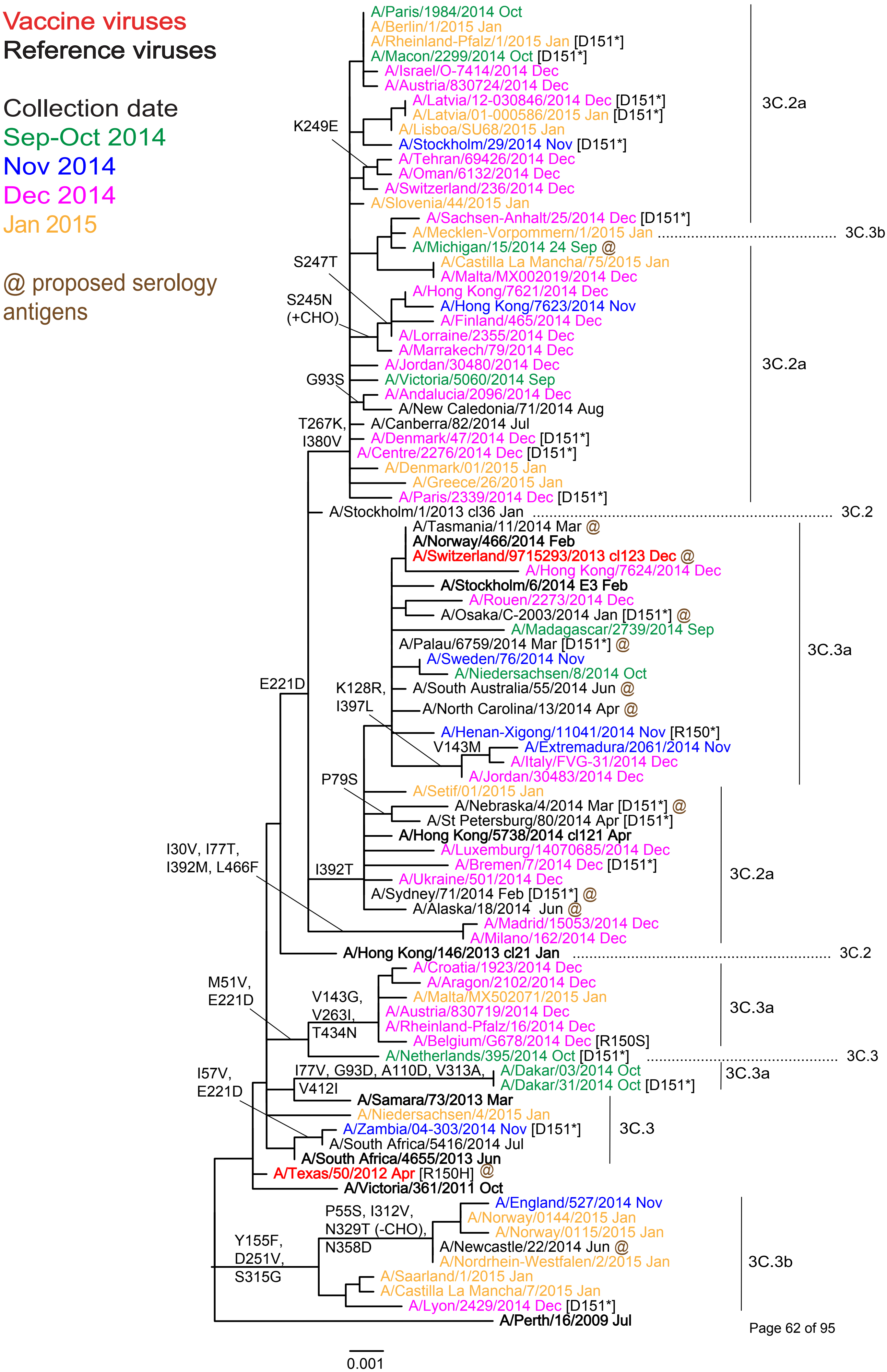


Figure 15. NA protein alignment for A(H3N2) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

```
                                       10        20        30        40        50        60        70        80        90       100
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009               MNPNQKIITIGSVSLTISTICFFMQIAILITTVTLHFKQYEFNSPPNNQVMLCEPTIIERNITEIVYLTNTTIEKEICPKLAEYRNWSKPQCDITGFAPF
A/Victoria/361/2011           ................................................................................P...........G....... 3C.1
A/Texas/50/2012               ................................................................................P...........G....... 3C.1
A/Samara/73/2013              ...............................................................K................P...........G....... 3C.3
A/Niedersachsen/4/2015        ................................................................................P...........G....... 3C.3
A/Castilla La Mancha/7/2015   ................................................................................P...........G....... 3C.3
A/Milano/162/2014             .............................V..............................................T...P...........G....... 3C.3
A/Norway/0144/2015            ......................................................S.........................P...........G....... 3C.3b
A/Malta/MX502071/2015         ...........................................F......V.............................P...........G....... 3C.3b
A/Croatia/1923/2014           ..................................................V.............................P...........G....... 3C.3b
A/South Africa/4655/2013      ................................................................................P...........G....... 3C.3
A/Stockholm/6/2014            ................................................................................P...........G....... 3C.3a
A/Jordan/30483/2014           ................................................................................P...........G....... 3C.3a
A/Hong Kong/7624/2014         ................................................................................P...........S....... 3C.3a
A/Norway/466/2014             ................................................................................P...........G....... 3C.3a
A/Sweden/76/2014              ................................................................................P...........G....... 3C.3a
A/Rouen/2273/2014             ................................................................................P...........G....... 3C.3a
A/Switzerland/9715293/2013    ................................................................................P...........G....... 3C.3a
A/Hong Kong/146/2013          ................................................................................P...........G....... 3C.2
A/Hong Kong/5738/2014         ................................................................................P...........G.K..... 3C.2a
A/Setif/01/2015               ................................................................................P...........G....... 3C.2a
A/Ukraine/501/2014            ................................................................................PS..........G....... 3C.2a
A/Paris/2339/2014             .......................................................I........................P...........G....... 3C.2a
A/Latvia/01-000586/2015       ................................................................................P...........G....... 3C.2a
A/Denmark/01/2015             ................................................................................P................... 3C.2a
A/Victoria/5060/2014          ................................................................................P...........G....... 3C.2a
A/Hong Kong/7621/2014         ................................................................................P...........G....... 3C.2a
A/Marrakech/79/2014           ................................................................................P...........G....... 3C.2a
A/Israel/O-7414/2014          ................................................................................P...........G....... 3C.2a
A/Rheinland-Pfalz/1/2015      ................................................................................P...........G....... 3C.2a

                                      110       120       130       140       150       160       170       180       190       200
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009               SKDNSIRLSAGGDIWVTREPYVSCDPDKCYQFALGQGTTLNNVHSNNTVRDRTPYRTLLMNELGVPFHLGTKQVCIAWSSSSCHDGKAWLHVCITGDDKN
A/Victoria/361/2011           ........................................................X...........................................
A/Texas/50/2012               .................................................H..................................................
A/Samara/73/2013              ....................................................................................................
A/Niedersachsen/4/2015        ........................................S...........................................................
A/Castilla La Mancha/7/2015   ......................................................F.............................................
A/Milano/162/2014             ....................................................................................................
A/Norway/0144/2015            ......................................................F.............................................
A/Malta/MX502071/2015         ..........................................G.........................................................
A/Croatia/1923/2014           ..........................................G.........................................................
A/South Africa/4655/2013      ....................................................................................................
A/Stockholm/6/2014            ....................................................................................................
A/Jordan/30483/2014           ...........................R........................................................................
A/Hong Kong/7624/2014         ................................................I...................................................
A/Norway/466/2014             ....................................................................................................
A/Sweden/76/2014              ....................................................................................................
A/Rouen/2273/2014             ....................................................................................................
A/Switzerland/9715293/2013    ....................................................................................................
A/Hong Kong/146/2013          ....................................................................................................
A/Hong Kong/5738/2014         ....................................................................................................
A/Setif/01/2015               .......................................I........................................................E...
A/Ukraine/501/2014            ....................................................................................................
A/Paris/2339/2014             ..........................................M.......X.................................................
A/Latvia/01-000586/2015       ..................................................X.................................................
A/Denmark/01/2015             ....................................................................................................
A/Victoria/5060/2014          ....................................................................................................
A/Hong Kong/7621/2014         ...................................................X................................................
A/Marrakech/79/2014           ............G.......................................................................................
A/Israel/O-7414/2014          ....................................................................................................
A/Rheinland-Pfalz/1/2015      ..................................................X.................................................

                                      210       220       230       240       250       260       270       280       290       300
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009               ATASFIYNGRLVDSVVSWSKEILRTQESECVCINGTCTVVMTDGSASGKADTKILFIEEGKIVHTSTLSGSAQHVEECSCYPRYPGVRCVCRDNWKGSNR
A/Victoria/361/2011           .........................................................K..........................................
A/Texas/50/2012               ....................................................................................................
A/Samara/73/2013              ....................................................................................................
A/Niedersachsen/4/2015        .....................V..............................................................................
A/Castilla La Mancha/7/2015   ..................................................V.................................................
A/Milano/162/2014             ....................D...............................................................................
A/Norway/0144/2015            ..................................................V.................................................
A/Malta/MX502071/2015         ....................D.........................................I.....................................
A/Croatia/1923/2014           ....................D.........................................I.....................................
A/South Africa/4655/2013      ....................................................................................................
A/Stockholm/6/2014            ....................D...............................................................................
A/Jordan/30483/2014           ....................D...............................................................................
A/Hong Kong/7624/2014         ....................D.............................................K.................................
A/Norway/466/2014             ....................D...............................................................................
A/Sweden/76/2014              ....................D...............................................................................
A/Rouen/2273/2014             ....................D...............................................................................
A/Switzerland/9715293/2013    ....................D...............................................................................
A/Hong Kong/146/2013          ....................................................................................................
A/Hong Kong/5738/2014         ....................D...............................................................................
A/Setif/01/2015               ....................D...............................................................................
A/Ukraine/501/2014            ....................D...............................................................................
A/Paris/2339/2014             ....................D.............................................K.................................
A/Latvia/01-000586/2015       ....................D.............................................K.................................
A/Denmark/01/2015             ....................D.............................................K.................................
A/Victoria/5060/2014          ....................D.............................................K.................................
A/Hong Kong/7621/2014         ....................D.......................N.T...................K.................................
A/Marrakech/79/2014           ....................D.......................N.....................K.................................
A/Israel/O-7414/2014          ....................D.............................................K.................................
A/Rheinland-Pfalz/1/2015      ....................D.............................................K.................................
```

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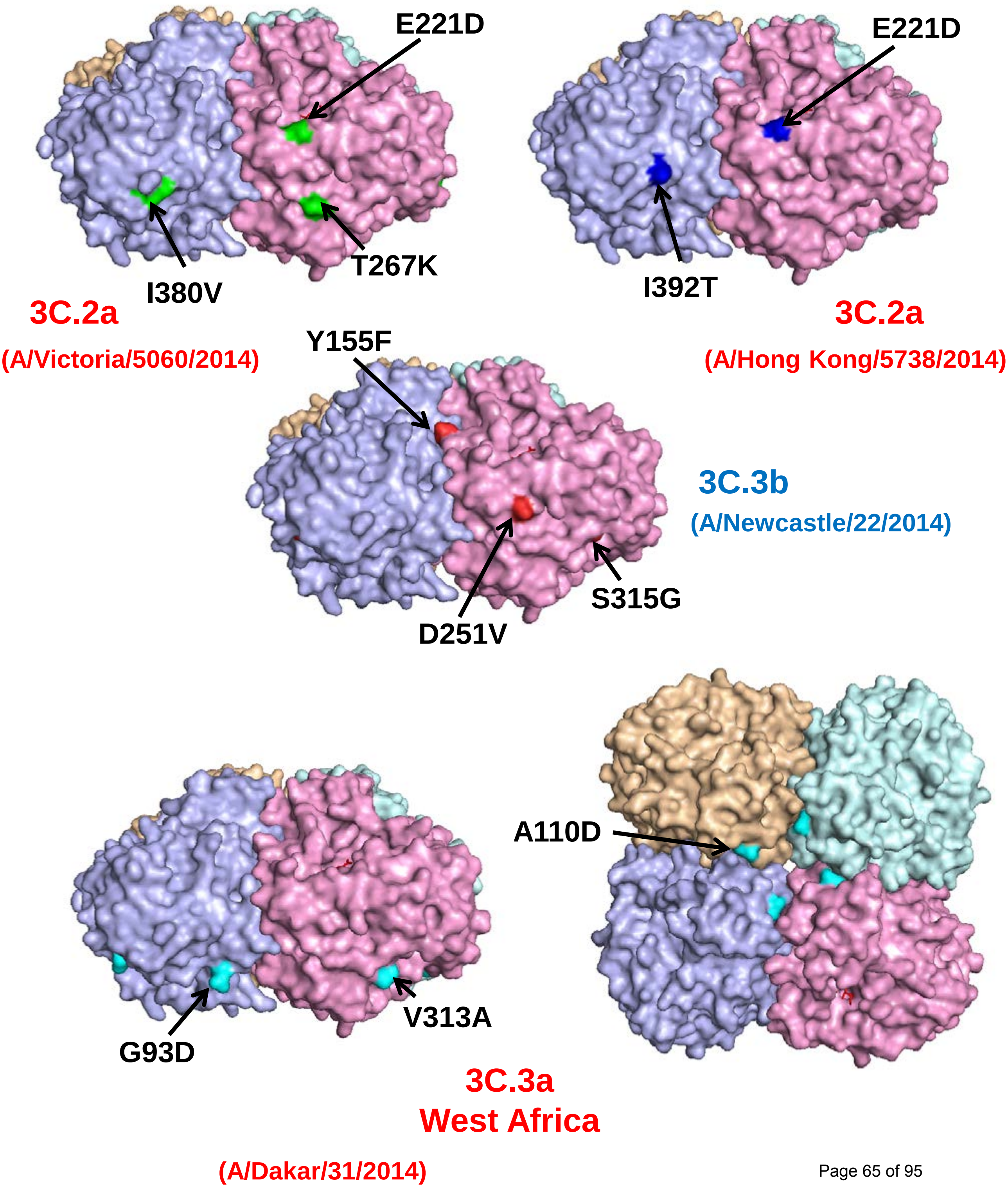
                                      310       320       330       340       350       360       370       380       390       400
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
A/Perth/16/2009               PIVDINIKDHSIVSSYVCSGLVGDTPRKNDSSSSSHCLDPNNEEGGHGVKGWAFDDGNDVWMGRTISEKSRLGYETFKVIEGWSNPKSKLQINRQVIVDR
A/Victoria/361/2011           ............................T......................................N.T..............................
A/Texas/50/2012               ...................................................................N.T..............................
A/Samara/73/2013              ...................................................................N.T..............................
A/Niedersachsen/4/2015        ..................................G...N............................N.T..............................
A/Castilla La Mancha/7/2015   ..............G....................................................N.T..............................
A/Milano/162/2014             ...................................................................N.T.....................M........
A/Norway/0144/2015            ...........V..G.............T............................D.........N.T..............................
A/Malta/MX502071/2015         ...................................................................N.T..............................
A/Croatia/1923/2014           ...................................................................N.T..............................
A/South Africa/4655/2013      ...................................................................N.T..............................
A/Stockholm/6/2014            ...................................................................N.T.....................T........
A/Jordan/30483/2014           ...................................................................N.T.....................T....L...
A/Hong Kong/7624/2014         ...................................................................N.T.....................T........
A/Norway/466/2014             ...................................................................N.T.....................T........
A/Sweden/76/2014              ...................................................................N.T.....................T........
A/Rouen/2273/2014             ...................................................................N.T.....................T........
A/Switzerland/9715293/2013    ...................................................................N.T.....................T........
A/Hong Kong/146/2013          ...................................................................N.T..............................
A/Hong Kong/5738/2014         ...................................................................N.T.....................T........
A/Setif/01/2015               ...................................................................N.T.....................T........
A/Ukraine/501/2014            ...................................................................N.T.....................T........
A/Paris/2339/2014             ...................................................................N.T.........V....................
A/Latvia/01-000586/2015       ...................................................................N.T.........V....................
A/Denmark/01/2015             ...................................................................N.A.........V....................
A/Victoria/5060/2014          ...................................................................N.T.........V....................
A/Hong Kong/7621/2014         ...................................................................N.T.........V....................
A/Marrakech/79/2014           ...................................................................N.T.........V....................
A/Israel/O-7414/2014          ...................................................................N.T.........V....................
A/Rheinland-Pfalz/1/2015      ...................................................................N.T.........V....................

                                      410       420       430       440       450       460
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....
A/Perth/16/2009               GNRSGYSGIFSVEGKSCINRCFYVELIRGRKEETEVLWTSNSIVVFCGTSGTYGTGSWPDGADINLMPI
A/Victoria/361/2011           .D.............................................................L.....
A/Texas/50/2012               .D.............................................................L.....
A/Samara/73/2013              .D........................T....................................L.....
A/Niedersachsen/4/2015        .D.............................................................L.....
A/Castilla La Mancha/7/2015   .D.............................................................L.....
A/Milano/162/2014             .D.............................................................L.F...
A/Norway/0144/2015            .D.............................................................L.....
A/Malta/MX502071/2015         .D...............................N.............................L.....
A/Croatia/1923/2014           .D...............................N.............................L.....
A/South Africa/4655/2013      .D.............................................................L.....
A/Stockholm/6/2014            .D.............................................................L.....
A/Jordan/30483/2014           .D.............................................................L.....
A/Hong Kong/7624/2014         .D.............................................................L.....
A/Norway/466/2014             .D.............................................................L.....
A/Sweden/76/2014              .D.............................................................L.....
A/Rouen/2273/2014             .D.............................................................L.....
A/Switzerland/9715293/2013    .D.............................................................L.....
A/Hong Kong/146/2013          .D.............................................................L.....
A/Hong Kong/5738/2014         .D.............................................................L.....
A/Setif/01/2015               .D.............................................................L.....
A/Ukraine/501/2014            .D.............................................................L.....
A/Paris/2339/2014             .D.............................................................L.....
A/Latvia/01-000586/2015       .D.............................................................L.....
A/Denmark/01/2015             .D.............................................................L.....
A/Victoria/5060/2014          .D.............................................................L.....
A/Hong Kong/7621/2014         .D.............................................................L.....
A/Marrakech/79/2014           .D.............................................................L.....
A/Israel/O-7414/2014          .D.............................................................L.....
A/Rheinland-Pfalz/1/2015      .D.............................................................L.....
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

Figure 16. NA amino acid substitutions in HA-defined genetic subgroups 3C.2a, 3C.3a and 3C.3b relative to A/Texas/50/2012



Influenza B viruses.

Influenza B viruses represented just over 27% of samples received with collection dates after 2014-08-31 (**Table 2**). B/Yamagata lineage viruses predominated over those of the B/Victoria lineage at a ratio of approximately 7:1.

B/Victoria lineage.

Influenza B viruses of the B/Victoria lineage were received from Cameroon, Germany and Ghana. The collection details and summary results are shown in **Table 11**.

HI results for the propagated viruses are shown in **Tables 12-1** to **12-4**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 12-2** to **12-4** are those with collection dates after 2014-08-31. The HA genetic group is indicated for those viruses that have been sequenced and those included in phylogenetic trees **(Figures 17 & 19)** are highlighted. All but a single cell-propagated test virus (B/Rheinland-Pfalz/1/2014) showed ≥8-fold reductions in HI titres compared with the titres for the homologous virus with post-infection ferret antisera raised against the recommended vaccine virus for quadrivalent live and inactivated vaccines, B/Brisbane/60/2008. Similarly, the test viruses were also poorly recognised by post-infection ferret antisera raised against the egg-propagated reference viruses B/Malta/636714/2011, B/South Australia/81/2012, and B/Johannesburg/3964/2012. In contrast, a higher proportion of the test viruses showed reactivity within 4-fold of the titres for the corresponding homologous viruses with antisera raised against viruses genetically closely related to B/Brisbane/60/2008 but propagated in cell culture. As shown in **Tables 12-1 to 12-4**, these antisera were raised against B/Paris/1762/2009, B/Hong Kong/514/2009, B/Odessa/3886/2010 and B/Formosa/V2367/2012. These viruses are considered to be surrogate cell-propagated antigens representing the egg-propagated prototype strain. However, a group of viruses from Cameroon, collected in November 2014, showed poorer recognition by antisera raised against these cell-propagated viruses.

Phylogenetic analysis of the HA gene of representative B/Victoria lineage viruses is shown in **Figure 17** and an alignment of the HA glycoproteins of a selection of these viruses is shown in **Figure 18**. Sequence analysis of the viruses from Cameroon and Ghana are still in progress. **Figure 19** shows a phylogenetic tree for the NA gene and **Figure 20** shows a corresponding amino acid sequence alignment for a selection of NA glycoproteins.

B/Rheinland-Pfalz/1/2014, which clusters closely with B/Brisbane/60/2008 in the phylogenetic trees, was isolated from a child recently vaccinated with a Live Attenuated Influenza Vaccine.

B/Yamagata lineage.

Influenza B viruses of the B/Yamagata lineage were received from countries in Europe, Africa/Indian Ocean (Algeria, Egypt, Ghana, Madagascar, Morocco, Senegal, Tunisia and Zambia), the Middle East (Iran and Jordan) and Hong Kong SAR. The collection details and results are summarised in **Table 13**.

The results of HI analyses for propagated viruses of the B/Yamagata lineage are shown in **Tables 14-1** to **14-8**. Test viruses in each table are sorted by date of collection and those below the red horizontal lines in **Tables 14-3** to **14-8** are viruses with collection dates after 2014-08-31. The clade into which the HA falls is shown, where known, and viruses for which gene sequences are included in phylogenetic trees are highlighted. Post-infection ferret antiserum raised against the egg-propagated vaccine virus B/Massachusetts/02/2012, recommended for use in the Northern Hemisphere for 2014/2015, recognised approximately 30% of viruses collected after 2014-08-31 at titres within 4-fold of the titre with the homologous virus. A ferret antiserum raised against a cell-propagated cultivar of B/Massachusetts/02/2012 recognised only 20% of viruses collected since 2014-08-31 at titres within 4-fold of its titre with the homologous virus. Another antiserum used in all HI assays, raised against B/Estonia/55669/2011 and belonging to the B/Massachusetts/02/2012 clade (clade 2), also recognised approximately 20% of the test viruses at a titre within 4-fold of the titre of the antiserum with the homologous virus. An antiserum raised against a previously recommended vaccine virus B/Wisconsin/1/2010 (clade 3) recognised all but one of the 110 test viruses at titres within 4-fold of the titre of the antiserum with the homologous virus. Approximately 97% of the test viruses were recognised similarly by antiserum raised against egg-propagated B/Stockholm/12/2011, a virus belonging to clade 3 represented by B/Wisconsin/1/2010 and B/Phuket/3073/2013. All test viruses were recognised at titres within 4-fold of the titre for the homologous virus, with 80% of titres being within 2-fold, by an antiserum raised against the egg-propagated vaccine virus B/Phuket/3073/2013 which was recommended for use in the Southern Hemisphere 2015 influenza season. Similarly, antiserum raised against another exclusively egg-propagated virus, B/Hong Kong/3417/2014, recognised all but one of the 91 test viruses (99%) collected since 2014-08-31 at titres within 2-fold of the titre for the homologous virus **(Tables 14-5** to **14-8)**.

An antiserum raised against a cell-propagated cultivar of B/Phuket/3073/2013 recognised just over 55% of the test viruses at titres within 4-fold of the titre for the homologous virus.

Figure 21 shows a phylogenetic analysis of the HA gene of representative B/Yamagata lineage viruses and an amino acid sequence alignment of a selection of the HA glycoproteins is shown in **Figure 22**. **Figure 23** shows a phylogenetic tree for NA genes of the same viruses represented in the HA phylogeny and **Figure 24** shows a corresponding amino acid sequence alignment for selected NA glycoproteins. The HA and NA genes of recently collected viruses fell into the B/Wisconsin/1/2010 - B/Phuket/3073/2013 clade (clade 3).

Antiviral Analysis.

No evidence for reduced sensitivity to the neuraminidase inhibitors zanamivir and oseltamivir was seen in the 114 influenza B viruses collected after 2014-08-31 **(Table 15).**

Conclusion.

Viruses of the B/Yamagata lineage have predominated. For viruses of the B/Victoria lineage most were antigenically similar to viruses closely related to B/Brisbane/60/2008. Viruses of the B/Yamagata lineage fell into genetic clade 3, the B/Wisconsin/10/2010 - B/Phuket/3073/2013 genetic clade, and were recognised well by antisera raised against egg-propagated vaccine/reference viruses from genetic clade 3.

Table 11. Summary of influenza B (Victoria-lineage) viruses analysed, collected after 2014-08-31

MONTH Country	B Victoria lineage				
	Number received[1]	Number propagated[2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]
2014					
SEPTEMBER					
Cameroon	1	1	1		
OCTOBER					
Cameroon	2	2	2		
Germany	1	1	1		
NOVEMBER					
Cameroon	6	6	2		4
Ghana	3	2	2		
DECEMBER					
Germany	3	1		1	
Ghana	3	1	1		
3 Countries	19	14	9	1	4
			64.3%	7.1%	28.6%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared

2. Propagated to sufficient titre to perform HI assay

3. Compared to homologous titres for ferret antisera raised against at least one of three tissue-culture grown B/Brisbane/60/2008-like equivalents (B/Paris/1762/2008, B/Hong Kong/514/2009 & B/Odessa/3886/2010)

Table 12-1. Antigenic analyses of influenza B viruses (Victoria lineage) 2014-09-17 + 2014-10-01

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre / Post infection ferret antisera: B/Bris[1,3] 60/08 Sh 522	B/Mal[2] 2506/04 F37/11	B/Bris[2] 60/08 F22/12	B/Paris[2] 1762/09 F07/11	B/HK[2] 514/09 F9/13	B/Odessa[2] 3886/10 F19/11	B/Malta[2] 636714/11 F33/11	B/Jhb[1] 3964/12 F01/13	B/For[2] V2367/12 F04/13	B/Sth Aus[2] 81/12 F41/13
	Genetic group			1A		1A	1A	1B	1B	1A	1A	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E4	2560	1280	80	10	10	<	80	160	40	160
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	80	320	80	80	80	320	320	160	640
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	5120	10	40	160	160	160	40	40	80	160
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	5120	20	80	160	160	320	80	160	160	320
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK4	5120	10	40	160	320	160	40	80	160	160
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	80	80	80	640	320	320	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	1280	1280	320	640	320	1280	1280	1280	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	80	320	80	80	80	160	160	320	640
B/South Australia/81/2012	1A	2012-11-28	E4/E1	2560	80	320	40	80	80	320	320	320	640
TEST VIRUSES													
B/Buenos Aires/10986152/2014	1A	2014-06-03	MDCK1/MDCK1	5120	10	20	80	160	160	40	80	160	160
B/Santa Fe/1211/2014	1A	2014-06-19	MDCK2/MDCK1	5120	20	10	160	160	160	40	<	160	160
B/Buenos Aires/10987831/2014	1A	2014-06-19	MDCK1/MDCK1	5120	80	320	80	80	80	320	320	320	1280
B/Yunnan-Gucheng/1948/2014	1A	2014-06-27	C2/MDCK1	2560	10	20	80	40	80	20	20	40	80
B/Buenos Aires/1115374/2014		2014-07-01	MDCK1/MDCK1	5120	20	40	160	160	320	80	80	160	320
B/Buenos Aires/10978558/2014		2014-07-04	MDCK1/MDCK1	5120	20	40	160	160	160	80	80	160	160
B/Buenos Aires/10978573/2014	1A	2014-07-07	MDCK1/MDCK1	5120	20	20	160	160	160	40	80	160	160

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Vaccine*

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 12-2. Antigenic analyses of influenza B viruses (Victoria lineage) 2015-01-14

Viruses	Genetic group	Collection date	Passage History	Haemagglutination inhibition titre – Post infection ferret antisera: B/Bris[1,3] 60/08 Sh 522	B/Mal[2] 2506/04 F37/11	B/Bris[2] 60/08 F22/12	B/Paris[2] 1762/09 F07/11	B/HK[2] 514/09 F19/13	B/Odessa[2] 3886/10 F19/11	B/Malta[2] 636714/11 F33/11	B/Jhb[4] 3964/12 F01/13	B/For[2] V2367/12 F04/13	B/Sth Aus[2] 81/12 F41/13	
	Genetic group			1A		1A	1A	1B	1B	1A	1A	1A	1A	
REFERENCE VIRUSES														
B/Malaysia/2506/2004		2004-12-06	E3/E4	2560	1280	80	10	10	<	80	160	40	160	
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	2560	80	320	80	80	80	320	320	160	640	
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	5120	10	40	160	160	160	40	40	80	160	
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	5120	20	80	160	160	320	80	160	160	320	
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK4	5120	10	40	160	320	160	40	80	160	160	
B/Malta/636714/2011	1A	2011-03-07	E4/E1	2560	80	320	80	80	80	640	320	320	640	
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	1280	1280	320	640	320	1280	1280	1280	1280	
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	5120	80	320	80	80	80	160	160	320	640	
B/South Australia/81/2012	1A	2012-11-28	E4/E1	2560	80	320	40	80	80	320	320	320	640	
TEST VIRUSES														
B/Baden-Wurttemberg/3/2014	1A	2014-10-24	C2/MDCK1	1280	<	<	160	80	80	40	<	40	160	
B/Rheinland-Pfalz/1/2014	1A	2014-12-16	C2/MDCK1	2560	80	320	40	80	40	320	320	320	1280	197/199 DET

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. <=<20

Vaccine*

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Sequences in phylogenetic trees

Table 12-3. Antigenic analyses of influenza B viruses (Victoria lineage) 2015-02-04

Viruses	Genetic group	Collection date	Passage History	B/Bris[1,3] 60/08 Sh 522 1A	B/Mal[2] 2506/04 F37/11	B/Bris[2] 60/08 F22/12 1A	B/Paris[2] 1762/09 F07/11 1A	B/HK[2] 514/09 F19/13 1B	B/Odessa[2] 3886/10 F19/11 1B	B/Malta[2] 636714/11 F33/11 1A	B/Jhb[4] 3964/12 F01/13 1A	B/For[2] V2367/12 F04/13 1A	B/Sth Aus[2] 81/12 F41/13 1A
				Haemagglutination inhibition titre									
				Post infection ferret antisera									
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	1280	640	80	<	10	<	80	160	80	160
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	1280	80	320	40	80	80	320	320	160	640
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	2560	<	20	80	80	160	20	<	80	80
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	1280	<	20	80	80	160	20	<	40	80
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK2	2560	<	20	80	80	160	40	<	80	160
B/Malta/636714/2011	1A	2011-03-07	E4/E1	1280	80	320	40	80	80	320	320	320	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	5120	640	1280	160	160	160	1280	1280	1280	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	40	160	40	40	40	160	160	320	640
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	80	320	80	80	80	320	320	320	1280
TEST VIRUSES													
B/Cameroon/14V-6811/2014		2014-09-26	SIAT1	2560	<	10	80	80	160	20	<	80	80
B/Cameroon/14V-7837/2014	1B	2014-10-27	MDCK2/MDCK1	1280	<	10	80	80	160	20	<	80	80
B/Cameroon/14V-7839/2014	1B	2014-10-30	MDCK2/MDCK1	1280	<	10	80	80	160	20	<	40	80
B/Ghana/DILI-14-0878/2014	1A	2014-11-04	C3/MDCK1	1280	<	10	40	40	80	10	<	40	80
B/Ghana/DILI-14-1032/2014	1A	2014-11-10	C2/MDCK1	1280	<	10	40	40	80	10	<	40	80

1. < = <40;   2. < = <10; 3. hyperimmune sheep serum; 4. <=<20

Vaccine*

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Table 12-4. Antigenic analyses of influenza B viruses (Victoria lineage) 2015-02-11

Viruses	Genetic group	Collection date	Passage History	B/Bris[1,3] 60/08 Sh 522	B/Mal[2] 2506/04 F37/11	B/Bris[2] 60/08 F22/12	B/Paris[2] 1762/09 F07/11	B/HK[2] 514/09 F19/13	B/Odessa[2] 3886/10 F19/11	B/Malta[2] 636714/11 F33/11	B/Jhb[4] 3964/12 F01/13	B/For[4] V2367/12 F04/13	B/Sth Aus[2] 81/12 F41/13
				Haemagglutination inhibition titre – Post infection ferret antisera									
	Genetic group			1A		1A	1A	1B	1B	1A	1A	1A	1A
REFERENCE VIRUSES													
B/Malaysia/2506/2004		2004-12-06	E3/E7	640	320	80	<	<	<	80	80	40	160
B/Brisbane/60/2008	1A	2008-08-04	E4/E3	640	80	320	40	80	80	320	320	160	1280
B/Paris/1762/2008	1A	2009-02-09	C2/MDCK2	1280	<	10	80	40	80	20	<	80	80
B/Hong Kong/514/2009	1B	2009-10-11	MDCK1/MDCK2	1280	<	10	80	40	160	10	<	40	80
B/Odessa/3886/2010	1B	2010-03-19	MDCK2/MDCK2	1280	<	20	40	40	80	20	<	80	80
B/Malta/636714/2011	1A	2011-03-07	E4/E1	640	80	320	40	80	40	320	320	320	640
B/Johannesburg/3964/2012	1A	2012-08-03	E1/E2	2560	320	1280	80	160	80	1280	1280	1280	1280
B/Formosa/V2367/2012	1A	2012-08-06	MDCK1/MDCK3	1280	40	160	40	40	40	160	160	160	640
B/South Australia/81/2012	1A	2012-11-28	E4/E2	1280	80	320	40	40	80	320	320	320	640
TEST VIRUSES													
B/Cameroon/14V-8365/2014	1A	2014-11-14	MDCK2	640	<	10	<	20	10	20	<	<	<
B/Cameroon/14V-8584/2014		2014-11-22	MDCK2	1280	<	10	80	40	80	20	<	40	80
B/Cameroon/14V-8787/2014		2014-11-24	MDCK2	1280	<	20	80	80	80	40	<	<	160
B/Cameroon/14V-8586/2014	1A	2014-11-24	MDCK2	640	<	10	<	20	10	20	<	<	<
B/Cameroon/14V-8640/2014		2014-11-25	MDCK1	640	<	10	<	20	10	10	<	<	<
B/Cameroon/14V-8639/2014	1A	2014-11-25	MDCK1	640	<	20	<	20	10	10	<	<	<
B/Ghana/DARI-14-0303/2014	1A	2014-12-29	MDCK1	1280	10	10	80	40	80	40	<	40	80

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. <=<20

Vaccine*

* B/Victoria-lineage virus recommended for use in quadravalent vaccines

Figure 17. Phylogenetic comparison of influenza B (Victoria-lineage) HA genes

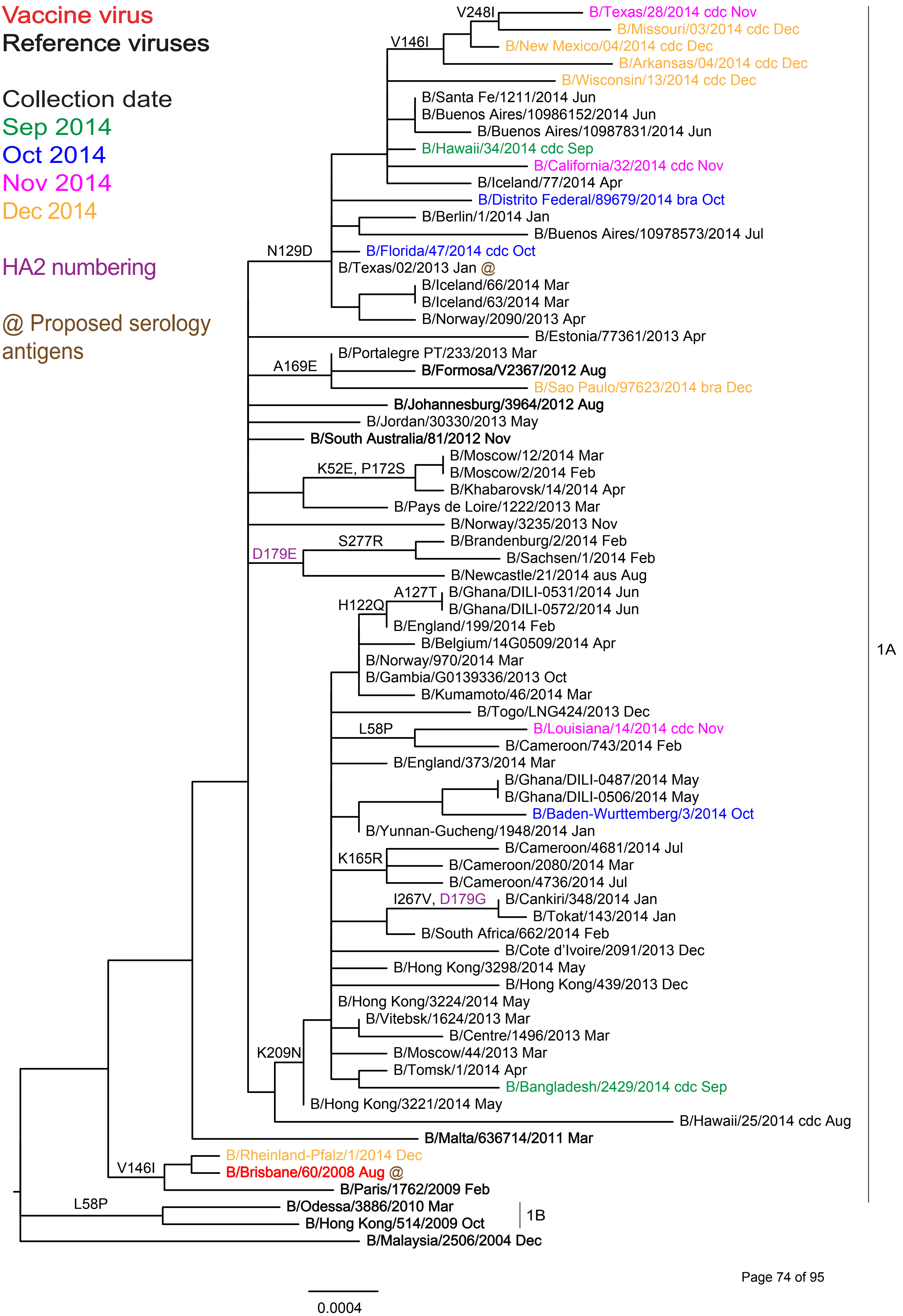


Figure 18. HA protein alignment for B (Victoria-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

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                                    10        20        30        40        50        60        70        80        90       100
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSHFANLKGTETRGKLCPKCLNCTDLDVALGRPKCTGNIPSARVSILHEVRPVTSGCFPIMHD
B/Hong Kong/514/2009         .........................................................P................K......................... 1B
B/Odessa/3886/2010           .........................................................P................K......................... 1B
B/Paris/1762/2009            ..........................................................................K......................... 1A
B/Brisbane/60/2008           ..........................................................................K......................... 1A
B/Malta/636714/2011          ..........................................................................K......................... 1A
B/Bangladesh/2429/2014       ..........................................................................K......................... 1A
B/Tomsk/1/2014 Apr           ..........................................................................K......................... 1A
B/Hong Kong/3298/2014        ........................................................................M.K......................... 1A
B/South Africa/662/2014      ..........................................................................K......................... 1A
B/Tokat/143/2014             ..........................................................................K......................... 1A
B/Cameroon/4681/2014         ..........................................................................K......................... 1A
B/Baden-Wurttemberg/3/2014   ..........................................................................K......................... 1A
B/Cameroon/743/2014          .........................................................P................K......................... 1A
B/Louisiana/14/2014          .........................................................P................K......................... 1A
B/Belgium/14G0509/2014       ..........................................................................K......................... 1A
B/Ghana/DILI-0572/2014       ..........................................................................K......................... 1A
B/Brandenburg/2/2014         ..........................................................................K......................... 1A
B/Khabarovsk/14/2014         ...................................................E......................K......................... 1A
B/South Australia/81/2012    ..........................................................................K......................... 1A
B/Johannesburg/3964/2012     ..............................................A...........................K......................... 1A
B/Formosa/V2367/2012         .................................A........................................K......................... 1A
B/Iceland/63/2014            ..........................................................................K......................... 1A
B/Texas/02/2013              ..........................................................................K......................... 1A
B/Buenos Aires/10978573/2014 ..........................................................................K......................... 1A
B/Iceland/77/2014            ..........................................................................K......................... 1A
B/Buenos Aires/10987831/2014 ..........................................................................K......................... 1A
B/Santa Fe/1211/2014         ..........................................................................K......................... 1A
B/Wisconsin/13/2014          ..........................................................................K......................... 1A
B/Missouri/03/2014           ..........................................................................K......................... 1A
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                                   110       120       130       140       150       160       170       180       190       200
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         RTKIRQLPNLLRGYEHIRLSTHNVINAENAPGGPYKIGTSGSCPNVTNGNGFFATMAWAVPKNDNNKTATNSLTIEVPYICTEGEDQITVWGFHSDNEIQ
B/Hong Kong/514/2009         ................................................................K......P..........................T.
B/Odessa/3886/2010           ................................................................K......P..........................T.
B/Paris/1762/2009            .............................................I..................K......P.................I........T.
B/Brisbane/60/2008           .............................................I..................K......P........................S.T.
B/Malta/636714/2011          ................................................................K......P........................S.T.
B/Bangladesh/2429/2014       ................V...............................................K......P..........................A.
B/Tomsk/1/2014 Apr           ................................................................K......P..........................T.
B/Hong Kong/3298/2014        ................................................................K......P..........................T.
B/South Africa/662/2014      ................................................................K......P..........................T.
B/Tokat/143/2014             ................................................................K......P..........................T.
B/Cameroon/4681/2014         ................................................................R......P..........................T.
B/Baden-Wurttemberg/3/2014   ................................................................KS.....P..........................T.
B/Cameroon/743/2014          ................................................................K......P..........................T.
B/Louisiana/14/2014          ................................................................K......P.................I........T.
B/Belgium/14G0509/2014       ................................................................K......P..........................T.
B/Ghana/DILI-0572/2014       .....................Q....T.....................................K......P..........................T.
B/Brandenburg/2/2014         ................................................................K......P..........................T.
B/Khabarovsk/14/2014         .................K..............................................K.................................T.
B/South Australia/81/2012    ................................................................K......P.......................X..X.
B/Johannesburg/3964/2012     ................................................................K......P........................X.T.
B/Formosa/V2367/2012         ................................................................K...E..P..........................T.
B/Iceland/63/2014            ............................D...................................K......P..........................T.
B/Texas/02/2013              ............................D...................................K......P........................X.X.
B/Buenos Aires/10978573/2014 ............................D...................................K......P..........................T.
B/Iceland/77/2014            ............................D...................................K......P..........................T.
B/Buenos Aires/10987831/2014 ............................D...................................K......P........................T.T.
B/Santa Fe/1211/2014         ............................D...................................K......P..........................T.
B/Wisconsin/13/2014          ............................D...................................K......P..........................T.
B/Missouri/03/2014           ............................D................I..................K......P..........................T.
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                                   210       220       230       240       250       260       270       280       290       300
                             ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004         MAKLYGDSKPQKFTSSANGVTTHYVSQIGGFPNQTEDGGLPQSGRIVVDYMVQKSGKTGTITYQRGILLPQKVWCASGRSKVIKGSLPLIGEADCLHEKY
B/Hong Kong/514/2009         ..............................................................................K.....................
B/Odessa/3886/2010           ....................................................................................................
B/Paris/1762/2009            ........................I.............................P.............................................
B/Brisbane/60/2008           ....................................................................................................
B/Malta/636714/2011          ....................................................................................................
B/Bangladesh/2429/2014       ........N.....................................................................K.....................
B/Tomsk/1/2014 Apr           ........N...........................................................................................
B/Hong Kong/3298/2014        ........N...........................................................................................
B/South Africa/662/2014      ........N...........................................................................................
B/Tokat/143/2014             ........N.........................................................V.................................
B/Cameroon/4681/2014         ........N...........................................................................................
B/Baden-Wurttemberg/3/2014   ........N...........................................................................................
B/Cameroon/743/2014          ........N..........................................M................................................
B/Louisiana/14/2014          ........N...........................................................................................
B/Belgium/14G0509/2014       ........N...........................................................................................
B/Ghana/DILI-0572/2014       ........N...........................................................................................
B/Brandenburg/2/2014         ............................................................................R.......................
B/Khabarovsk/14/2014         ....................................................................................................
B/South Australia/81/2012    ....................................................................................................
B/Johannesburg/3964/2012     ....................................................................................................
B/Formosa/V2367/2012         ....................................................................................................
B/Iceland/63/2014            ....................................................................................................
B/Texas/02/2013              ....................................................................................................
B/Buenos Aires/10978573/2014 ....................................................................................................
B/Iceland/77/2014            ....................................................................................................
B/Buenos Aires/10987831/2014 ....................................................................................................
B/Santa Fe/1211/2014         ....................................................................................................
B/Wisconsin/13/2014          ....................................................................................................
B/Missouri/03/2014           ...............................................I....................................................
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                                     310       320       330       340       350       360       370       380       390       400
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          GGLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLN
B/Hong Kong/514/2009          ....................................................................................................
B/Odessa/3886/2010            ....................................................................................................
B/Paris/1762/2009             ....................................................................................................
B/Brisbane/60/2008            ....................................................................................................
B/Malta/636714/2011           ....................................................................................................
B/Bangladesh/2429/2014        ....................................................................................................
B/Tomsk/1/2014 Apr            ....................................................................................................
B/Hong Kong/3298/2014         ....................................................................................................
B/South Africa/662/2014       ....................................................................................................
B/Tokat/143/2014              ....................................................................................................
B/Cameroon/4681/2014          ....................................................................................................
B/Baden-Wurttemberg/3/2014    ....................................................................................................
B/Cameroon/743/2014           ....................................................................................................
B/Louisiana/14/2014           ....................................................................................................
B/Belgium/14G0509/2014        ..............................................................................I.....................
B/Ghana/DILI-0572/2014        ....................................................................................................
B/Brandenburg/2/2014          ....................................................................................................
B/Khabarovsk/14/2014          ....................................................................................................
B/South Australia/81/2012     ....................................................................................................
B/Johannesburg/3964/2012      ....................................................................................................
B/Formosa/V2367/2012          ....................................................................................................
B/Iceland/63/2014             ....................................................................................................
B/Texas/02/2013               ....................................................................................................
B/Buenos Aires/10978573/2014  ....................................................................................................
B/Iceland/77/2014             ....................................................................................................
B/Buenos Aires/10987831/2014  ....................................................................................................
B/Santa Fe/1211/2014          ....................................................................................................
B/Wisconsin/13/2014           ....................................................................................................
B/Missouri/03/2014            .................V......................S...........................................................

                                     410       420       430       440       450       460       470       480       490       500
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          SLSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIA
B/Hong Kong/514/2009          ....................................................................................................
B/Odessa/3886/2010            ....................................................................................................
B/Paris/1762/2009             ............................D.......................................................................
B/Brisbane/60/2008            ....................................................................................................
B/Malta/636714/2011           ....................................................................................................
B/Bangladesh/2429/2014        ....................................................................................................
B/Tomsk/1/2014 Apr            ....................................................................................................
B/Hong Kong/3298/2014         ....................................................................................................
B/South Africa/662/2014       ....................................................................................................
B/Tokat/143/2014              ....................................................................................................
B/Cameroon/4681/2014          ....................................................................................................
B/Baden-Wurttemberg/3/2014    ....................................................................................................
B/Cameroon/743/2014           ....................................................................................................
B/Louisiana/14/2014           ....................................................................................................
B/Belgium/14G0509/2014        ....................................................................................................
B/Ghana/DILI-0572/2014        ....................................................................................................
B/Brandenburg/2/2014          ....................................................................................................
B/Khabarovsk/14/2014          ....................................................................................................
B/South Australia/81/2012     ....................................................................................................
B/Johannesburg/3964/2012      ....................................................................................................
B/Formosa/V2367/2012          ....................................................................................................
B/Iceland/63/2014             ....................................................................................................
B/Texas/02/2013               ....................................................................................................
B/Buenos Aires/10978573/2014  ....................................................................................................
B/Iceland/77/2014             .............................................................................I......................
B/Buenos Aires/10987831/2014  ....................................................................................................
B/Santa Fe/1211/2014          ....................................................................................................
B/Wisconsin/13/2014           ....................................................................................................
B/Missouri/03/2014            ....................................................................................................

                                     510       520       530       540       550       560       570
                              ....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004          AGTFDAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMIAIFVVYMVSRDNVSCSICL
B/Hong Kong/514/2009          ......................................................................
B/Odessa/3886/2010            ......................................................................
B/Paris/1762/2009             ......................................................................
B/Brisbane/60/2008            ......................................................................
B/Malta/636714/2011           ......................................................................
B/Bangladesh/2429/2014        ......................................................................
B/Tomsk/1/2014 Apr            ......................................................................
B/Hong Kong/3298/2014         ......................................................................
B/South Africa/662/2014       ....N.................................................................
B/Tokat/143/2014              .........................G............................................
B/Cameroon/4681/2014          ......................................................................
B/Baden-Wurttemberg/3/2014    ......................................................................
B/Cameroon/743/2014           ......................................................................
B/Louisiana/14/2014           ......................................................................
B/Belgium/14G0509/2014        ......................................................................
B/Ghana/DILI-0572/2014        ......................................................................
B/Brandenburg/2/2014          .........................E............................................
B/Khabarovsk/14/2014          ......................................................................
B/South Australia/81/2012     ......................................................................
B/Johannesburg/3964/2012      ......................................................................
B/Formosa/V2367/2012          ......................................................................
B/Iceland/63/2014             ......................................................................
B/Texas/02/2013               ......................................................................
B/Buenos Aires/10978573/2014  ......................................................................
B/Iceland/77/2014             ......................................................................
B/Buenos Aires/10987831/2014  ......................................................................
B/Santa Fe/1211/2014          ......................................................................
B/Wisconsin/13/2014           .....V................................................................
B/Missouri/03/2014            ......................................................................
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Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

Figure 19. Phylogenetic comparison of influenza B (Victoria-lineage) NA genes

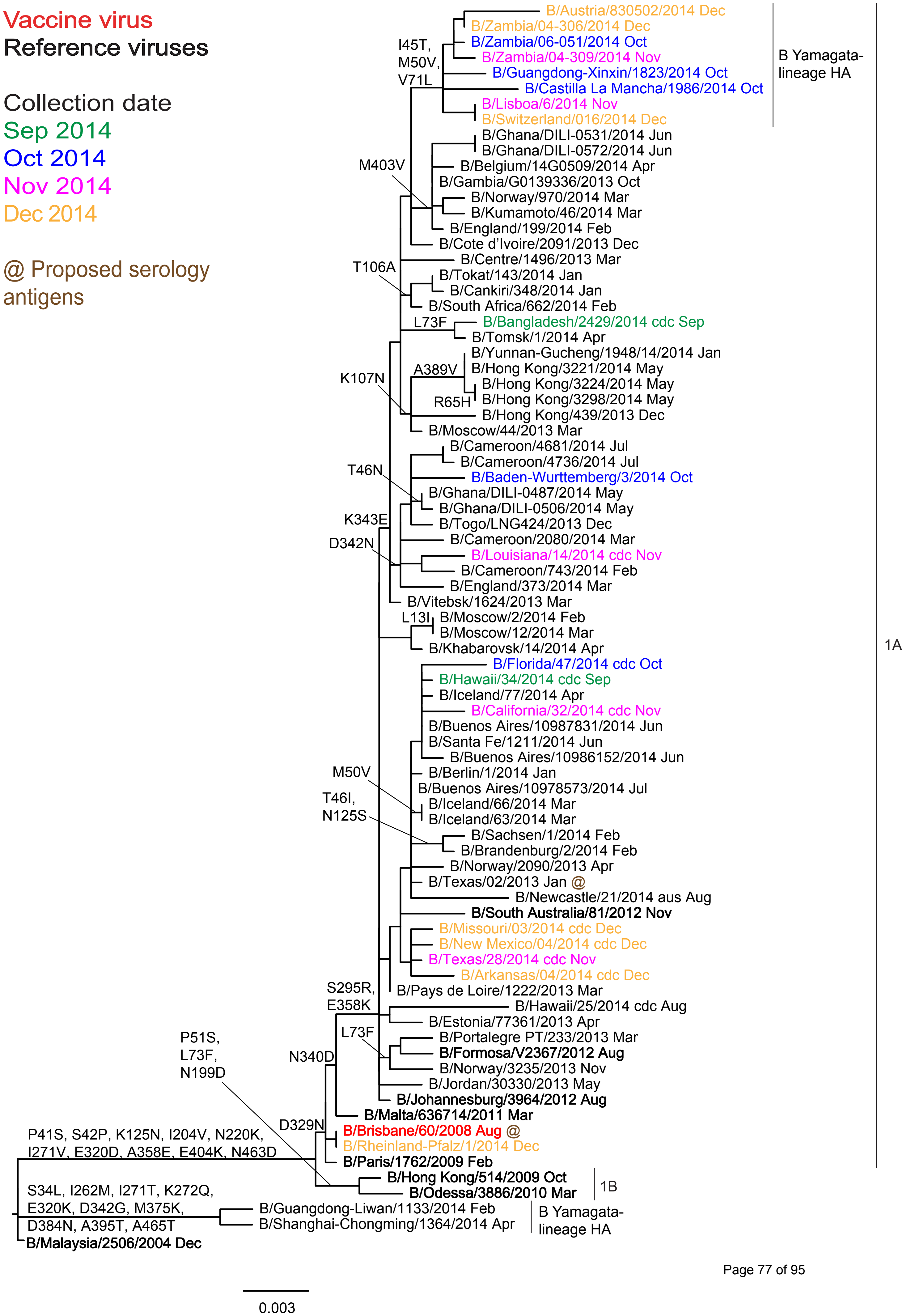

Figure 20. NA protein alignment for B (Victoria-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

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                                     10        20        30        40        50        60        70        80        90       100
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004            MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFPSTEITAPTMPLDCANASNVQAVNRSATKGVTLLLPEPEWTYPRLSCPGSTFQKALLISP
B/Hong Kong/514/2009            ........................................SP........S.....................S........................... 1B
B/Odessa/3886/2010              ..........................L.............SP........S.....................F........................... 1B
B/Paris/1762/2009               ........................................SP.......T.................................................. 1A
B/Brisbane/60/2008              ........................................SP.......................................................... 1A
B/Malta/636714/2011             ........................................SP......A................................................... 1A
B/Bangladesh/2429/2014          ........................................SP..............................F........................... 1A
B/Tomsk/1/2014                  ........................................SP..............................F........................... 1A
B/Hong Kong/3298/2014           ........................................SP......................H................................... 1A
B/South Africa/662/2014         ........................................SP.......................................................... 1A
B/Tokat/143/2014                ........................................SP.......................................................... 1A
B/Cameroon/4681/2014            ........................................SP.......................................................... 1A
B/Baden-Wurttemberg/3/2014      ........................................SP........Q................................................. 1A
B/Cameroon/743/2014             .......................................LSP.......................................................... 1A
B/Louisiana/14/2014             .......................M................SP.......................................................... 1A
B/Belgium/14G0509/2014          ........................................SP.......................................................... 1A
B/Ghana/DILI-0572/2014          ........................................SP.......................................................... 1A
B/Brandenburg/2/2014            ........................................SP...I...................................................... 1A
B/Khabarovsk/14/2014            ........................................SP.......................................................... 1A
B/South Australia/81/2012       ........................................SP........................................................N. 1A
B/Johannesburg/3964/2012        ........................................SP.......................................................... 1A
B/Formosa/V2367/2012            ........................................SP..............................F........................... 1A
B/Iceland/63/2014               ........................................SP.......V.................................................. 1A
B/Texas/02/2013                 ........................................SP.......................................................... 1A
B/Buenos Aires/10978573/2014    ........................................SP.......................................................... 1A
B/Iceland/77/2014               ........................................SP.......................................................... 1A
B/Buenos Aires/10987831/2014    ........................................SP.......................................................... 1A
B/Santa Fe/1211/2014            ........................................SP.......................................................... 1A
B/Missouri/03/2014              ........................................SP...............................R.......................... 1A
B/Switzerland/016/2014          ........................................SP..T....V....................L............................. C3Y

                                    110       120       130       140       150       160       170       180       190       200
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004            HRFGETKGNSAPLIIREPFIACGPKECKHFALTHYAAQPGGYYNGTRGDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDNNA
B/Hong Kong/514/2009            ........................N.........................................................................D.
B/Odessa/3886/2010              ........................N.........................................................................D.
B/Paris/1762/2009               ........................N...........................................................................
B/Brisbane/60/2008              ........................N...........................................................................
B/Malta/636714/2011             ........................N...........................................................................
B/Bangladesh/2429/2014          ........................N...........................................................................
B/Tomsk/1/2014                  ........................N...........................................................................
B/Hong Kong/3298/2014           ......N.................N...........................................................................
B/South Africa/662/2014         .....A..................N...........................................................................
B/Tokat/143/2014                .....A..................N...........................................................................
B/Cameroon/4681/2014            ........................N...........................................................................
B/Baden-Wurttemberg/3/2014      ........................N..........................R................................................
B/Cameroon/743/2014             ........................N...........................................................................
B/Louisiana/14/2014             ........................N...........................................................................
B/Belgium/14G0509/2014          ........................N...........................................................................
B/Ghana/DILI-0572/2014          ........................N...........................................................................
B/Brandenburg/2/2014            ........................S...........................................................................
B/Khabarovsk/14/2014            ........................N...........................................................................
B/South Australia/81/2012       ........................N...........................................................................
B/Johannesburg/3964/2012        ........................N...........................................................................
B/Formosa/V2367/2012            ........................N...........................................................................
B/Iceland/63/2014               ........................N...........................................................................
B/Texas/02/2013                 ........................N...........................................................................
B/Buenos Aires/10978573/2014    ........................N...........................................................................
B/Iceland/77/2014               ........................N...........................................................................
B/Buenos Aires/10987831/2014    ........................N...........................................................................
B/Santa Fe/1211/2014            ........................N...........................................................................
B/Missouri/03/2014              ........................N...........................................................................
B/Switzerland/016/2014          ........................N...........................................................................

                                    210       220       230       240       250       260       270       280       290       300
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004            LLKIKYGEAYTDTYHSYANNILRTQESACNCIGGNCYLMITDGSASGVSECRFLKIREGRIIKEIFPTGRIKHTEECTCGFASNKTIECACRDNSYTAKR
B/Hong Kong/514/2009            ...V...............K..................................................V.............................
B/Odessa/3886/2010              ...V...............K..................................................V.............................
B/Paris/1762/2009               ...V...............K..................................................V.............................
B/Brisbane/60/2008              ...V...............K..................................................V.............................
B/Malta/636714/2011             ...V...............K..................................................V.............................
B/Bangladesh/2429/2014          ...V...............K..................................................V.......................R.....
B/Tomsk/1/2014                  ...V...............K..................................................V.......................R.....
B/Hong Kong/3298/2014           ...V...............K..................................................V.......................R.....
B/South Africa/662/2014         ...V...............K..................................................V.......................R.....
B/Tokat/143/2014                ...V...............K..................................................V.......................R.....
B/Cameroon/4681/2014            ...V...............K..................................................V.......................R.....
B/Baden-Wurttemberg/3/2014      ...V...............K...................V..............................V.......................R.....
B/Cameroon/743/2014             ...V...............K..................................................V.......................R.....
B/Louisiana/14/2014             ...V...............K..................................................V.......................R.....
B/Belgium/14G0509/2014          ...V...............K..................................................V.......................R.....
B/Ghana/DILI-0572/2014          ...V...............K..................................................V.......................R.....
B/Brandenburg/2/2014            ...V...............K..................................................V.......................R.....
B/Khabarovsk/14/2014            ...V...............K..................................................V.......................R.....
B/South Australia/81/2012       ...V...............K..................................................V.......................R.....
B/Johannesburg/3964/2012        ...V...............K..................................................V.......................R.....
B/Formosa/V2367/2012            ...V...............K..................................................V.......................R.....
B/Iceland/63/2014               ...V...............K..................................................V.......................R.....
B/Texas/02/2013                 ...V...............K..................................................V.......................R.....
B/Buenos Aires/10978573/2014    ...V...............K..................................................V.......................R.....
B/Iceland/77/2014               ...V...............K..................................................V.......................R.....
B/Buenos Aires/10987831/2014    ...V...............K..................................................V.......................R.....
B/Santa Fe/1211/2014            ...V...............K..................................................V.......................R.....
B/Missouri/03/2014              ...V...............K..................................................V.......................R.....
B/Switzerland/016/2014          ...V...............K..................................................V.......................R.....
```

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                                        310       320       330       340       350       360       370       380       390       400
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Malaysia/2506/2004           PFVKLNVETDTAEIRLMCTETYLDTPRPDDGSITGPCESNGDKGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWADSDALAFSGVM
B/Hong Kong/514/2009           ...................D.....................................E..............E...........................
B/Odessa/3886/2010             ...................D.....................................E..............E...........................
B/Paris/1762/2009              ...................D.........N...........................E..............E...........................
B/Brisbane/60/2008             ...................D.........N...........................E..............E...........................
B/Malta/636714/2011            ...................D.........N.........D.................E..............E...........................
B/Bangladesh/2429/2014         ...................D.........N.........D..E..............K..............E...........................
B/Tomsk/1/2014                 ...................D.........N.........D..E..............K..............E...........................
B/Hong Kong/3298/2014          ...................D.........N.........D..E..............K..............E...............V...........
B/South Africa/662/2014        ...................D.........N.........D..E..............K..............E.....................V.....
B/Tokat/143/2014               ...................D.........N.........D..E..............K..............E...........................
B/Cameroon/4681/2014           ...................D.........N.........D.NE..............K..............E...........................
B/Baden-Wurttemberg/3/2014     ...................D.........N.........D.NE..............K..............E...........................
B/Cameroon/743/2014            ...................D.........N.........D.NE..............K..........I...E...........................
B/Louisiana/14/2014            ...................D.........N.........D.NE..............K..............E.....................T.....
B/Belgium/14G0509/2014         ...................D.........N.........D..E..............K..............E...........................
B/Ghana/DILI-0572/2014         ...................D.........N.........D..E..............K..............E...........................
B/Brandenburg/2/2014           ...................D.........N.........D.................K..............E...........................
B/Khabarovsk/14/2014           ...................D.........N.........D.................K..............E...........................
B/South Australia/81/2012      ...................D.........N.........D.................K..............E...........................
B/Johannesburg/3964/2012       ...................D.........N.........D.................K..............E...........................
B/Formosa/V2367/2012           ...................D.........N.........D.................K..............E...........................
B/Iceland/63/2014              ...................D.........N.........D.................K..............E...........................
B/Texas/02/2013                ...................D.........N.........D.................K..............E...........................
B/Buenos Aires/10978573/2014   ...................D.........N.........D.................K..............E...........................
B/Iceland/77/2014              ...................D.........N.........D.................K..............E...........................
B/Buenos Aires/10987831/2014   ...................D.........N.........D.................K..............E...........................
B/Santa Fe/1211/2014           ...................D.........N.........D.................K..............E...........................
B/Missouri/03/2014             ...................D.........N.........D.................K..............E...........................
B/Switzerland/016/2014         ...................D.........N.........D..E..............K..............E...........................

                                        410       420       430       440       450       460
                               ....|....|....|....|....|....|....|....|....|....|....|....|....|.
B/Malaysia/2506/2004           VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKETWHSAATAIYCLMGSGQLLWDTVTGVNMAL
B/Hong Kong/514/2009           ...K..........................................................D...
B/Odessa/3886/2010             ...K..........................................................D...
B/Paris/1762/2009              ...K..........................................................D...
B/Brisbane/60/2008             ...K..........................................................D...
B/Malta/636714/2011            ...K..........................................................D...
B/Bangladesh/2429/2014         ...K..........................................................D...
B/Tomsk/1/2014                 ...K..........................................................D...
B/Hong Kong/3298/2014          ...K..........................................................D...
B/South Africa/662/2014        ...K..........................................................D...
B/Tokat/143/2014               ...K..........................................................D...
B/Cameroon/4681/2014           ...K..........................................................D...
B/Baden-Wurttemberg/3/2014     ...K..........................................................D...
B/Cameroon/743/2014            ...K..........................................................D...
B/Louisiana/14/2014            ...K..........................................................D...
B/Belgium/14G0509/2014         ..VK.......................................................A..D...
B/Ghana/DILI-0572/2014         ..VK..........................................................D...
B/Brandenburg/2/2014           ...K..........................................................D...
B/Khabarovsk/14/2014           ...K..........................................................D...
B/South Australia/81/2012      ...K..........................................................D...
B/Johannesburg/3964/2012       I..K..........................................................D...
B/Formosa/V2367/2012           ...K..........................................................D...
B/Iceland/63/2014              ...K..........................................................D...
B/Texas/02/2013                ...K..........................................................D...
B/Buenos Aires/10978573/2014   ...K..........................................................D...
B/Iceland/77/2014              ...K..........................................................D...
B/Buenos Aires/10987831/2014   ...K..........................................................D...
B/Santa Fe/1211/2014           ...K..........................................................D...
B/Missouri/03/2014             .P.K..........................................................D...
B/Switzerland/016/2014         ...K..........................................................D...
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site
C3Y: Reassortant virus, B/Yamagata clade 3 HA and B/Victoria NA

Table 13. Summary of influenza B (Yamagata-lineage) viruses analysed, collected after 2014-08-31

MONTH	B Yamagata lineage				
Country	Number received[1]	Number propagated[2]	≤2 fold[3]	4 fold[3]	≥8 fold[3]
2014					
SEPTEMBER					
France	1	1	1		
Senegal	1	1	1		
OCTOBER					
Algeria	2	2	2		
Egypt	1	1		1	
France	1	1	1		
Madagascar	2	2	2		
Netherlands	1	1	1		
Spain	3	3	3		
United Kingdom	1	1	1		
Zambia	4	4	4		
NOVEMBER					
Belgium	2	1		1	
France	2	2	1	1	
Germany	1	1		1	
Hong Kong	3	3	1	2	
Norway	8	3	1	2	
Portugal	2	2	2		
Spain	1	1	1		
United Kingdom	1	1	1		
Zambia	1	1	1		
DECEMBER					
Austria	1	1	1		
Belgium	1	1		1	
Croatia	2	2	2		
Finland	2	2	2		
France	7	7	3	4	
Germany	1	1		1	
Greece	1	1	1		
Iran	2	2		2	
Italy	7	7	7		
Jordan	2	2	2		
Latvia	2	2	2		
Morocco	4	4	4		
Norway	7	4	3	1	
Portugal	7	7	7		
Spain	7	7	4	3	
Switzerland	2	2	2		
Tunisia	4	4	3	1	
Ukraine	10	10	10		
United Kingdom	1	1	1		
Zambia	1	1	1		
2015					
JANUARY					
Ghana	1	1		1	
Morocco	2	2	2		
Portugal	2	1	1		
Slovenia	1	1	1		
Spain	1	1	1		
Ukraine	1	1	1		
	117	107	85	22	0
28 Countries			79.4%	20.6%	0.0%

1. Numbers shown in red indicate that not all of the specimens received had been propagated at the time this report was prepared
2. Propagated to sufficient titre to perform HI assay
3. Reduction in HI titre with ferret antiserum raised against B/Phuket/3073/2013 (egg-grown vaccine virus)

Table 14-1. Antigenic analyses of influenza B viruses (Yamagata lineage) 2014-09-16 + 2014-10-03 + 2014-10-16

Viruses	Genetic Group	Collection date	Passage History	Haemagglution Inhibition Titre Post infection ferret antisera B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 02/12 Egg F2/13	B/Mass[2] 02/12 T/C F15/13	
	Genetic Group			1	1	2	3	3	2	3	2	2	2	
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	1280	1280	320	640	160	40	320	1280	320	
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	1280	160	40	640	1280	320	
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	320	320	320	640	10	40	80	640	80	
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	320	160	160	640	10	40	80	320	80	
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	320	80	160	160	1280	80	1280	320	1280	
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	2560	320	160	640	640	320	640	640	320	1280	
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	320	160	160	320	1280	160	1280	320	1280	
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	1280	1280	320	1280	160	40	640	1280	640	
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	640	640	80	640	1280	1280	
TEST VIRUSES														
B/Egypt/ALX6527/2014	3	unknown	MDCK0	2560	160	160	320	320	320	640	320	320	320	
B/Guangdong-Liwan/1133/2014	3	2014-02-12	E2/E2	160	40	20	40	40	<	10	20	80	40	196/198 NKA
B/Estonia/86059/2014	3	2014-04-01	MDCK1/MDCK1	1280	160	80	320	320	40	160	160	320	160	
B/Moscow/23/2014	3	2014-04-16	MDCK1/MDCK1	2560	320	160	640	1280	320	320	640	640	1280	
B/Santa Fe/1209/2014		2014-06-03	MDCK2/MDCK1	2560	160	80	80	160	640	40	640	320	640	
B/Buenos Aires/455/2014	3	2014-06-19	MDCK2/MDCK1	1280	160	80	320	320	40	160	160	320	160	
B/Buenos Aires/10974742/2014		2014-07-01	MDCK2/MDCK1	1280	160	80	320	320	40	160	160	320	160	
B/Santa Fe/1252/2014	2	2014-07-01	MDCK1/MDCK1	2560	160	160	160	160	640	40	640	320	640	
B/Chaco/5277/2014	2	2014-07-02	MDCK1/MDCK1	2560	160	160	160	320	640	40	640	320	640	
B/Santa Fe/1079/2014	2	2014-07-02	MDCK1/MDCK1	2560	160	80	80	160	320	20	640	320	320	
B/Santa Fe/1149/2014	3	2014-07-02	MDCK1/MDCK1	1280	160	80	640	320	40	80	160	320	160	
B/Buenos Aires/1109549/2014		2014-07-04	MDCK2/MDCK1	2560	320	160	320	640	80	160	320	320	320	
B/Buenos Aires/1114825/2014	3	2014-07-07	MDCK1/MDCK1	1280	160	80	320	640	80	160	160	320	160	

Vaccine
NH2013/14
SH 2014
NH 2014/15

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 14-2. Antigenic analyses of influenza B viruses (Yamagata lineage) 2014-10-22 + 2014-11-14

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Novo[2] 1/12 F31/12	B/HK[2] 3577/12 F33/12	B/Mass[2] 02/12 Egg F2/13	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 F33-34/14
				Haemagglution Inhibition Titre										
				Post infection ferret antisera										
Genetic Group				1	1	2	3	3	2	3	2	2	2	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	160	640	160	40	320	1280	160	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	320	40	320	1280	320	640
B/Wisconsin/1/2010	3	2010-02-20	E3/E2	1280	160	160	160	320	<	40	40	640	80	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	80	160	10	40	40	160	80	160
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	160	80	80	80	1280	80	1280	320	1280	160
B/Novosibirsk/1/2012	3	2012-02-14	C2/MDCK3	2560	160	160	320	320	320	320	320	320	320	640
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	1280	160	80	80	160	1280	80	1280	320	640	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	640	640	160	640	160	40	320	1280	320	640
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	640	640	80	640	1280	640	640
B/Phuket/3073/2013	3	2013-11-21	E4/E2	1280	320	80	320	320	20	40	40	320	80	320
TEST VIRUSES														
B/Egypt/ALX3964/2014	2	unknown	MDCK1	2560	320	160	160	320	1280	80	1280	320	1280	160
B/Egypt/ALX6641/2014	3	unknown	MDCK1	640	160	80	160	320	40	160	160	320	160	160
B/Egypt/GAB3930/2014	2	unknown	MDCK1	2560	320	320	160	320	1280	80	1280	640	1280	160
B/Jordan/30244/2014	3	2014-05-14	MDCK1/MDCK1	640	160	40	160	160	20	160	80	160	80	80
B/Jordan/20356/2014	3	2014-05-22	MDCK1/MDCK1	1280	160	40	160	160	20	160	80	160	80	80
B/Jordan/20358/2014	3	2014-05-22	MDCK1/MDCK1	1280	160	40	160	320	40	160	80	320	80	160
												Vaccine NH2013/14 SH 2014 NH 2014/15		Vaccine SH2015

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Table 14-3. Antigenic analyses of influenza B viruses (Yamagata lineage) 2014-12-19

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre										
				Post infection ferret antisera										
				B/Fl[1,3] 4/06	B/Fl[1] 4/06	B/Bris[2] 3/07	B/Wis[2] 1/10	B/Stock[2] 12/11	B/Estonia[2] 55669/11	B/HK[2] 3577/12	B/Mass[2] 02/12	B/Mass[2] 02/12	B/Phuket[2] 3073/13	B/Phuket[2] 3073/13
				SH479	F1/10	F21/12	F10/13	F12/12	F26/11	F33/12	Egg F2/13	T/C F15/13	Egg F33-34/14	T/C F35/14
	Genetic Group			1	1	2	3	3	2	2	2	2	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	160	320	160	160	640	160	160	20
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	320	640	320	320	1280	320	320	20
B/Wisconsin/1/2010	3	2010-02-20	E3/E1	1280	640	320	640	640	20	80	640	160	320	80
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	80	160	<	20	160	40	80	20
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	160	80	40	160	640	640	160	640	40	80
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	320	80	80	160	640	640	160	640	80	80
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	640	640	160	640	160	320	1280	320	320	10
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	320	640	640	640	640	160	80
B/Phuket/3073/2013	3	2013-11-21	E4/E2	1280	320	160	160	320	<	40	320	40	160	20
B/Phuket/3073/2013	3	2013-11-21	M2/M2	2560	320	160	320	320	160	320	320	320	160	640
TEST VIRUSES														
B/England/515/2014	3	2014-10-22	SIAT1/MDCK1	2560	320	160	320	640	320	320	320	320	160	1280
B/England/522/2014	3	2014-11-14	MDCK1/MDCK1	2560	160	80	160	320	160	320	160	320	160	640
											Vaccine NH2013/14 SH 2014 NH 2014/15		Vaccine SH2015	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 14-4. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-01-14

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/HK[2] 3577/12 F33/12	B/Mass[2] 02/12 Egg F28/13	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F33-34/14	B/Phuket[2] 3073/13 T/C F35/14
				Haemagglutination inhibition titre – Post infection ferret antisera										
Genetic Group				1	1	2	3	3	2	2	2	2	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	640	160	640	160	320	640	320	320	10
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	640	320	640	160	320	1280	320	320	10
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	1280	320	80	160	320	<	20	320	40	160	10
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	160	80	80	320	<	20	320	40	80	10
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	2560	80	40	20	80	640	640	160	640	40	40
B/Hong Kong/3577/2012	2	2012-06-13	MDCK4	2560	80	40	40	80	640	640	160	640	40	40
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	640	640	160	640	160	320	1280	320	320	10
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	320	160	320	640	640	320	640	160	20
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	320	80	160	320	10	40	160	40	160	20
B/Phuket/3073/2013	3	2013-11-21	M2/M2	1280	160	80	160	320	80	320	160	320	160	320
TEST VIRUSES														
B/Paris/2058/2014	3	2014-09-25	MDCK1	640	80	40	160	160	40	80	80	80	80	40
B/Asturias/1952/2014	3	2014-10-10	SIAT1/MDCK2	640	80	40	160	160	80	160	40	160	80	80
B/Pais Vasco/1978/2014	3	2014-10-21	SIAT1/MDCK2	640	80	20	80	160	40	80	40	160	80	160
B/Castilla La Mancha/1986/2014	3	2014-10-25	MDCK1	5120	320	160	320	640	320	640	320	640	320	80
B/Paris/2052/2014	3	2014-10-27	MDCK1	640	80	20	80	80	20	80	40	80	80	80
B/Thuringen/3/2014	3	2014-11-08	C2/MDCK1	640	80	20	80	80	20	40	40	40	40	160
B/Belgium/G634/2014	3	2014-11-17	MDCK1/MDCK1	640	80	20	80	160	20	80	40	80	40	40
B/Bretagne/2216/2014	3	2014-11-24	MDCK1/MDCK1	1280	160	40	160	160	80	80	80	160	80	40
B/Norway/3085/2014	3	2014-11-24	MDCK1	320	80	20	40	80	20	40	20	40	40	40
B/Centre/2213/2014	3	2014-11-25	MDCK1/MDCK1	640	80	20	80	80	20	80	40	80	40	40
B/Norway/3108/2014	3	2014-11-29	MDCK1	640	80	40	160	160	80	160	80	160	80	1280
B/Baden-Wurttemberg/2/2014	3	2014-12-01	C2/MDCK1	640	160	40	80	160	20	80	40	80	40	80
B/Paris/2232/2014	3	2014-12-02	MDCK1/MDCK1	320	80	20	80	80	20	40	40	80	40	320
B/Belgium/G668/2014	3	2014-12-02	MDCK1/MDCK1	640	80	20	80	160	20	40	40	40	40	80
B/Madrid/SO12928/2014	3	2014-12-05	MDCK1	320	80	20	80	80	10	40	40	40	40	40
B/Castilla La Mancha/2082/2014	3	2014-12-09	MDCK1	320	40	20	40	40	10	40	20	40	40	40
B/Castilla La Mancha/2083/2014	3	2014-12-11	MDCK1	320	40	20	40	40	10	40	20	40	40	40
											Vaccine NH2013/14 SH 2014 NH 2014/15		Vaccine SH2015	

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum

Sequences in phylogenetic trees

Table 14-5. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-01-21

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Mass[2] 02/12 Egg F28/13	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F33-34/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[2,4] 3417/14 St Judes F715/14
				Haemagglutination inhibition titre – Post infection ferret antisera										
Genetic Group				1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	1280	640	160	640	160	1280	320	320	<	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	640	320	640	160	1280	320	320	<	320
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	1280	320	160	160	320	<	320	80	320	20	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	1280	160	80	80	320	<	160	40	80	<	160
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	80	40	20	80	640	80	640	80	40	160
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	1280	640	160	640	160	1280	320	320	<	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	2560	640	320	160	320	320	320	640	160	40	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	320	160	160	320	<	160	80	160	20	320
B/Phuket/3073/2013	3	2013-11-21	M2/M2	1280	160	80	160	320	80	80	320	160	320	160
B/Hong Kong/3417/2014	3	2014-06-04	E5	160	80	40	80	80	<	40	40	20	<	160
TEST VIRUSES														
B/Guangdong-Xinxin/1823/2014	3	2014-10-08	MDCK2/SIAT1/SIAT1	2560	320	160	320	320	320	320	640	320	160	160
B/Netherlands/382/2014	3	2014-10-16	MDCK1/SIAT1	1280	160	40	160	160	80	80	160	80	640	320
A/Egypt/15/2014	3	2014-10-21	E2 /E1	640	80	20	80	80	<	40	40	40	1240	320
B/Anhui-Guichi/1736/2014		2014-11-03	MDCK2/SIAT1/SIAT1	2560	160	80	320	320	320	80	320	160	160	160
B/Hong Kong/3783/2014	3	2014-11-17	MDCK1/SIAT1	320	80	20	80	80	20	40	80	40	80	160
B/Hong Kong/3787/2014	3	2014-11-20	MDCK1/SIAT1	640	80	20	80	80	20	80	80	40	40	160
B/Hong Kong/3788/2014	3	2014-11-24	MDCK1/SIAT1	1280	160	40	160	160	40	160	160	80	160	160
B/Norway/3078/2014		2014-11-24	MDCK3	640	80	40	80	160	20	40	80	40	80	160
B/Extremadura/2067/2014	3	2014-11-27	MDCKx/MDCK2	640	160	40	160	320	40	80	320	160	80	160
B/Lyon-CHU/23.76/2014		2014-12-14	MDCK2/SIAT1	640	160	40	160	160	40	80	80	80	80	160
B/Lyon/2423/2014	3	2014-12-15	MDCK2/SIAT1	640	80	20	80	80	10	40	40	40	40	160
B/Lyon/2433/2014		2014-12-15	MDCK2/SIAT1	2560	160	80	160	160	640	160	640	80	80	160
B/Lyon-CHU/23.84/2014	3	2014-12-15	MDCK2/SIAT1	640	160	40	160	160	40	80	80	80	80	160
B/Lyon/2455/2014	3	2014-12-16	MDCK2/SIAT1	640	80	20	80	80	20	40	80	40	80	160
B/Clermont-Ferrand/2467/2014	3	2014-12-17	MDCK2/SIAT1	640	80	20	80	80	20	40	80	40	40	160
B/Ghom/90195/2014	3	2014-12-20	MDCK1/SIAT1	640	80	20	80	160	20	40	80	40	80	160
B/Ghom/90199/2014	3	2014-12-28	MDCK1/SIAT1	640	80	20	40	160	10	40	80	40	<	<
										Vaccine NH2013/14 SH 2014 NH 2014/15		Vaccine SH2015		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE-treated antiserum pre-absorbed with TRBC

Sequences in phylogenetic trees

Table 14-6. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-01-28

Viruses	Genetic Group	Collection date	Passage History	B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Mass[2] 02/12 Egg F28/13	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[2,4] 3417/14 St Judes F715/14
				Haemagglutination inhibition titre										
				Post infection ferret antisera										
	Genetic Group			1	1	2	3	3	2	2	2	3	3	3
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	1280	320	640	160	1280	320	160	<	320
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	640	640	320	1280	320	320	<	640
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	1280	320	320	640	320	<	160	80	160	20	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	160	160	160	320	<	160	80	80	<	160
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	160	160	80	80	1280	40	640	80	40	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	640	1280	320	320	160	1280	160	160	<	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	640	320	640	640	640	320	40	640
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	320	320	640	640	80	320	80	160	20	320
B/Phuket/3073/2013	3	2013-11-21	M2/M2	2560	320	320	640	640	160	160	320	320	640	320
B/Hong Kong/3417/2014	3	2014-06-04	E5	320	80	80	160	80	<	40	40	40	<	320
TEST VIRUSES														
B/Lisboa/5/2014		2014-11-17	SIAT1/MDCK1	1280	160	160	320	320	40	80	160	320	80	320
B/Lisboa/6/2014	3	2014-11-20	SIAT1/MDCK1	1280	320	320	640	640	80	160	320	640	40	640
B/Lisboa/7/2014		2014-12-01	SIAT1/MDCK1	1280	160	160	320	160	80	40	160	320	80	320
B/Andalucía/2092/2014	3	2014-12-02	SIAT2/MDCK1	1280	160	320	320	320	160	160	320	320	320	320
B/Khmelnitsky/15/2015	3	2014-12-04	MDCKx/MDCK1	640	160	160	160	160	40	40	80	160	80	640
B/Latvia/12-016677/2014	3	2014-12-06	C2/MDCK1	640	160	160	160	160	40	40	80	160	320	320
B/Ukraine/490/2014	3	2014-12-08	MDCKx/MDCK1	640	160	160	160	160	40	80	80	160	160	640
B/Milano/1/2014	3	2014-12-09	SIAT1/MDCK1	1280	160	160	320	160	80	80	160	160	320	640
B/Austria/830502/2014	3	2014-12-09	SIAT1/MDCK1	5120	320	320	640	640	320	320	640	640	320	320
B/Khmelnitsky/19/2015		2014-12-11	MDCKx/MDCK1	2560	320	320	640	640	80	160	320	640	160	320
B/Khmelnitsky/18/2015		2014-12-14	MDCKx/MDCK1	640	160	160	160	160	40	80	80	160	320	640
B/Madrid/15046/2014	3	2014-12-15	SIAT2/MDCK1	1280	160	320	320	320	160	160	320	320	320	640
B/Italy/FVG-01/2014	3	2014-12-15	SIAT1/MDCK1	640	80	160	160	80	20	40	80	80	80	320
B/Switzerland/016/2014	3	2014-12-16	MDCK1	640	160	160	320	160	20	80	80	80	1280	640
B/Khmelnitsky/14/2015	3	2014-12-16	MDCKx/MDCK1	640	160	160	160	160	40	80	80	160	320	320
B/Latvia/12-043751/2014	3	2014-12-16	C1/MDCK1	640	80	160	160	80	20	40	80	80	640	640
B/Switzerland/457/2014	3	2014-12-17	MDCK1	640	80	160	160	80	20	40	80	80	640	640
B/Lisboa/9/2014		2014-12-17	SIAT2/MDCK2	1280	160	160	320	160	160	160	160	320	40	320
B/Khmelnitsky/17/2015		2014-12-17	MDCKx/MDCK1	1280	160	160	320	320	40	80	160	320	40	320
B/Lisboa/10/2014	3	2014-12-18	SIAT1/MDCK1	1280	160	160	320	160	80	80	160	320	80	320
B/Khmelnitsky/16/2015		2014-12-19	MDCKx/MDCK1	1280	160	320	320	320	80	160	320	320	80	320
B/Ukraine/497/2014	3	2014-12-20	MDCKx/MDCK1	640	80	160	160	80	20	40	80	80	160	320
B/Roma/3/2014	3	2014-12-20	SIAT1/MDCK1	5120	320	320	640	640	320	320	640	640	160	320
B/Lisboa/12/2014		2014-12-22	SIAT2/MDCK1	1280	160	160	320	160	80	80	160	320	80	640
B/Roma/1/2014	3	2014-12-22	SIAT1/MDCK1	640	80	160	160	80	20	40	80	80	80	320
B/Roma/2/2014		2014-12-22	SIAT1/MDCK1	640	80	160	160	160	40	40	160	160	320	640
B/Khmelnitsky/13/2015	3	2014-12-24	MDCKx/MDCK1	1280	160	160	160	160	40	80	160	160	80	320
B/Lisboa/18/2014		2014-12-29	SIAT1/MDCK1	2560	160	160	640	320	160	160	320	640	80	320
B/Navarra/15036/2014	3	2014-12-29	SIAT2/MDCK1	1280	160	160	320	320	160	80	320	320	640	320
B/Italy/FVG-02/2014	3	2014-12-29	SIAT1/MDCK1	640	80	160	160	160	40	40	160	160	80	320
B/Lisboa/14/2014		2014-12-30	SIAT1/MDCK1	1280	160	160	640	320	80	160	320	640	40	320
B/Lisboa/17/2014	3	2014-12-30	SIAT1/MDCK1	1280	160	160	320	160	40	80	160	160	160	320
B/Greece/8/2014	3	2014-12-30	MDCK1	640	80	160	320	80	20	40	80	160	1280	640
B/Madrid/052/2015	3	2014-12-30	SIAT2/MDCK1	5120	320	320	640	640	320	320	640	1280	40	320
B/Ukraine/6/2015	3	2014-12-30	MDCKx/MDCK1	640	160	160	160	160	20	40	80	80	320	320
B/Roma/4/2014	3	2014-12-30	SIAT1/MDCK1	2560	320	320	320	320	160	160	640	640	1280	640
B/Ukraine/1/2015	3	2015-01-03	MDCKx/MDCK1	640	80	160	80	80	20	40	80	80	80	320
B/Lisboa/1/2015	3	2015-01-04	SIAT1/MDCK1	640	80	160	320	80	40	40	80	160	80	640
										Vaccine NH2013/14 SH 2014 NH 2014/15		Vaccine SH2015		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE-treated antiserum pre-absorbed with TRBC

Sequences in phylogenetic trees

Table 14-7. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-02-04

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre Post infection ferret antisera B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F26/11	B/Mass[2] 02/12 Egg F28/13	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[2,4] 3417/14 St Judes F715/14	B/Utah[2] 09/14 MDCK CDC F8/15
Genetic Group				1	1	2	3	3	2	2	2	3	3	3	3
REFERENCE VIRUSES															
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	1280	320	640	160	1280	320	320	40	320	640
B/Brisbane/3/2007	2	2007-09-03	E2/E2	5120	1280	1280	640	1280	160	1280	320	640	40	640	1280
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	1280	320	320	320	320	20	160	80	160	40	320	640
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	320	160	160	320	20	160	80	160	40	320	640
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	160	160	160	160	640	160	640	160	160	320	1280
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	640	1280	320	640	80	1280	160	320	20	320	640
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	320	640	320	640	1280	320	160	640	1280
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	320	320	320	320	40	320	80	160	40	320	640
B/Phuket/3073/2013	3	2013-11-21	M2/M2	2560	320	320	640	640	640	640	320	640	1280	320	1280
B/Hong Kong/3417/2014	3	2014-06-04	E5	320	160	80	80	80	20	80	40	80	<	160	320
TEST VIRUSES															
B/Utah/09/2014	3	2014-05-29	C2/MDCK1	2560	320	320	640	320	80	320	160	320	320	640	640
B/Dakar/02/2014	3	2014-09-29	C1/MDCK1	2560	320	160	320	320	160	320	160	160	320	320	ND
B/Zambia/04-294/2014	3	2014-10-11	MDCK2/MDCK1	1280	160	160	320	320	80	160	160	320	320	640	ND
B/Madagascar/3124/2014	3	2014-10-22	MDCK2/MDCK1	2560	320	320	640	640	320	640	640	640	640	640	ND
B/Madagascar/3129/2014	3	2014-10-24	MDCK1/MDCK1	2560	320	320	640	640	640	640	640	640	1280	640	ND
B/Zambia/02-348/2014		2014-10-24	MDCK2/MDCK1	2560	160	320	160	160	320	320	640	160	80	640	ND
B/Zambia/06-051/2014	3	2014-10-29	MDCK2/MDCK1	1280	160	160	320	320	80	160	160	320	160	640	ND
B/Zambia/06-046/2014	2	2014-10-30	MDCK2/MDCK1	2560	160	160	160	160	160	320	640	80	80	320	ND
B/Zambia/04-309/2014	3	2014-11-15	MDCK2/MDCK1	1280	160	160	320	320	160	320	320	320	320	320	ND
B/Jordan/30442/2014	3	2014-12-06	MDCK1/MDCK1	640	160	160	160	160	80	80	80	160	160	320	ND
B/Finland/470/2014	3	2014-12-10	MDCK1/MDCK1	1280	80	80	160	80	80	80	160	160	160	320	ND
B/Tunisia/14192/2014	3	2014-12-11	MDCK1	1280	160	160	320	320	80	160	160	160	160	320	ND
B/Zambia/04-306/2014	3	2014-12-11	MDCK2/MDCK1	1280	160	160	320	320	160	320	320	320	320	640	ND
B/Finland/471/2014	3	2014-12-15	MDCK1/MDCK1	1280	80	80	160	160	80	160	160	160	160	320	ND
B/Tunisia/14933/2014		2014-12-16	MDCK1	1280	160	160	320	160	80	160	160	160	160	640	ND
B/Meknes/32/2014	3	2014-12-16	MDCK1/MDCK1	1280	160	160	320	160	80	80	160	160	80	320	ND
B/Tunisia/14475/2014	3	2014-12-17	MDCK1	640	80	80	160	160	40	80	80	160	80	320	ND
B/Jordan/20555/2014	3	2014-12-17	MDCK1/MDCK1	1280	160	160	320	160	160	160	320	320	320	640	ND
B/Norway/3238/2014	3	2014-12-19	MDCK1	1280	160	160	160	160	80	160	160	160	160	320	ND
B/Oujda/96/2014	3	2014-12-23	MDCK1/MDCK1	1280	160	160	320	320	160	320	160	320	320	320	ND
B/Meknes/51/2014		2014-12-23	MDCK1/MDCK1	640	160	160	320	160	80	160	80	160	160	320	ND
B/Meknes/53/2014		2014-12-23	MDCK1/MDCK1	1280	160	160	320	320	160	160	80	160	320	320	ND
B/Croatia/1988/2014	3	2014-12-29	MDCKx/MDCK1	1280	160	160	320	160	80	160	80	160	160	640	ND
B/Croatia/1989/2014	3	2014-12-29	MDCKx/MDCK1	1280	160	160	320	320	80	320	160	320	320	640	ND
B/Meknes/102/2015	3	2015-01-07	MDCK1/MDCK1	1280	160	160	320	320	80	320	160	320	320	320	ND
B/Meknes/106/2015		2015-01-07	MDCK1/MDCK1	1280	160	160	320	320	160	160	160	320	320	320	ND
B/Valladolid/1/2015	3	2015-01-08	MDCK1	640	80	160	320	160	80	160	160	160	160	320	ND
B/Slovenia/135/2015	3	2015-01-09	MDCKx/MDCK1	1280	160	160	320	320	80	160	160	160	160	640	ND
										Vaccine NH2013/14 SH 2014 NH 2014/15		Vaccine SH2015			

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE-treated antiserum pre-absorbed with TRBC; ND = Not Done

Sequences in phylogenetic trees

Table 14-8. Antigenic analyses of influenza B viruses (Yamagata lineage) 2015-02-11

Viruses	Genetic Group	Collection date	Passage History	Haemagglutination inhibition titre Post infection ferret sera B/Fl[1,3] 4/06 SH479	B/Fl[1] 4/06 F1/10	B/Bris[2] 3/07 F21/12	B/Wis[2] 1/10 F10/13	B/Stock[2] 12/11 F12/12	B/Estonia[2] 55669/11 F32/12	B/Mass[2] 02/12 Egg F2/13	B/Mass[2] 02/12 T/C F15/13	B/Phuket[2] 3073/13 Egg F36/14	B/Phuket[2] 3073/13 T/C F35/14	B/HK[2,4] 3417/14 Egg St Judes F715/14
			Genetic Group	1	1	2	3	3	2	2	2	3	3	
REFERENCE VIRUSES														
B/Florida/4/2006	1	2006-12-15	E7/E1	5120	640	1280	160	640	80	1280	160	160	<	320
B/Brisbane/3/2007	2	2007-09-03	E2/E3	2560	320	640	80	320	80	640	160	160	<	160
B/Wisconsin/1/2010	3	2010-02-20	E3/E3	1280	320	320	160	320	20	320	40	160	<	320
B/Stockholm/12/2011	3	2011-03-28	E4/E1	2560	160	160	80	320	10	320	40	80	<	160
B/Estonia/55669/2011	2	2011-03-14	MDCK1/MDCK1	1280	80	160	80	80	320	160	640	80	80	320
B/Massachusetts/02/2012	2	2012-03-13	E3/E4	5120	640	640	80	320	80	640	160	160	<	320
B/Massachusetts/02/2012	2	2012-03-13	MDCK1/C2/MDCK3	5120	640	640	160	320	160	640	640	160	<	320
B/Phuket/3073/2013	3	2013-11-21	E4/E3	1280	160	160	160	320	20	320	40	160	<	320
B/Phuket/3073/2013	3	2013-11-21	M2/M2	2560	160	160	320	320	320	320	640	320	320	320
B/Hong Kong/3417/2014	3	2014-06-04	E5	160	80	80	80	80	<	160	20	40	<	160
TEST VIRUSES														
B/Wakayama/130/2014	3	2014-09-30	MDCK1/MDCK1/MDCK1	640	80	80	160	80	40	160	40	80	80	320
B/Algiers/09/2014	3	2014-10-12	C0/MDCK1	640	80	80	80	80	40	160	40	80	40	320
B/Boumerdes/21/2014	3	2014-10-12	C0/MDCK1	640	80	80	160	160	80	160	80	160	160	320
B/Tunisia/13757/2014		2014-12-03	MDCK1	320	40	40	40	40	20	80	40	40	40	160
B/Norway/3136/2014	3	2014-12-07	MDCK1	640	80	80	160	160	40	160	40	80	80	320
B/Norway/3147/2014		2014-12-10	MDCK1	640	80	80	80	80	40	160	80	80	80	160
B/Norway/3322/2014	3	2014-12-15	MDCK1	640	40	80	80	80	40	160	40	40	80	160
B/England/598/2014	3	2014-12-31	SIAT1/MDCK1	1280	80	80	160	160	80	160	80	160	160	160
B/Ghana/DILI-15-0011/2015	3	2015-01-06	MDCK1	640	80	80	160	80	20	160	40	40	80	160
										Vaccine NH2013/14 SH 2014 NH 2014/15		Vaccine SH2015		

1. < = <40; 2. < = <10; 3. hyperimmune sheep serum; 4. RDE-treated antiserum pre-absorbed with TRBC; ND = Not Done

Sequences in phylogenetic trees

Figure 21. Phylogenetic comparison of influenza B (Yamagata-lineage) HA genes

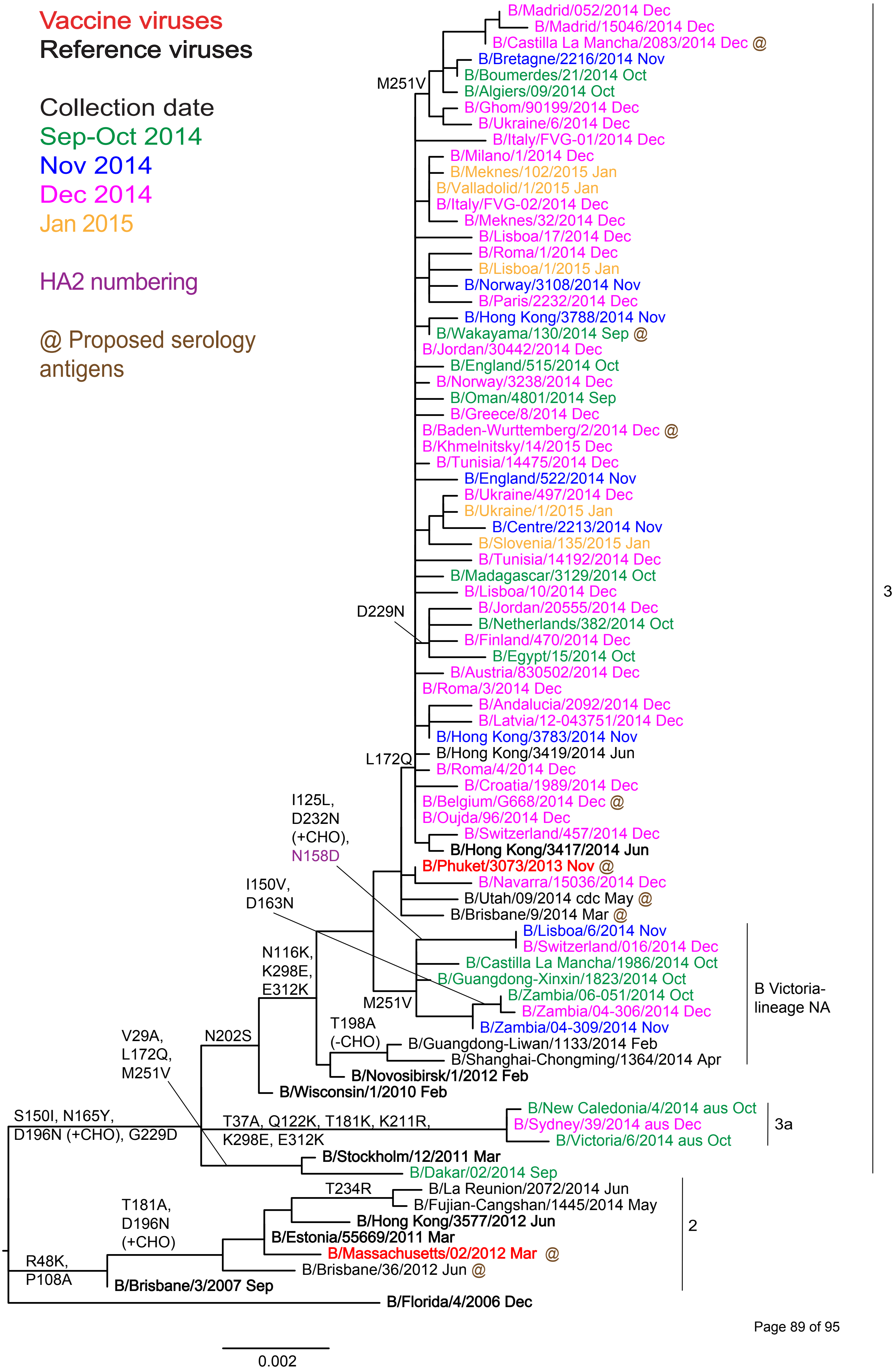

Figure 22. HA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

3a: these viruses form a distinct subgroup within clade 3 and are antigenically distinguishable. 3VN: these viruses are B/Yamagata HA clade 3 with B/Victoria NA.

```
                                        10        20        30        40        50        60        70        80        90       100
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006                DRICTGITSSNSPHVVKTATQGEVNVTGVIPLTTTPTKSYFANLKGTRTRGKLCPDCLNCTDLDVALGRPMCVGTTPSAKASILHEVKPVTSGCFPIMHD
B/Brisbane/3/2007               ...............................................K.......................................R............ 2
B/Massachusetts/02/2012         ...............................................K.......................................R............ 2
B/Estonia/55669/2011            ...............................................K.......................................R............ 2
B/Hong Kong/3577/2012           ...............................................K.......................................R............ 2
B/Dakar/02/2014                 ............................A..........................................................R............ 3
B/Stockholm/12/2011             ............................A..........................................................R............ 3
B/Victoria/6/2014               ....................................A..................................................R............ 3a
B/New Caledonia/4/2014          ....................................A..................................................R............ 3a
B/Wisconsin/1/2010              .......................................................................................R............ 3
B/Novosibirsk/1/2012            .......................................................................................R............ 3
B/Zambia/04/306/2014            .......................................................................................R............ 3VN
B/Castilla La Mancha/1986/2014  .......................................................................................R............ 3VN
B/Switzerland/016/2014          .......................................................................................R............ 3VN
B/Navarra/15036/2014            .......................................................................................R............ 3
B/Phuket/3073/2013              .......................................................................................R............ 3
B/Hong Kong/3417/2014           .......................................................................................R............ 3
B/Hong Kong/3783/2014           .......................................................................................R............ 3
B/Latvia/12/043751/2014         ..................V....................................................................R............ 3
B/Finland/470/2014              .......................................................................................R............ 3
B/Jordan/20555/2014             .......................................................................................R............ 3
B/Slovenia/135/2015             .......................................................................................R............ 3
B/Ukraine/1/2015                .......................................................................................R............ 3
B/Paris/2232/2014               .......................................................................................R............ 3
B/Lisboa/1/2015                 .......................................................................................R............ 3
B/Italy/FVG-02/2014             .......................................................................................R............ 3
B/Meknes/102/2015               .......................................................................................R............ 3
B/Ukraine/6/2014                .......................................................................................R............ 3
B/Bretagne/2216/2014            .......................................................................................R............ 3
B/Madrid/052/2014               .......................................................................................R............ 3

                                       110       120       130       140       150       160       170       180       190       200
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006                RTKIRQLPNLLRGYENIRLSTQNVIDAEKAPGGPYRLGTSGSCPNATSKSGFFATMAWAVPKDNNKNATNPLTVEVPYICTEGEDQITVWGFHSDDKTQM
B/Brisbane/3/2007               .......A............................................................................................
B/Massachusetts/02/2012         .......A........................................................................A...................
B/Estonia/55669/2011            .......A........................................................................A..............N....
B/Hong Kong/3577/2012           .......A........................................................................A..............N....
B/Dakar/02/2014                 .....................KT..........................I..............Y......Q.......................N....
B/Stockholm/12/2011             .................................................I..............Y......Q............................
B/Victoria/6/2014               .....................K...........................I..............Y...............K..............N....
B/New Caledonia/4/2014          .....................K...........................I..............Y...............K..............N....
B/Wisconsin/1/2010              .................................................I..............Y..............................N....
B/Novosibirsk/1/2012            ...............K.................................I..............Y..............................N....
B/Zambia/04/306/2014            ...............K.................................V............N.Y..............................N....
B/Castilla La Mancha/1986/2014  ...............K.................................I..............Y..............................N....
B/Switzerland/016/2014          ...............K........L........................I..............Y..............................N....
B/Navarra/15036/2014            ...............K.................................I..............Y..............................N....
B/Phuket/3073/2013              ...............K.................................I..............Y..............................X.X..
B/Hong Kong/3417/2014           ...............K.................................I..............Y......Q.......................N.N..
B/Hong Kong/3783/2014           ...............K.................................I..............Y......Q.......................N....
B/Latvia/12/043751/2014         ...............K.................................I..............Y......Q.......................N....
B/Finland/470/2014              ...............K.................................I..............Y......Q.......................N....
B/Jordan/20555/2014             ...............K.................................I..............Y......Q.......................N....
B/Slovenia/135/2015             ...............K.................................I..............Y......Q.......................N....
B/Ukraine/1/2015                ...............K.................................I..............Y......Q.......................N....
B/Paris/2232/2014               ...............K.................................I..............Y......Q.......................N....
B/Lisboa/1/2015                 ...............K.................................I..............Y......Q.......................N....
B/Italy/FVG-02/2014             ...............X.................................I..............Y......Q.......................N....
B/Meknes/102/2015               ...............K.................................I..............Y......Q.......................N....
B/Ukraine/6/2014                ...............K.................................I..............Y......Q.......................N....
B/Bretagne/2216/2014            ...............K.............................V...I..............Y......Q.......................N....
B/Madrid/052/2014               ...............K.................................I..............Y......Q.......................N....

                                       210       220       230       240       250       260       270       280       290       300
                                ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006                KNLYGDSNPQKFTSSANGVTTHYVSQIGSFPDQTEDGGLPQSGRIVVDYMMQKPGKTGTIVYQRGVLLPQKVWCASGRSKVIKGSLPLIGEADCLHEKYG
B/Brisbane/3/2007               ............................G.......................................................................
B/Massachusetts/02/2012         ............................G.......................................................................
B/Estonia/55669/2011            ............................G.......................................................................
B/Hong Kong/3577/2012           ............................G.......................................................................
B/Dakar/02/2014                 ............................D.....................V.................................................
B/Stockholm/12/2011             ............................D.....................V.................................................
B/Victoria/6/2014               ..........R.................D.....................................................................E..
B/New Caledonia/4/2014          ..........R.................D.....................................................................E..
B/Wisconsin/1/2010              .S..........................D.......................................................................
B/Novosibirsk/1/2012            .S..........................D.....................................................................E..
B/Zambia/04/306/2014            .S..........................D.....................V...............................................E..
B/Castilla La Mancha/1986/2014  .S..........................D.....................V...............................................E..
B/Switzerland/016/2014          .S..........................D..N..................V...............................................E..
B/Navarra/15036/2014            .S..........................D....K................................................................E..
B/Phuket/3073/2013              .S..........................D.....................................................................E..
B/Hong Kong/3417/2014           .S..........................D.....................................................................E..
B/Hong Kong/3783/2014           .S..........................D.....................................................................E..
B/Latvia/12/043751/2014         .S..........................D.....................................................................E..
B/Finland/470/2014              .S..........................N.....................................................................E..
B/Jordan/20555/2014             .S..........................N.....................................................................E..
B/Slovenia/135/2015             .S..........................D.....................................................................E..
B/Ukraine/1/2015                .S..........................D.....................................................................E..
B/Paris/2232/2014               .S..........................D.....................................................................E..
B/Lisboa/1/2015                 .S..........................D.....................................................................E..
B/Italy/FVG-02/2014             .S..........................D.....................................................................E..
B/Meknes/102/2015               .S..........................D.....................................................................E..
B/Ukraine/6/2014                .S..........................D.....................V...............................................E..
B/Bretagne/2216/2014            .S..........................D.....................V...............................................E..
B/Madrid/052/2014               .S..........................D.....................V...............................................E..
```

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                                          310       320       330       340       350       360       370       380       390       400
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006                 GLNKSKPYYTGEHAKAIGNCPIWVKTPLKLANGTKYRPPAKLLKERGFFGAIAGFLEGGWEGMIAGWHGYTSHGAHGVAVAADLKSTQEAINKITKNLNS
B/Brisbane/3/2007                ....................................................................................................
B/Massachusetts/02/2012          ....................................................................................................
B/Estonia/55669/2011             ....................................................................................................
B/Hong Kong/3577/2012            ....................................................................................................
B/Dakar/02/2014                  ....................................................................................................
B/Stockholm/12/2011              ....................................................................................................
B/Victoria/6/2014                ...........K........................................................................................
B/New Caledonia/4/2014           ...........K........................................................................................
B/Wisconsin/1/2010               ....................................................................................................
B/Novosibirsk/1/2012             ...........K........................................................................................
B/Zambia/04/306/2014             ...........K........................................................................................
B/Castilla La Mancha/1986/2014   ...........K........................................................................................
B/Switzerland/016/2014           ...........K........................................................................................
B/Navarra/15036/2014             ...........K........................................................................................
B/Phuket/3073/2013               ...........K........................................................................................
B/Hong Kong/3417/2014            ...........K........................................................................................
B/Hong Kong/3783/2014            ...........K........................................................................................
B/Latvia/12/043751/2014          ...........K........................................................................................
B/Finland/470/2014               ...........K........................................................................................
B/Jordan/20555/2014              ...........K........................................................................................
B/Slovenia/135/2015              ...........K........................................................................................
B/Ukraine/1/2015                 ...........K........................................................................................
B/Paris/2232/2014                ...........K........................................................................................
B/Lisboa/1/2015                  ...........K........................................................................................
B/Italy/FVG-02/2014              ...........K........................................................................................
B/Meknes/102/2015                ...........K........................................................................................
B/Ukraine/6/2014                 ...........K........................................................................................
B/Bretagne/2216/2014             ...........K........................................................................................
B/Madrid/052/2014                ...........K........................................................................................

                                          410       420       430       440       450       460       470       480       490       500
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006                 LSELEVKNLQRLSGAMDELHNEILELDEKVDDLRADTISSQIELAVLLSNEGIINSEDEHLLALERKLKKMLGPSAVEIGNGCFETKHKCNQTCLDRIAA
B/Brisbane/3/2007                .............................................................................D......................
B/Massachusetts/02/2012          .............................................................................D......................
B/Estonia/55669/2011             .............................................................................D......................
B/Hong Kong/3577/2012            .............................................................................D......................
B/Dakar/02/2014                  .............................................................................D......................
B/Stockholm/12/2011              .............................................................................D......................
B/Victoria/6/2014                .............................................................................D......................
B/New Caledonia/4/2014           .............................................................F...............D......................
B/Wisconsin/1/2010               .............................................................................D......................
B/Novosibirsk/1/2012             .............................................................................D......................
B/Zambia/04/306/2014             .............................................................................D......................
B/Castilla La Mancha/1986/2014   .............................................................................D......................
B/Switzerland/016/2014           .............................................................................D......................
B/Navarra/15036/2014             .............................................................................D......................
B/Phuket/3073/2013               .............................................................................D......................
B/Hong Kong/3417/2014            .............................................................................D......................
B/Hong Kong/3783/2014            .............................................................................D......................
B/Latvia/12/043751/2014          .............................................................................D......................
B/Finland/470/2014               .............................................................................D......................
B/Jordan/20555/2014              .............................................................................D......................
B/Slovenia/135/2015              .............................................................................D......................
B/Ukraine/1/2015                 ...I.........................................................................D......................
B/Paris/2232/2014                .............................................................................D......................
B/Lisboa/1/2015                  .............................................................................D......................
B/Italy/FVG-02/2014              .............................................................................D......................
B/Meknes/102/2015                .............................................................................D......................
B/Ukraine/6/2014                 .............................................................................D......................
B/Bretagne/2216/2014             .............................................................................D......................
B/Madrid/052/2014                .............................................................................D......................

                                          510       520       530       540       550       560
                                 ....|....|....|....|....|....|....|....|....|....|....|....|....|....
B/Florida/4/2006                 GTFNAGEFSLPTFDSLNITAASLNDDGLDNHTILLYYSTAASSLAVTLMLAIFIVYMVSRDNVSCSICL
B/Brisbane/3/2007                .....................................................................
B/Massachusetts/02/2012          .....................................................................
B/Estonia/55669/2011             .....................................................................
B/Hong Kong/3577/2012            .....................................................................
B/Dakar/02/2014                  .............................S.......................................
B/Stockholm/12/2011              .....................................................................
B/Victoria/6/2014                .....................................................................
B/New Caledonia/4/2014           .....................................................................
B/Wisconsin/1/2010               .....................................................................
B/Novosibirsk/1/2012             .....................................................................
B/Zambia/04/306/2014             .....................................................................
B/Castilla La Mancha/1986/2014   .....................................................................
B/Switzerland/016/2014           ...D.................................................................
B/Navarra/15036/2014             .....................................................................
B/Phuket/3073/2013               .....................................................................
B/Hong Kong/3417/2014            .....................................................................
B/Hong Kong/3783/2014            .....................................................................
B/Latvia/12/043751/2014          .....................................................................
B/Finland/470/2014               .....................................................................
B/Jordan/20555/2014              .................................................................P...
B/Slovenia/135/2015              .....................................................................
B/Ukraine/1/2015                 .....................................................................
B/Paris/2232/2014                .....................................................................
B/Lisboa/1/2015                  .....................................................................
B/Italy/FVG-02/2014              .....................................................................
B/Meknes/102/2015                .....................................................................
B/Ukraine/6/2014                 .....................................................................
B/Bretagne/2216/2014             .....................................................................
B/Madrid/052/2014                .....................................................................
```

Vaccine virus: Reference virus: Genetic group: HA1/HA2 boundary: Potential N-linked glycosylation site

3a: these viruses form a distinct subgroup within clade 3 and are antigenically distinguishable. 3VN: these viruses are B/Yamagata HA clade 3 with B/Victoria NA.

Figure 23. Phylogenetic comparison of influenza B (Yamagata-lineage) NA genes

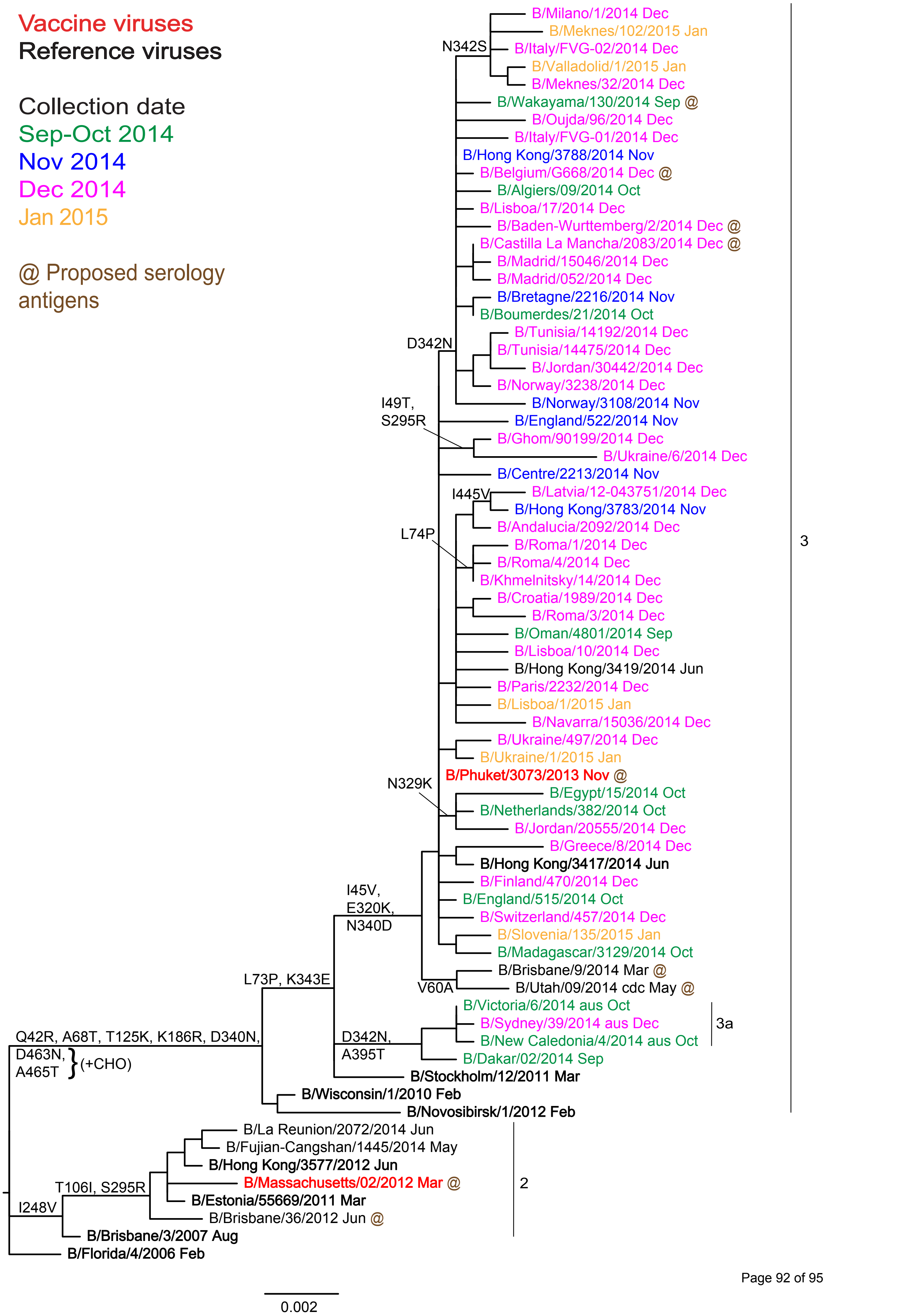

Figure 24. NA protein alignment for B (Yamagata-lineage) viruses.

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site
3a: these viruses form a distinct subgroup within clade 3 and are antigenically distinguishable.

```
                              10        20        30        40        50        60        70        80        90       100
                      ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006      MLPSTIQTLTLFLTSGGVLLSLYVSASLSYLLYSDILLKFSQTEITAPIMPLDCANASNVQAVNRSAAKGVTLLLPEPEWTYPRLSCPGSTFQKALLISP
B/Brisbane/3/2007     .......K............................................................................................ 2
B/Massachusetts/02/2012 ................................................................H................................... 2
B/Estonia/55669/2011  .................................................................................................... 2
B/Hong Kong/3577/2012 .................................................................................................... 2
B/La Reunion/2072/2014 .................................................................................................... 2
B/Dakar/02/2014       .........................................R.........................T....P........................... 3
B/Stockholm/12/2011   .........................................R.........................T....P........................... 3
B/Victoria/6/2014     .........................................R.........................T....P........................... 3a
B/New Caledonia/4/2014 .........................................R.........................T....P........................... 3a
B/Wisconsin/1/2010    .........................................R.........................T................................ 3
B/Novosibirsk/1/2012  .........................................R.........................T................................ 3
B/Navarra/15036/2014  .........................................R..V.T.........T..........T....P........................... 3
B/Phuket/3073/2013    .........................................R..V......................T....P........................... 3
B/Hong Kong/3417/2014 .........................................R..V......................T....P........................... 3
B/Croatia/1989/2014   .........................................R..V......................T....P........................... 3
B/Hong Kong/3783/2014 .........................................R..V......................T....P........................... 3
B/Latvia/12-043751/2014 .........................................R..V......................T....P........................... 3
B/Finland/470/2014    .........................................R..V......................T....P........................... 3
B/Jordan/20555/2014   ............I............................R..V......................T....P........................... 3
B/Slovenia/135/2015   .........................................R..V...................H..T....P........................... 3
B/Ukraine/1/2015      .........................................R..V......................T....P........................... 3
B/Tunisia/14475/2014  .........................................R..V......................T....P........................... 3
B/Paris/2232/2014     .........................................R..V......................T....P........................... 3
B/Lisboa/1/2015       .........................................R..V......................T....P..S........................ 3
B/Italy/FVG-02/2014   .........................................R..V......................T....P........................... 3
B/Meknes/102/2015     .........................................R..V.........V............T....P........................... 3
B/Ukraine/6/2014      .........................................R..V...T..................T....P........................... 3
B/Bretagne/2216/2014  .........................................R..V......................T....P........................... 3
B/Madrid/052/2014     .........................................R..V......................T....P........................... 3

                             110       120       130       140       150       160       170       180       190       200
                      ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006      HRFGETKGNSAPLIIREPFIACGPTECKHFALTHYAAQPGGYYNGTREDRNKLRHLISVKLGKIPTVENSIFHMAAWSGSACHDGKEWTYIGVDGPDSNA
B/Brisbane/3/2007     ....................................................................................................
B/Massachusetts/02/2012 .....I..............................................................................................
B/Estonia/55669/2011  .....I..............................................................................................
B/Hong Kong/3577/2012 .....I..............................................................................................
B/La Reunion/2072/2014 .....I..............................................................................................
B/Dakar/02/2014       ........................K............................................................R..............
B/Stockholm/12/2011   ........................K............................................................R..............
B/Victoria/6/2014     ........................K............................................................R..............
B/New Caledonia/4/2014 ........................K............................................................R..............
B/Wisconsin/1/2010    ........................K............................................................R..............
B/Novosibirsk/1/2012  ........................K............................................................R..............
B/Navarra/15036/2014  ........................K............................................................R..............
B/Phuket/3073/2013    ........................K............................................................R..............
B/Hong Kong/3417/2014 ........................K............................................................R..............
B/Croatia/1989/2014   ........................K............................................................R..............
B/Hong Kong/3783/2014 ........................K............................................................R..............
B/Latvia/12-043751/2014 ........................K............................................................R..............
B/Finland/470/2014    ........................K............................................................R..............
B/Jordan/20555/2014   ........................K............................................................R..............
B/Slovenia/135/2015   ........................K............................................................R..............
B/Ukraine/1/2015      ........................K............................................................R..............
B/Tunisia/14475/2014  ........................K............................................................R..............
B/Paris/2232/2014     ........................K...........T................................................R..............
B/Lisboa/1/2015       ........................K............................................................R..............
B/Italy/FVG-02/2014   ........................K............................................................R..............
B/Meknes/102/2015     ........................K............................................................R..............
B/Ukraine/6/2014      ........................K............................................................R..............
B/Bretagne/2216/2014  ........................K............................................................R..............
B/Madrid/052/2014     ........................K............................................................R..............

                             210       220       230       240       250       260       270       280       290       300
                      ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006      LLKIKYGEAYTDTYHSYAKNILRTQESACNCIGGDCYLMITDGPASGISECRFLKIREGRIIKEIFPTGRVKHTEECTCGFASNKTIECACRDNSYTAKR
B/Brisbane/3/2007     ...............................................V....................................................
B/Massachusetts/02/2012 ...............................................V..............................................R.....
B/Estonia/55669/2011  ...............................................V..............................................R.....
B/Hong Kong/3577/2012 ...............................................V..............................................R.....
B/La Reunion/2072/2014 ...............................................V..............................................R.....
B/Dakar/02/2014       ....................................................................................................
B/Stockholm/12/2011   ....................................................................................................
B/Victoria/6/2014     ....................................................................................................
B/New Caledonia/4/2014 ....................................................................................................
B/Wisconsin/1/2010    ....................................................................................................
B/Novosibirsk/1/2012  ....................................................................................................
B/Navarra/15036/2014  ....................................................................................................
B/Phuket/3073/2013    ....................................................................................................
B/Hong Kong/3417/2014 ....................................................................................................
B/Croatia/1989/2014   ....................................................................................................
B/Hong Kong/3783/2014 ....................................................................................................
B/Latvia/12-043751/2014 ...................................................................I................................
B/Finland/470/2014    ....................................................................................................
B/Jordan/20555/2014   ....................................................................................................
B/Slovenia/135/2015   ....................................................................................................
B/Ukraine/1/2015      ....................................................................................................
B/Tunisia/14475/2014  ....................................................................................................
B/Paris/2232/2014     ....................................................................................................
B/Lisboa/1/2015       ....................................................................................................
B/Italy/FVG-02/2014   ....................................................................................................
B/Meknes/102/2015     ....................................................................................................
B/Ukraine/6/2014      ..............................................................................................R.....
B/Bretagne/2216/2014  ....................................................................................................
B/Madrid/052/2014     ....................................................................................................
```

```
                                 310       320       330       340       350       360       370       380       390       400
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|....|
B/Florida/4/2006          PFVKLNVETDTAEIRLMCTETYLDTPRPNDGSITGPCESDGDKGSGGIKGGFVHQRMASKIGRWYSRTMSKTKRMGMGLYVKYDGDPWTDSEALALSGVM
B/Brisbane/3/2007         ....................................................................................................
B/Massachusetts/02/2012   ....................................................................................................
B/Estonia/55669/2011      ....................................................................................................
B/Hong Kong/3577/2012     ....................................................................................................
B/La Reunion/2072/2014    .................................................................................R..................
B/Dakar/02/2014           .......................................N.NE...................................................T.....
B/Stockholm/12/2011       .......................................N..E.........................................................
B/Victoria/6/2014         .......................................N.NE...................................................T.....
B/New Caledonia/4/2014    .......................................N.NE...................................................T.....
B/Wisconsin/1/2010        .......................................N............................................................
B/Novosibirsk/1/2012      .......................................N............................................................
B/Navarra/15036/2014      ...................K......................E.........................................................
B/Phuket/3073/2013        ...................K......................E.........................................................
B/Hong Kong/3417/2014     ...................K......................E.........................................................
B/Croatia/1989/2014       ...................K......................E.........................................................
B/Hong Kong/3783/2014     ...................K......................E.........................................................
B/Latvia/12-043751/2014   ...................K......................E.........................................................
B/Finland/470/2014        ...................K......................E.........................................................
B/Jordan/20555/2014       ...................K........K............EE.........................................................
B/Slovenia/135/2015       ...................K......................E.........................................................
B/Ukraine/1/2015          ...................K......................E.........................................................
B/Tunisia/14475/2014      ...................K.....................NE.........................................................
B/Paris/2232/2014         ...................K......................E.................T.......................................
B/Lisboa/1/2015           ...................K......................E.........................................................
B/Italy/FVG-02/2014       ...................K.....................SE.........................................................
B/Meknes/102/2015         ...................K.....................SE.........................................................
B/Ukraine/6/2014          ..............K....K......................E........................................G................
B/Bretagne/2216/2014      ...................K.....................NE.........................................................
B/Madrid/052/2014         ...................K.....................NE.........................................................

                                 410       420       430       440       450       460
                          ....|....|....|....|....|....|....|....|....|....|....|....|....|.
B/Florida/4/2006          VSMEEPGWYSFGFEIKDKKCDVPCIGIEMVHDGGKTTWHSAATAIYCLMGSGQLLWDTVTGVDMAL
B/Brisbane/3/2007         ..................................................................
B/Massachusetts/02/2012   ..................................................................
B/Estonia/55669/2011      ..................................................................
B/Hong Kong/3577/2012     ..................................................................
B/La Reunion/2072/2014    ..................................................................
B/Dakar/02/2014           ..............................................................N.T.
B/Stockholm/12/2011       ..............................................................N.T.
B/Victoria/6/2014         ..............................................................N.T.
B/New Caledonia/4/2014    ..............................................................N.T.
B/Wisconsin/1/2010        ..............................................................N.T.
B/Novosibirsk/1/2012      ..............................................................N.T.
B/Navarra/15036/2014      ..............................................................N.T.
B/Phuket/3073/2013        ..............................................................N.T.
B/Hong Kong/3417/2014     ...........................................V..................N.T.
B/Croatia/1989/2014       ..............................................................N.T.
B/Hong Kong/3783/2014     ............................................V.................N.T.
B/Latvia/12-043751/2014   ............................................V.................N.T.
B/Finland/470/2014        ..............................................................N.T.
B/Jordan/20555/2014       ..............................................................N.T.
B/Slovenia/135/2015       ..............................................................N.T.
B/Ukraine/1/2015          ..............................................................N.T.
B/Tunisia/14475/2014      ..............................................................N.T.
B/Paris/2232/2014         ..............................................................N.T.
B/Lisboa/1/2015           ..............................................................N.T.
B/Italy/FVG-02/2014       ..............................................................N.T.
B/Meknes/102/2015         ..............................................................N.T.
B/Ukraine/6/2014          ...........................................T..................N.T.
B/Bretagne/2216/2014      ..............................................................N.T.
B/Madrid/052/2014         ..............................................................N.T.
```

Vaccine virus: Reference virus: Genetic group: Potential N-linked glycosylation site

3a: these viruses form a distinct subgroup within clade 3 and are antigenically distinguishable.

Table 15. Antiviral (NAI) testing of isolates with collection dates in 2014 and 2015

Month of Collection	Number of viruses tested						
	A(H1N1)pdm09		A(H3N2)		Flu B		Total
	NI	RI/HRI	NI	RI/HRI	NI	RI/HRI	
2014							
January	131 (3)	1[a] + 1	183 (6)	1[d]	15		332
February	120 (12)	1[b]	142 (14)	5[e,f,g,h,i]	20 (2)		288
March	72 (19)		108 (21)	1[j]	29 (7)		210
April	26 (3)		60 (14)		9 (2)		95
May	10 (7)		21 (12)		17 (4)		48
June	8	1[c]	38 (30)		16 (5)		63
July	8 (6)		40 (32)		15 (10)		63
August			15 (12)				15
Totals	375 (50)	4	607 (141)	7	121 (30)		1114 (221)
September	4		6		3		13
October	8		27		17		52
November	5		27		21		53
December	52		76		66		194
2015							
January	11		15		7		33
Totals	80		151		114		345
Overall Totals	455 (50)*	4	758 (141)	7	235 (30)	0	1459 (221)

Viruses were subjected to phenotypic testing against oseltamivir and zanamivir

a A(H1N1)pdm09 virus (A/Belgium/14G0006/2014) carries I223R	Fold difference = os 40 zan 4	
b A(H1N1)pdm09 virus (A/Cyprus/F79/2014) carries S247N	Fold difference = os 7 zan 4	Outlier
c A(H1N1)pdm09 virus (A/Paris/1823/2014) carries H275Y	Fold difference = os 446 zan 3	
A(H1N1)pdm09 virus (A/Hungary/366/2014) carries H275Y	Not tested	
d A(H3N2) virus (A/Serbia/NS-613/2014) carries Q391H	Fold difference = os 8 zan 9	Outlier
e A(H3N2) virus (A/Milano/84/2014) carries V215I and S331R (-CHO)	Fold difference = os 10 zan 8	
f A(H3N2) virus (A/Ukraine/86/2014) carries S331R (-CHO)	Fold difference = os 10 zan 8	
g A(H3N2) virus (A/Stockholm/14/2014) carries I194V, S331R (-CHO) and E381G	Fold difference = os 26 zan 9	
h A(H3N2) virua (A/Israel/Z-1213/2014) carries E74K, I312T and N329K (-CHO)	Fold difference = os 13 zan 3	
i A(H3N2) virus (A/Ukraine/125/2014) carries I312T and S331R (-CHO)	Fold difference = os 10 zan 1	
j A(H3N2) virus (A/Calarasi/162804/2014) carries S331R (-CHO) and D339N	Fold difference = os 17 zan 1	

* Numbers in parentheses indicate the number of viruses assayed since the September 2014 VCM